

# Orchestron User Documentation

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# Orchestron User Documentation

This manual documents the current Orchestron user-facing functionality as implemented in the repository (frontend, backend API behavior visible to users, and feature history from the git log) as of **2026-09-07**, including the DAW audio-routing and persistent Perform mixer changes in commit `b487abd`.

It is organized by chapter and subchapter so you can read it front-to-back or jump directly to a workflow.

When Orchestron is served locally from the backend, opening `http://localhost:8000/` redirects to the client application at `/client`.

## Scope

- Visual instrument design (opcode graph editor, formulas, GEN tables, compile/runtime testing)
- Live performance workflow (instrument rack, audio mixer, Master, inserts, sends, sequencers, piano rolls, MIDI controllers, import/export)
- Configuration and runtime behavior (audio engine settings, MIDI setup, browser-clock audio, persistence)

## Reading Order

- Start with **Instrument Design** if you are creating patches.
- Continue with **Performance** to build and perform a multi-instrument setup.
- Use **Configuration** for audio engine tuning, MIDI setup, UI language/help, and browser-clock audio.

## Instrument Design

- [Patch Toolbar and Instrument Tabs](#)
- [Opcode Catalog and Integrated Documentation](#)
- [Graph Editor](#)
- [Input Formula Assistant](#)
- [GEN Table Editor \(GEN Meta-Opcode\)](#)
- [Runtime Panel and Compilation Workflow](#)
- [Instrument Import / Export and CSD Export](#)
- [Supported Opcodes](#)

## Performance

- [Instrument Rack and Engine Transport](#)
- [Audio Mixer and Routing](#)
- [Melodic Sequencers and Step Editing](#)
- [Pattern Pads, Queued Switching, and Pad Looper](#)
- [Controller Sequencers](#)
- [Arpeggiators](#)
- [Piano Rolls](#)
- [MIDI Controllers](#)
- [Performance Import / Export](#)
- [Live Status and Safety Controls](#)

## Configuration

- [GUI Language and Integrated Help](#)
- [Audio Engine Settings \(Config Page\)](#)
- [MIDI Setup and Inputs](#)
- [Browser-Clock Latency](#)



- [Persistence and Defaults](#)

## Feature Coverage Map

| Feature                                                                       | Where It Is Documented                                                                                                                                 |
|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Multi-language UI (EN/DE/FR/ES)                                               | <a href="#">GUI Language and Integrated Help</a>                                                                                                       |
| Integrated help pages and opcode docs modal                                   | <a href="#">GUI Language and Integrated Help</a> , <a href="#">Opcode Catalog and Integrated Documentation</a>                                         |
| Visual graph editor, zoom, selection, deletion                                | <a href="#">Graph Editor</a>                                                                                                                           |
| Stereo audio interfaces, built-in patch starters and draft audition           | <a href="#">Patch Toolbar</a> , <a href="#">Graph Editor</a> , <a href="#">Audition</a>                                                                |
| Persistent audio mixer, Master, inserts, sends and routing repair             | <a href="#">Audio Mixer and Routing</a>                                                                                                                |
| Input Formula Assistant (multi-input formulas + functions)                    | <a href="#">Input Formula Assistant</a>                                                                                                                |
| GEN meta-opcode editor (GEN01/GENpadsynth/etc.)                               | <a href="#">GEN Table Editor (GEN Meta-Opcode)</a>                                                                                                     |
| Instrument import/export bundles + CSD export                                 | <a href="#">Instrument Import / Export and CSD Export</a>                                                                                              |
| Performance rack, multi-instrument assignments, engine transport              | <a href="#">Instrument Rack and Engine Transport</a>                                                                                                   |
| Melodic sequencers, pattern pads, pad looper, transposition                   | <a href="#">Melodic Sequencers and Step Editing</a> , <a href="#">Pattern Pads</a> , <a href="#">Queued Switching</a> , and <a href="#">Pad Looper</a> |
| Controller sequencers and CC curve editor                                     | <a href="#">Controller Sequencers</a>                                                                                                                  |
| Backend arpeggiator virtual instruments and presets                           | <a href="#">Arpeggiators</a>                                                                                                                           |
| Piano rolls, scale/mode following, mixed-mode highlighting                    | <a href="#">Piano Rolls</a>                                                                                                                            |
| Manual MIDI controller panel (up to 6 knobs)                                  | <a href="#">MIDI Controllers</a>                                                                                                                       |
| Performance bundle import/export with conflict resolution                     | <a href="#">Performance Import / Export</a>                                                                                                            |
| Offline performance render export ( .csd + .mid ZIP or inline-score .csd ZIP) | <a href="#">Performance Import / Export</a>                                                                                                            |
| Audio engine settings ( sr , control_rate , ksmps , buffers)                  | <a href="#">Audio Engine Settings (Config Page)</a>                                                                                                    |
| Browser audio mode ( browser_clock PCM)                                       | <a href="#">Browser-Clock Latency</a>                                                                                                                  |
| Browser-clock latency tuning for unified browser audio                        | <a href="#">Browser-Clock Latency</a>                                                                                                                  |
| App-state persistence and defaults                                            | <a href="#">Persistence and Defaults</a>                                                                                                               |
| Supported opcode catalog (144 opcodes)                                        | <a href="#">Supported Opcodes</a>                                                                                                                      |

## File Format Quick Reference

- Instrument definition export: `.orch.instrument.json` or `.orch.instrument.zip`
- Performance export: `.orch.json` or `.orch.zip`
- Offline performance Csound export: `.csd.zip` (Export CSD (MIDI) includes a `.mid`; Export CSD (SCORE) embeds notes and controller sweeps in the score)
- Csound export from Instrument Design: `.csd`
- ZIP exports are used automatically when referenced GEN01 audio assets are included.



# Instrument Design

This chapter covers the complete **Instrument Design** workflow on the [Instrument Design](#) page.

Patches have an editable Instrument Type. Load Patch browses collapsed type groups and searches names and descriptions after four characters and a 500 ms pause. Type is retained when saving and importing/exporting instruments.

## What You Can Do Here

- Build instruments visually from Csound opcodes using the graph editor.
- Maintain multiple instrument tabs (parallel drafts or different patches).
- Use localized integrated help and opcode-level documentation without leaving the app.
- Choose built-in instrument, effect and output starters and define grouped stereo audio interfaces.
- Compile the current graph, inspect generated ORC, and audition drafts in isolation or in a temporary performance snapshot.
- Define advanced input-combine formulas when multiple signals feed the same input.
- Configure function tables with the GEN meta-opcode (including GEN01 audio-file tables and GENpadsynth).
- Export instruments as Orchestron bundle files and as .csd.

## Chapter Contents

- [Patch Toolbar and Instrument Tabs](#)
- [Opcode Catalog and Integrated Documentation](#)
- [Graph Editor](#)
- [Input Formula Assistant](#)
- [GEN Table Editor \(GEN Meta-Opcode\)](#)
- [Runtime Panel and Compilation Workflow](#)
- [Instrument Import / Export and CSD Export](#)
- [Supported Opcodes](#)

## Recommended Workflow

1. Create or load a patch in the patch toolbar.
2. Add opcodes from the catalog (click or drag-and-drop).
3. Connect signals and set constant values.
4. If needed, define an Input Formula on a socket with multiple inputs.
5. Compile and inspect the runtime panel output.
6. Save the patch.
7. Export a .csd or an Orchestron instrument bundle.

## Important Behavior

- The compile status badge is per active instrument tab/patch snapshot (compiled, pending changes, errors).
- Runtime starts collapsed to give more graph space; open it with Show runtime. Collapse Patch controls to hide metadata and file actions while keeping instrument tabs visible. Both choices survive view switches until browser reload.
- Saving a normal patch performs compile validation first; a failing compile prevents a bad save.
- Saving a patch marked Template? skips compile validation so incomplete starter graphs can be stored and reused. Template patches remain loadable in Instrument Design but are not selectable as Perform rack instruments.

## Screenshots



Orchestron Visual opcode patching with realtime CSound sessions and macOS MIDI loopback support.

GUI LANGUAGE  
EN DE FR ES

INSTRUMENT DESIGN

PERFORM

CONFIG

FM BELL

CLARINETTE

DEMO INSTRUMENT

PAD WITH VCOS AND LFOS

+

PATCH NAME

DESCRIPTION

LOAD PATCH

PAD with VCOS and LFOS

Stereo PAD with using vco2 opcode and vcomb

Current

NEW

CLONE

SAVE

COMPILE

EXPORT

IMPORT

EXPORT CSD

DELETE

OPCODE CATALOG

GRAPH EDITOR (37 NODES, 49 CONNECTIONS) (pending changes)

SHOW RUNTIME

Search opcode

Selected: 0 opcode(s), 0 connection(s)

**deltap**  
DELAY

**deltap3**  
DELAY

**fLanger**  
DELAY

**vcomb**  
DELAY

**vdelay3**  
DELAY

**vdelaysx**  
DELAY

**clip**  
DISTORTION

**distort1**  
DISTORTION

**fold**  
DISTORTION

**powershape**  
DISTORTION

**dam**  
DYNAMICS

**limit**  
DYNAMICS

**PAD with VCOS and LFOS**

Instrument Design page overview with catalog, graph editor, and runtime panel.



# Patch Toolbar and Instrument Tabs

The patch toolbar is the main control surface for patch metadata, patch library operations, and instrument-level file actions.

## Collapse Patch Controls

Use Patch controls to collapse metadata and load/save/compile/import/export actions. Instrument tabs, including add and close, stay visible. Controls start expanded; your choice survives view switches until browser reload.

## Patch Toolbar Layout

The toolbar includes:

- Instrument tabs (multiple editable patch workspaces)
- Patch metadata fields (Patch Name, Description)
- Instrument Type selector
- Grouped, searchable patch picker (Load Patch)
- Action buttons (New, Clone, Delete, Save, Compile, Export, Import, Export CSD)

## Instrument Tabs

- Use the + button to add a new instrument tab.
- Each tab stores its own editable patch snapshot (graph + metadata).
- Tabs can be switched without losing the current edit state.
- Tabs can be closed with the x button.
- Tabs are persisted in app state (see Configuration chapter), so drafts can be restored after reload.

## Patch Metadata

- Patch Name is the display name used in the patch library and performance rack dropdowns for normal patches.
- Description is a three-line patch note field and accepts up to 2048 characters. Hover over the field to see the full description in a tooltip.
- Template? marks the patch as a reusable starter graph. Template patches show a TEMPLATE token beside their name and are excluded from Perform rack instrument choices.
- Instrument Type offers Percussion, Melody, Bass, Effects / Noise, and Continuous (always-on). Choosing Continuous enables always-on scheduling; every other type uses MIDI activation. Continuous patches run when added to the Perform rack and started. They may generate audio without an inlet. Receiving audio requires an actual inlet, independently of activation. Musical role is separate interface metadata.
- Metadata updates affect the current tab immediately, but they are not stored in the patch library until you save.

## Patch Library Loading

- Load Patch opens collapsed type groups with counts. Expand a group to see patch names in alphabetical order; template patches retain their marker. Patches already open in instrument tabs are excluded.
- Search matches names and descriptions across all groups, ignoring case and surrounding spaces. Enter at least **four characters**; results update after **500 ms without typing**. Matching groups expand automatically. Shortening the search below four characters restores browsing immediately.
- Use Tab or Up/Down to move through controls and Enter to expand a group or load a patch. Enter in an active search loads its first result. Escape or clicking outside closes the picker; closing resets search and group expansion.
- Loading replaces the active tab with the selected saved patch.

## Actions



## New

- Opens the built-in creation chooser: Playable instrument, Drumset, Audio effect, Audio Output, or Empty patch.
- Creates an unsaved draft with the selected starter graph and default engine settings.
- This draft is not yet stored in the patch library until `Save` is used.

## New from Template

- Opens the same chooser with the five built-in starters plus any saved templates.
- Creates a new unsaved normal patch with the chosen template's graph and description.
- Built-in choices remain available when the library has no saved templates.

## Clone

- Creates a new saved patch by duplicating the current patch graph and metadata.
- Preserves the `Template?` flag and Instrument Type from the source patch.
- If the name already exists, Orchestron generates a `(... copy)` style name.
- The cloned patch is loaded automatically after creation.

## Delete

- Deletes the currently saved patch from the patch library.
- Requires confirmation.
- After delete, Orchestron loads another available patch or creates a new draft if none remain.

## Compile

- Compiles the current instrument graph into generated Csound ORC/CSD artifacts.
- Updates compile status badge state.
- Does not save the patch or start audio automatically. Use [Audition](#) in the graph header to hear an unsaved draft.

## Save

- Performs compile validation first for normal patches.
- If compile succeeds, persists the patch to the backend patch library.
- If compile fails, save is blocked and the patch remains unsaved.
- Template patches skip compile validation so incomplete starter graphs can be saved.
- Always-on patches still save as normal patch definitions; they run continuously only after they are added to a started Perform rack. A continuous source need not be an effect or have audio inlets.

## Export / Import

- `Export` writes an Orchestron instrument definition bundle (`.orch.instrument.json` or `.orch.instrument.zip`).
- `Import` loads an instrument definition bundle and supports conflict handling (overwrite / rename / skip).
- Details are in [Instrument Import / Export and CSD Export](#).

## Export CSD

- Triggers compile and downloads the compiled `.csd` file.
- Useful for running the generated instrument outside Orchestron.

## Practical Notes

- `Compile` is the fast iteration action while designing.
- `Save` is the persistence action for the patch library.
- `Export` is the sharing/transfer action for Orchestron-specific instrument bundles.
- `Export CSD` is the interoperability action for raw Csound usage.
- The integrated `?` help for this toolbar now focuses on tab-local draft state versus saved library state, plus the



difference between `Compile` , `Save` , `Clone` , and `export` actions.

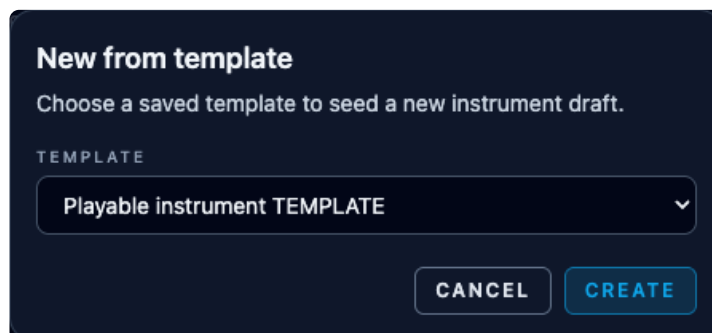
## Built-in creation choices

New and New from template offer Playable instrument (MIDI pitch/velocity, sine oscillator and madsr), Audio effect (stereo pass-through), Audio Output (stereo input to direct output), and Empty patch. These choices work without saved templates. Guided instruments connect to an fixed internal Master when added to a performance.

### Choose a starter

| Choice              | Type, activation and audio interface                                                   | Next step                                                                 |
|---------------------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Playable instrument | Melody; MIDI notes; named stereo output; pitch, velocity, sine oscillator and envelope | Save, add to Perform, and play its MIDI channel.                          |
| Drumset             | Percussion; MIDI notes; named stereo output; analog drum voices                        | Save, add to Perform, and sequence its drum notes.                        |
| Audio effect        | Continuous; named stereo input and output, initially pass-through                      | Add processing between the input and output before using it as an effect. |
| Audio Output        | Continuous; stereo input to Direct Audio Output ( outs )                               | Save for use as a custom output processor.                                |
| Empty patch         | Melody; empty graph for custom design                                                  | Build and compile the required signal path.                               |

Role describes the patch's audio interface; Instrument Type determines whether it runs continuously or on MIDI notes. Audio effect and Audio Output starters use Continuous; use Effects / Noise for MIDI-triggered sound effects. Every performance has a fixed internal Master. It has no editable patch or rack assignment; add processing through its mixer inserts.



*New opens the starter chooser even when no saved templates exist.*

## Older patches and saved metadata

Existing always-on patches become Continuous without changing playback. Other older patches are classified from their names first, then descriptions: drum/percussion terms precede bass terms, followed by melodic and sound-effect terms. Unmatched patches default to Melody. Review the inferred Instrument Type and change it when needed; later name or description edits do not reclassify a patch.

Types are saved with the patch and retained in clones, templates, restored tabs, and instrument/performance JSON or ZIP bundles. New playable and empty patches default to Melody, drumsets to Percussion, and effect/output starters to Continuous.

## Screenshots



STEREO RETURN × +

?

PATCH NAME

Stereo Return

DESCRIPTION

Warm filter-swept lead

LOAD PATCH

Current

TEMPLATE? ☐

ACTIVATION

Continuous

NEW

NEW FROM TEMPLATE

CLONE

SAVE

COMPILE

EXPORT

IMPORT

EXPORT CSD

DELETE

Earlier toolbar with a continuous stereo-effect draft and library actions. Instrument Type now replaces Activation; an updated screenshot is pending.



# Runtime Panel and Compilation Workflow

The Runtime panel supports live testing of the current patch and exposes compile/runtime diagnostics for the unified browser-clock runtime.

## Runtime Panel Sections

- MIDI Input selector
- Compile Output (generated ORC)
- Browser Audio status for browser-clock mode
- Session Events log (recent backend/runtime events)

## MIDI Input Binding

Use the MIDI Input dropdown to bind an external MIDI input to the active runtime session.

- This is how external hardware/DAW MIDI reaches the running instrument session.
- `internal:loopback` is always available as the built-in app-only loopback.
- Internal sequencers, piano rolls, and manual controller lanes always go straight to the session engine and do not depend on this binding.
- The selected input is session-specific and also reflected on the Performance page status footer.

See [MIDI Setup and Inputs](#) for OS-level MIDI setup guidance.

## Compile Output (Generated ORC)

After compiling a patch, the panel shows the generated ORC text.

This is useful for:

- Verifying the compiled signal chain
- Understanding how formulas/GEN/meta-opcodes are rendered
- Debugging compile issues with concrete generated code

A compilable patch must contain at least one audio output sink: either `outs` for direct stereo output or `outleta` for named audio routing into another instrument. This allows source instruments to feed always-on effects without also writing directly to the final stereo output.

## Session Events

The event list shows recent session events (most recent first, limited window in the UI).

Typical uses:

- Confirm that start/stop events happened
- Inspect runtime payloads and state transitions
- Diagnose issues during live testing

## Browser Audio

The runtime now always uses `browser_clock` audio mode, so the Runtime panel can show:

- Browser PCM queue/runtime status ( `connecting` , `live` , `error` )
- Error message when the controller socket or AudioWorklet path fails
- No `<audio>` player, because playback is owned directly by the browser `AudioContext`



See [Browser-Clock Latency](#) for the browser-clock mode guide and tuning workflow.

## Runtime Panel Collapse / Show

Runtime starts collapsed. Open it with Show runtime in the graph header and close it with Hide. Your choice survives view switches until browser reload; compilation and playback continue while it is hidden.

To maximize graph editor space:

- Click `Hide` in the Runtime panel to collapse it
- Use `Show runtime` in the graph header to bring it back

This is especially useful on smaller displays or complex patches.

## Compile Status Badge (Graph Header)

The instrument page header above the graph displays a compile-state badge for the current patch/tab snapshot:

- `compiled` - the current graph snapshot matches the last successful compile
- `pending changes` - the graph changed since the last successful compile
- `errors` - the last compile for this exact snapshot failed

This helps you avoid assuming a graph is live-valid after edits.

## Recommended Test Loop

1. Edit the graph
2. Compile
3. Read compile output / diagnostics
4. Open Audition, choose a preview context, then Prepare audition
5. Use Stop and audition to hear the draft; close the preview when finished
6. Save only after compile is clean

The integrated `?` help for this panel now focuses on session-specific external MIDI binding, generated ORC inspection, runtime event feedback, and browser-clock PCM playback.

## Audition a draft

Compile validates the graph without starting audio. Audition offers Isolated and In current performance. An isolated effect receives the built-in test instrument or a selected saved source. Current-performance audition replaces one selected rack instance in a temporary snapshot; choose the instance explicitly when a patch appears more than once.

Prepare compiles the temporary preview before interrupting the performance. Stop and audition explicitly stops the original performance and starts the preview. Closing deletes the temporary session and restores the original performance state and playhead with transport stopped. Draft patches, test sources and preview routes are never saved into the library.

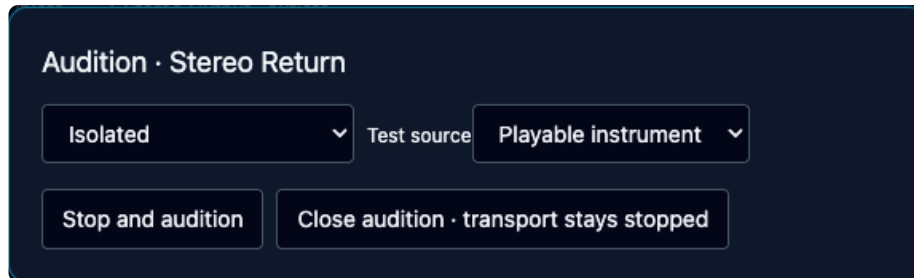
### Preview step by step

1. Click **Audition** in the graph header with the draft you want to hear.
2. Choose **Isolated** to test only this patch. For an effect, select **Test source**: the built-in Playable instrument or a saved source with suitable outputs. Isolated preview needs a stereo main output; source outputs must match the effect input count.
3. Alternatively choose **In current performance** and select the rack instance in **Source**. The short ID distinguishes instances of the same patch, including a return and a separate insert. The draft replaces only that selected instance in the temporary performance snapshot.
4. Click **Prepare audition**. Compilation errors appear in the dialog; fix the draft and prepare again. Preparation does not interrupt the original performance.

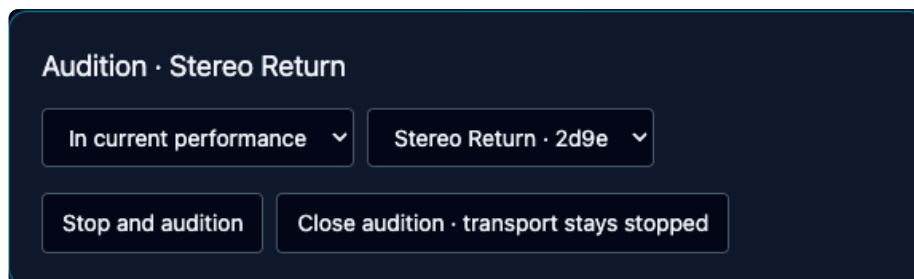


5. Click **Stop and audition** to stop the original engine and start the compiled preview. Note-triggered preview sources receive a test note (MIDI 60, velocity 96); current-performance mode also uses the performance sequencer configuration.
6. Click **Close audition · transport stays stopped**. The preview session is deleted and the original performance state/playhead is restored with transport stopped. Restart the performance explicitly when ready.

If you edit the draft after preparing, prepare again before listening. Preview is temporary: use Save separately to update the library patch.



*Isolated effect preview after successful preparation, ready for Stop and audition.*



*Current-performance preview targets one explicitly selected rack instance.*

## Screenshots



RUNTIME

HIDE

?

MIDI INPUT

Internal Loopback

COMPILE OUTPUT

```
sr = 48000
ksmps = 32
nchnls = 2
0dbfs = 1.0
opcode vcs_mixer_ramp, a, ki
setksmps 1
kTarget, iInitial xin
kValue init iInitial
kPrevious init iInitial
kRemaining init 0
if kTarget != kPrevious then
    kRemaining = 0.02
    kPrevious = kTarget
```

BROWSER AUDIO

Browser-owned PCM runtime active

|                    |        |
|--------------------|--------|
| PCM queue          | 159 ms |
| Pending render     | 0 ms   |
| Underruns          | 0      |
| Overruns           | 0      |
| Render/audio ratio | 13%    |

SESSION EVENTS

```
arpeggiators_configured
2:01:12 PM
{"tempo_bpm":120,"arpeggiators":0}

started
2:01:12 PM
{"backend":"ctcsound","detail":"started with
CSound browser-clock PCM runtime (rtmidi
fallback:
```

Runtime panel showing MIDI input selection, generated ORC, active browser-owned PCM runtime, queue diagnostics and recent session events.

## Audition notes for branching instruments

The **Test MIDI note** field chooses which note audition sends. A Switch driven by `notnum` starts with its first MIDI-number case (36 for Drumset); other patches start at 60. Change it to 38 or 42 to test the other drums, or to an unmatched note to check Default Silence. [Branching guide \(EN/DE/FR/ES\)](#).



# MIDI Setup and Inputs

This page covers practical MIDI setup and how external MIDI binding works in Orchestron's unified browser-clock runtime.

## External MIDI Input Binding In Orchestron

External MIDI input is bound in the **Instrument Design Runtime panel** (not on the Config page).

Workflow:

1. Open Instrument Design.
2. Use the Runtime panel MIDI Input dropdown.
3. Select the desired input:
  - `internal:loopback` for the built-in app-only loopback and default fallback binding
  - a `host_bridge` input published by `host-midi-helper` for external hardware or DAW MIDI
1. Start the instrument session and play from your hardware/DAW.

The selected MIDI input is reflected in the Perform page footer status line.

## Internal Performance Controls Do Not Require External MIDI

These features generate MIDI/control events internally and do not require an external MIDI input device or host bridge:

- melodic sequencers
- controller sequencers
- arpeggiators
- piano rolls (on-screen keyboard)
- manual MIDI controller knob lanes

External MIDI input is only needed when you want to play Orchestron from outside the app. Internal MIDI is always delivered directly into the session engine and ignores the external input binding.

That means:

- switching the session input does not disable sequencers, arpeggiators, piano rolls, or manual controller lanes
- binding a helper-provided external input adds another source into the same engine scheduler

## External MIDI Bridge

The backend no longer relies on direct OS MIDI enumeration for session playback. External MIDI arrives through the optional Rust host bridge in `host-midi-helper`.

General setup pattern:

1. Start the backend with `VISUALCSOUND_HOST_MIDI_TOKEN` set.
2. Start `host-midi-helper` on the same host OS with the same token.
3. Bind the helper-published input in the Runtime panel.

If no helper is running, `internal:loopback` remains available and internal performance tools still work.

The backend still performs backend-native MIDI enumeration where available, but live session playback now uses the helper-backed `host_bridge` path for external devices.

## macOS Loopback Setup (IAC Driver)

For DAW/app -> Orchestron MIDI routing on macOS:



1. Open **Audio MIDI Setup**.
2. Open **MIDI Studio**.
3. Open **IAC Driver**.
4. Enable **Device is online**.
5. Add/select an IAC bus (for example IAC Driver Bus 1).
6. Route your DAW/app MIDI output to that IAC bus.
7. Select that input in Orchestron's Runtime panel.

## Troubleshooting MIDI Inputs

### No External MIDI Inputs Listed

- Ensure your OS/device/virtual bus is enabled (for example IAC Driver on macOS, ALSA/JACK route on Linux, loopMIDI on Windows).
- Confirm `host-midi-helper` is running and uses the same token as the backend.
- Restart the helper after changing system MIDI configuration.

### Perform Page Footer Shows `midi input: internal:loopback`

- Internal app MIDI is working.
- External hardware/DAW MIDI will not reach the session until a helper-provided input is bound.

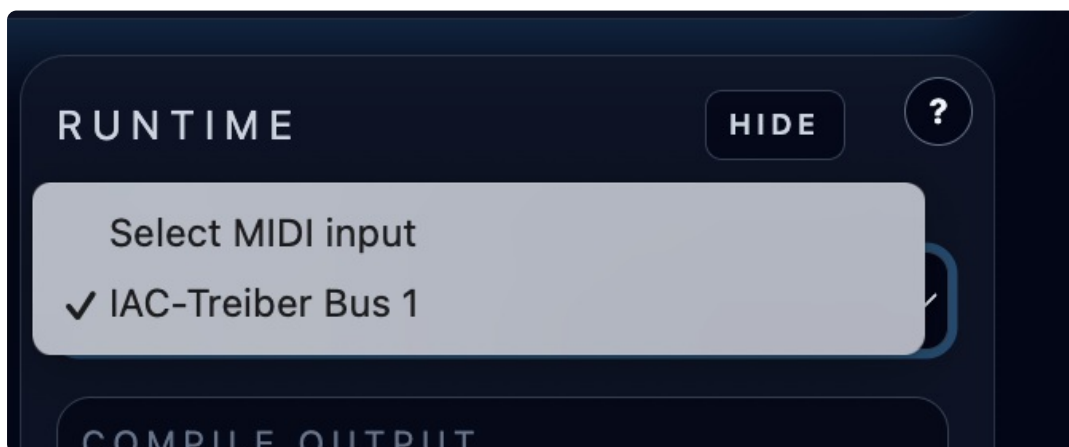
### Sequencer/Piano Roll Works But Hardware MIDI Does Not

- This usually means internal controls are working, but no external helper-provided MIDI input is bound.

## Related Pages

- [Runtime Panel and Compilation Workflow](#)
- [Live Status and Safety Controls](#)

## Screenshots



*Runtime panel MIDI input dropdown expanded with available MIDI inputs.*



## Live Status

The Perform page includes several status for live operation.

### Browser Audio Health

The Runtime panel shows queued PCM and pending-render milliseconds, underrun and overrun counters, and the latest backend render-time/audio-time percentage while a browser-clock session is active.

Audio refills and PCM writes run in a Dedicated Worker. Sequencer step and pad updates are carried with the rendered PCM and applied only when their target frame becomes audible. If the Perform page is hidden, visual animation pauses and resumes from the current audible snapshot instead of replaying missed steps.

Performances are limited to 128 flattened backend note tracks. Each drummer row becomes one backend note track, so the total includes melodic sequencers plus all drummer rows.

### Footer Status Bar

At the bottom of the Perform page, Orchestron shows live status values such as:

- playhead position (current step / step count)
- cycle count
- active midi input name (for example `internal:loopback` or a helper-provided device)

This footer is useful during performance because it gives quick feedback without opening the Instrument Design runtime panel.

### Track State Labels (Queued Changes)

A track's state badge can reflect queued actions that apply at the next cycle boundary, for example:

- starting @ step 1
- stopping @ step 1

This helps you understand why a track button press did not take effect immediately while transport is running.

### Error Banner (Transport / Runtime Errors)

The Perform page displays an error banner for problems such as:

- no active instrument session when trying to start sequencers
- trying to play piano roll notes before the instrument engine is ready
- pad queue request failures
- controller send failures
- sequencer config sync failures

### Editing During Playback

Notes, controller curves, tempo and arrangement edits keep the transport at its current position. Rapid edits are combined after an 80 ms pause; the previous configuration keeps playing while the update is prepared. The completed edit takes effect at the next engine block. Editing a displayed pad changes that pad, including when the arranger has selected it automatically.

Automatic pad changes, playhead movement and status updates do not resubmit the performance configuration. Queued pad launches still wait for their usual pad boundary. Stop cancels pending configuration application. Shortening a finite arrangement past the current position stops playback without rewinding.



If preparation fails, the error banner reports it and the previous configuration continues playing. Your edit remains in the editor. Make another edit or explicitly restart to try again; the app does not continuously retry a failed update. Saving and exporting retain the existing file formats.

## MIDI Input Reference

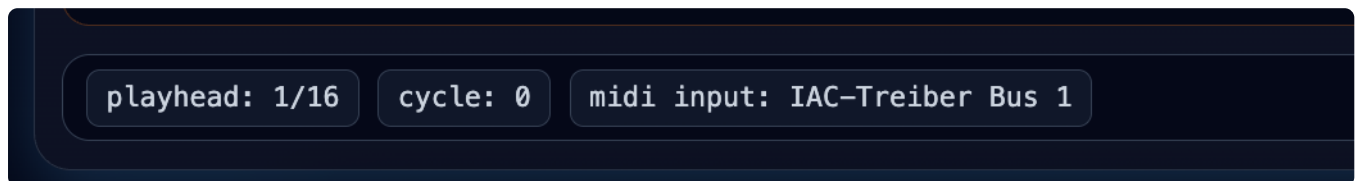
The footer's MIDI input display reflects the session's bound external MIDI input (selected in the Instrument Design Runtime panel). If it shows `internal:loopback`, internal app MIDI is still active, but external hardware/DAW MIDI will not reach the session until a helper-provided input is bound.

See [Runtime Panel and Compilation Workflow](#) and [MIDI Setup and Inputs](#).

## Live Performance Safety Checklist

1. Confirm instrument session is running.
2. Confirm expected MIDI input is bound.
3. Check track/controller/piano-roll enable states.
4. Save the performance after major routing changes.

## Screenshots



*Footer status bar with playhead/cycle/MIDI input.*

## Definition audition

Audition is a session-only track override on the shared transport. The sequencer's existing Audition control switches a running track at its next pad/rest boundary after preparation succeeds; failures preserve current playback and edited drafts. Other tracks continue. Return to arrangement locates the authored sequence at the then-current song position, including pauses and finite endings. Stop audition clears the override and stops only that track.

Arranger Play clears all auditions before applying normal playback-source behavior. Arranger Stop clears/stops auditions while preserving independently started manual tracks. Seeks and loop wraps restart active auditions from their first token at the destination. Removing devices, replacing the performance or engine session, and engine shutdown clear transient audition state. Runtime status never becomes authored configuration.

## Temporary lane output controls

Arranger Mute/Solo gates a lane's future MIDI output without disabling its sequencer or changing its playhead, audition, repeat or playback source. Existing notes release normally. Drummer rows change together; multiple solos are supported and explicit mute wins. These controls are separate from mixer-strip controls, so lanes sharing one instrument remain independently controllable. See [lane Mute/Solo](#) for arpeggiator dependencies, controller values and reset behavior.

Arranger speaker previews use a 250 ms hold delay, then switch at the next lane cycle boundary during arranger playback. While stopped, they start at the next available engine block after preparation without moving the arranger cursor or changing its highlighted Stop state. Play is highlighted only during song playback. Releasing before the queued boundary cancels the preview without disturbing the current notes. Release restores the previous arrangement/manual/audition mode at the current clock position; stopped lanes stay stopped. Gesture identities and revisions reject stale commands, including release-before-start. Pending attacks and controllers from replaced material are discarded; note releases retain source ownership so other lanes sharing a channel remain intact. Preview restoration state is temporary and never saved.



# Multitrack Arranger

The arranger is the only editor of song order. Each lane places references to its eight pads, reusable groups and supergroups, and rests. Sequencers edit pad contents; melodic and drummer devices assemble reusable phrases in their upper-right pattern workspace. In the arranger, edit phrases by ungrouping, rearranging, and regrouping timeline material. Editing a shared definition changes every occurrence; changing its duration shifts following material on that lane.

Pads are labelled **1–8** in arranger lanes and palettes, and **#1–#8** in sequencers and phrase editors. Pads are green, groups are red, and supergroups are violet throughout the sequencers, phrase libraries, palettes and arranger lanes. Rests keep their dark styling; the number in **Rest 1/2/4/8/16** is its duration in local beats, while a pad number identifies a reusable slot. Right-click a musical item or palette entry to Set colour or Reset colour. Custom colours apply to every occurrence of that definition within its lane and are saved with the performance. The picker offers swatches and a custom colour; text contrast adjusts automatically. Sequencer colours remain the defaults for now. Selection outlines and Playing/Queued indicators remain separate from these colours.

## Compact track headers

Each lane starts collapsed. Click the triangle or track title (Enter/Space when focused) to open or close its editing tools. Click anywhere in an occurrence box to select it and open the lane; the triangle can close it again while retaining the editing context. Clicking empty timeline space clears occurrence highlights and chooses the insertion position without changing panel expansion. Double-click a pad or choose Edit pattern to open its sequencer editor. Open in arranger opens the lane and focuses the definition in its palette.

The expanded lane shows **Position (master beats)** followed by the wrapping palette. There is no separate Patterns and phrases panel in the arranger; melodic and drummer sequencers provide their own pattern workspace. **Playback settings** contains Playback source and, for Arrangement lanes, At end, and starts collapsed. Disclosure choices survive view switches; New/Load/Import resets them. Closing panels preserves drafts, selections and scroll, cancels active gestures and momentary previews, and leaves normal playback running.

## Temporary lane Mute and Solo

The small **Mute** and **Solo** buttons control this sequencer lane, independently of the instrument mixer. Several lanes can drive the same instrument and still be controlled separately. A drummer's rows act together. Multiple Solo buttons may be active; explicitly muted lanes stay muted. A dashed amber Mute button indicates suppression by another lane's Solo.

Lanes keep advancing through pads, rests, loops and auditions while suppressed. Changes apply at the next available engine block; already buffered audio still plays. New attacks are suppressed, while sounding notes complete their scheduled releases and effects decay normally. Unmuting resumes future events at the current position without retriggering held notes. Controller lanes stop sending CC values while muted; the last applied value remains until an active source updates it.

Solo preserves the upstream inputs and downstream arpeggiator processing needed by the selected lane. Explicit mute overrides these dependencies. External MIDI and independent Live arpeggiators remain outside arranger solo selection. Instrument mixer mute/solo still applies independently.

Lane controls survive transport stop/start and engine restart within this performance. They reset on New/Load/Import and browser reload; deleted lanes lose their controls. Save, autosave, native bundles and both CSD exports exclude this temporary state.

## Place and move material

Expand a lane to show its palette of populated pattern pads, groups, supergroups and rests on the Position row. Melodic, drummer and arpeggiator pads appear only when they contain a non-rest step; their original numbers are preserved. Empty pads remain editable in the sequencer, and existing timeline occurrences remain intact. Controller curves remain available, including zero-valued curves. Drag a palette item to add an occurrence. Clicking an entry only focuses it. Empty phrase drafts cannot be placed or previewed. Add, Insert, Paste, Duplicate and number-key placement are removed from the arranger; sequencer editing shortcuts remain unchanged.



Drop at an element edge to insert and shift later material on this lane. Drop musical material inside a rest when it fits to replace that portion of silence. Drop after the final element to fill the gap with supported pause tokens and append. Dropping inside occupied musical material is rejected with a red preview. Rest palette items insert time at boundaries or extend the lane after its end.

The ruler numbers master bars and beats. Placement snaps to whole local beats using the lane's beat ratio; previews show the exact master-beat position and duration. Drag an occurrence body to move it, or move a contiguous selection together. Moves leave equivalent silence at the source; dropping at the original position changes nothing. Placement stays within the originating lane. Invalid or over-limit edits leave the whole arrangement unchanged.

Right-click to open lane actions. Right-clicking a selected occurrence preserves the selection; right-clicking another selects it. Cmd/Ctrl-click toggles additional selections. Grouping requires at least two adjacent selected elements. **Group** accepts pads/rests; **Supergroup** also accepts groups. **Ungroup** expands one level and retains the reusable definition in the palette.

After ungrouping, rearrange the material, then select and group it again. Choose **Create new definition** (the default) to leave other occurrences unchanged, or explicitly select **Update existing definition** to update all its references. The dialog lists uses and explains that duration changes shift following material. Groups retain letter IDs and supergroups Roman numeral IDs.

**Create variation** copies a pad into an unused slot, or copies a group/supergroup under a new ID, and retargets only the selected occurrence. It inherits the source colour. The eight-pad limit remains. Palette context menus allow deleting unused definitions; saved references list their uses and block deletion. Temporary workspace references expand into the saved contents, including retained drafts in closed panels; the deleted definition's own draft is removed.

Use Enter/Space to select a focused occurrence, Cmd/Ctrl-click for multiple selection, and Shift+F10 or the Context Menu key for its menu. Arrow keys navigate occurrences or menu actions; Escape closes menus and cancels gestures. Menus return focus to their originating item.

## Delete and resize rests

- **Delete/Backspace** and **Remove, leave gap** replace selected musical occurrences with equal-duration silence. Other occurrences keep their positions.
- **Remove and close gap** (also Shift+Delete) removes selected musical occurrences or rest spans and closes their time on this lane. Multiple selections form one edit.
- Consecutive pause tokens appear as one rest span. Drag its right edge or choose Duration from its context menu to resize it in whole local beats, shifting later items.
- Leading, internal, trailing and rest-only arrangements are preserved. Trailing silence contributes to Fit, playback duration, repetition and exports.

## Playback and audition

Each lane has **Playback source: Arrangement | Manual pads**. The same setting appears in the sequencer. Empty lanes use Manual pads. Adding the first occurrence enables Arrangement; closing all time returns to Manual pads. Ordinary deletion leaves rests and retains Arrangement.

For Arrangement lanes, **At end: Stop track | Repeat track sequence** controls the whole authored sequence. New devices stop at the end; imported repeat settings remain intact. **Loop selected range** is the separate shared transport loop. Click its ruler to seek and clear the loop, or drag a range to loop it. Rewind/Fast forward move one master beat. Fit shows the whole longest lane, including trailing rests.

**Play** ends all auditions, starts Arrangement lanes and stops Manual pads sequencers. It starts the instrument engine when needed. **Stop** clears and stops auditions and Arrangement lanes while preserving independently started manual sequencers. It leaves the rack engine running. Double-click Stop resets to the loop start or song start when no manual track keeps transport active.

Every musical occurrence and available pad/group/supergroup has a small speaker icon near its right edge. Hold it for **250 ms** to begin a momentary preview from its first token. Shorter clicks do nothing. While the arranger is running, prepared previews start at the lane's next pad/rest cycle boundary. While stopped, they start at the next available audio block without moving the arranger playhead; existing audio buffering still contributes latency. It repeats while held. Selecting content never launches it.



Release the speaker to end preview and restore the lane's previous playback at the current shared-clock position. An Arrangement lane can return into a rest or a finite track that has ended. Previously stopped lanes stay stopped. Independently playing manual pads and existing sequencer auditions are restored. Other lanes continue and lane Mute/Solo remains effective. Releases preserve other lanes sharing an instrument/channel.

Hold Space/Enter on a focused speaker for the same behaviour. Releasing outside it, cancelling the gesture, losing focus, collapsing, or leaving the view ends the preview. Preparation failures retain current playback; releasing before preparation finishes prevents a late launch. Transport Play/Stop cancels previews, and a late release cannot restart playback.

With the arranger stopped, preview starts only that lane through the normal instrument startup flow. The song cursor stays fixed and the Stop button stays highlighted. Play resumes the song from that cursor. During song playback, Play is highlighted instead. Seeks and loop wraps restart held previews at their first token on the shared clock. During song playback, song range/end remain authoritative. Arranger arpeggiators support previews; independent Live arpeggiators remain unchanged.

Melodic and drummer sequencers use a [pattern workspace](#) with a Play/Stop toggle and the same momentary speakers. A speaker held over workspace playback restores the workspace on release; stopping the workspace restores the prior lane state. Controller sequencers and arpeggiators retain their existing Audition/Return/Stop controls.

## Saved format

Performances still store root sequences, pads, groups, supergroups and supported pause tokens. Version 16 and imports of versions 1–16 remain supported; the app-state and native bundle envelope versions are unchanged. Old enabled arrangements that resolve to no tokens are imported as an explicit occurrence of their former active pad, preserving their repeat setting and sound.

Optional `definitionColors` metadata maps typed definition references (`pad:0`, `group:A`, `super:I`) to hexadecimal colours. It does not change sequence compilation or trigger audio preparation. Missing/invalid colours use type defaults. Save/load, autosave, native bundles and CLI handling retain valid colours. CSD audio ignores them.

Audition/preview commands and restoration state, selection, position, sounding pads and temporary enablement are session state. Save, autosave, native bundles and both CSD exports use authored data. Musical edits made while auditioning save normally.



# Pattern Pads and Reusable Phrases

Each sequencer has eight pattern slots, #1–#8. Click a pad to select it for editing. Playback and queued launches never change this editing selection. The Playing and Queued indicators show what is sounding separately.

Pads are labelled **1–8** in arranger lanes and palettes, and **#1–#8** in sequencers and phrase editors. In melodic and drummer sequencers, pads are green only when their playable steps contain at least one note or active hit with velocity above zero; empty, hold-only and zero-velocity pads stay dark. Selection and queued outlines still appear on empty pads. Groups are red and supergroups violet. Controller pads with only two zero-valued endpoints stay dark; extra points or nonzero values mark content. Zero remains valid CC output. Arpeggiator pads turn green for a note-producing step with positive velocity and probability; rest-only, tie-only and suppressed pads stay dark, regardless of held notes. Rests keep their dark styling; the number in **Rest 1/2/4/8/16** is its duration in local beats, while a pad number identifies a reusable slot. Selection outlines and Playing/Queued indicators remain separate from these type colours.

## Pad contents

Melodic pads store notes, chords, holds, timing offsets and theory settings. Drummer pads store hits and velocities for each row. Controller pads store automation curves. Arpeggiator pads store complete musical variations and rhythm steps.

Pad lengths are 1–8 local beats for melodic and drummer devices, and 1–8 or 16 beats for controller and arpeggiator devices. Arpeggiator durations use master beats. Meter, grid and beat ratio belong to the owning sequencer; tempo is shared.

Drag one pad onto another to copy it explicitly. Melodic pad edge buttons transpose by a scale degree on a short press, or change the diatonic key on a long press. Drummer and controller pads do not use these transpose controls.

## Pattern workspace

Melodic, drummer and controller sequencers and arpeggiators show an always-visible **Pattern workspace** in the upper-right header area. On narrow screens it stacks below the musical controls. The eight pads and step grid remain below the header.

Drag pads or saved groups/supergroups into **Free workspace** to assemble a phrase. A drop inserts at the visible marker; dragging an existing item or selection moves it. Dragging one of the eight pads onto another still copies that pad. Workspace references stay within their owning sequencer.

Click an item to select it. Cmd/Ctrl-click toggles extra items; Shift-click selects a range. **Group** gathers at least two selected pads/rests in left-to-right order, including separated selections, and places the new group at the first selected position. **Supergroup** also accepts groups. Unselected items retain their relative order. These actions create reusable definitions immediately; groups use letter IDs and supergroups Roman numeral IDs. Groups cannot contain groups; supergroups cannot contain supergroups.

Right-click an item, or use Shift+F10, for **Edit**, **Group**, **Supergroup**, **Ungroup**, and **Remove**. Ungroup expands one level locally and keeps the saved definition. Delete/Backspace removes selected workspace items and closes the gap. Use **Add rest...** for rests of 1, 2, 4, 8 or 16 local beats. Long assemblies scroll horizontally; the saved phrase palette has a bounded height.

Choose a saved phrase or double-click a workspace group/supergroup to edit its draft. **Apply** updates every saved use; **Save as new** creates an independent definition. **Discard changes** restores its saved contents. The editor lists arrangement/phrase uses and warns that changing duration shifts later song material. Workspace construction never edits song order; place saved phrases in the [arranger](#). **Open in arranger** focuses the selected definition there.

Each definition has a separate draft, and the free workspace is retained when switching between definitions. Drafts, selection and scroll survive panel collapse and view switches. New/Load/Import and browser reload clear temporary drafts. Loose workspace items and unapplied edits are excluded from save, autosave and export; Group/Supergroup creation and Apply save through the existing performance workflow.

**Delete definition** is available in saved phrase and arranger palettes. Saved arrangement or phrase references block deletion. Temporary references expand into the saved contents, preserving order and selection even in closed workspaces; the definition's own draft is removed. Empty definitions can remain drafts but cannot be placed or auditioned. Referenced



definitions cannot be applied empty. Containers and expanded sequences are limited to 256 tokens; invalid edits leave drafts unchanged.

All four device types use this workspace, including Live arpeggiators. Live mode hides arrangement source and navigation controls while retaining phrase creation and editing.

## Playback source and audition

**Playback source: Arrangement** follows the song timeline. Its Stop/Repeat-at-end option lives in the arranger lane. **Manual pads** enables a separate Launch pad action for jamming. Selection alone never starts sound.

The play button beside the workspace loops the **entire displayed assembly**, regardless of selection. An amber outline follows the currently playing pad, group, supergroup or rest, independently of the cyan editing selection. It follows audible playback, including repeated pads and loop wrap; a momentary speaker preview hides the workspace outline until release. Click it again to restore the track's previous arrangement, manual pad or stopped state. It works before saving a phrase. Workspace edits enter at the next cycle boundary after the normal 80 ms coalescing delay. Failed preparation retains the previous sound and the edited draft; another edit or explicit restart retries.

Every pad, group and supergroup has a separate speaker button. Hold for **250 ms** to begin a momentary preview; shorter clicks do nothing. Preview repeats until release. With the arranger running it enters at the next lane cycle boundary; with the arranger stopped it starts at an audio block without moving the song cursor. Space/Enter supports the same hold/release gesture. The speaker never selects, transposes or drags its item.

A speaker preview temporarily overrides workspace playback, then returns to that workspace at the current clock position. Stopping the workspace restores the original track state. Releasing before preparation or a queued boundary cancels the preview. Pointer cancellation, focus loss and leaving the view end speaker previews; collapse/navigation also end workspace playback. Arranger Play/Stop cancels both layers, and late replies/releases cannot restart them. Other tracks continue and lane Mute/Solo stays effective.

Live arpeggiator workspace Play and held speakers start at the configured next rhythm cycle or master bar when a Live clock is established; otherwise they start at an audio block. Pad duration determines phrase lengths. Incoming or held notes are required; no test chord is supplied. Hold and Active/Bypass/Mute remain effective. Chord changes during previews are retained, and release restores the underlying Live pulse. **Launch pad** explicitly launches the editing pad in Live/manual mode, separately from temporary previews. Saved definitions retain the same performance format.



# Browser-Clock Latency

Orchestron now uses backend audio mode `browser_clock` on every supported platform. The browser becomes the playback clock and requests PCM render chunks from the backend, so the most important latency and stability controls live in the Config page's `Browser-Clock Latency` section.

## When This Section Appears

The `Browser-Clock Latency` section is shown when the backend is running in `browser_clock` mode, for example:

- `docker compose up --build`
- CLI argument `--audio-output-mode browser_clock`
- environment variable `VISUALCSOUND_AUDIO_OUTPUT_MODE=browser_clock`

This is the standard runtime path for macOS, Linux, Windows, and Docker. In practice, that means this section describes the normal live-audio path of the app, not a Docker-only special case.

## Why These Settings Matter

On the browser-clock path, audio is not pushed to a local DAC on the backend host. Instead:

- the backend renders PCM blocks on demand
- a dedicated browser audio worker keeps the target queue of decoded PCM and requests more audio
- the AudioWorklet consumes that queue independently of React rendering

That makes the browser-clock queue settings the main tradeoff between low live-play latency and glitch resistance.

The queue producer no longer runs on the React/main thread. Main-thread stalls can delay visual updates, but they do not stop the worker from requesting or buffering PCM. Playback starts only after the startup high-water target is reached. The Runtime panel reports current queued and pending-render milliseconds, underrun/overrun counts, and the latest backend render-time/audio-time ratio.

The backend suppresses Csound performance-time logging by default in headless browser-clock sessions. This prevents stdout or Docker logging I/O from delaying `performKsmps()`. To diagnose Csound's own runtime messages, start the backend with `VISUALCSOUND_CSOUND_PERFORMANCE_LOGGING=true`; this is intended as a temporary diagnostic setting rather than a latency control.

Lower values usually feel more immediate, but they also leave less headroom for:

- backend scheduling jitter
- network / websocket round-trip jitter between browser and backend
- browser tab throttling or temporary main-thread stalls
- heavier patches that take longer to render

Higher values add safety margin, but manual playing and note feedback feel less direct.

## Main Control Groups

### Queue Targets

- `Steady Low Water` / `Steady High Water` define the normal browser PCM queue window after startup has stabilized.
- `Startup Low Water` / `Startup High Water` define the larger queue window used while the stream is priming or recovering.

If you hear underruns after connect or after transport restarts, start by increasing the steady or startup queue targets slightly.



## Recovery Headroom

- `Underrun Recovery Boost` adds extra queue headroom after an underrun is detected.
- `Max Underrun Boost` limits how large that temporary recovery buffer may grow.

These controls help the browser recover cleanly from brief dropouts without forcing a permanently larger steady-state latency.

## Render Request Size And Parallelism

- `Max Blocks Per Request` limits how much audio the browser asks for in one render request.
- `Steady Parallel Requests` , `Startup Parallel Requests` , and `Recovery Parallel Requests` control how many render requests may stay in flight in each state.

Smaller requests can improve responsiveness, while larger or more parallel requests can make it easier to keep the queue full on slower systems.

## Live Note Responsiveness

- `Immediate Note Render Blocks` sends a small urgent render burst after a live note-on event.
- `Immediate Note Render Cooldown` limits how often those urgent note-triggered bursts may happen.

These are the most direct controls for making manual keyboard or MIDI playing feel snappier in browser-clock mode.

## Manual MIDI Safety Limits

Browser-clock manual MIDI is bounded on the backend even when events are routed through an arpeggiator. Deployments can tune the maximum accepted future timestamp with `VISUALCSOUND_BROWSER_CLOCK_MANUAL_MIDI_MAX_FUTURE_MS` , the accepted manual MIDI rate with `VISUALCSOUND_BROWSER_CLOCK_MANUAL_MIDI_RATE_PER_SECOND` and `VISUALCSOUND_BROWSER_CLOCK_MANUAL_MIDI_BURST` , and the per-session arpeggiator pending input cap with `VISUALCSOUND_ARPEGGIATOR_PENDING_INPUT_MAX_EVENTS` .

## Session Resource Limits

Runtime sessions allocate backend Csound worker resources, so session creation and live session-event observers are also bounded. `VISUALCSOUND_SESSION_MAX_ACTIVE` limits total active sessions, `VISUALCSOUND_SESSION_MAX_ACTIVE_PER_CLIENT` limits sessions owned by one client address, `VISUALCSOUND_SESSION_CREATE_RATE_PER_MINUTE` and `VISUALCSOUND_SESSION_CREATE_RATE_BURST` limit repeated creation attempts, and `VISUALCSOUND_SESSION_IDLE_TIMEOUT_SECONDS` deletes stopped idle sessions so capacity is returned automatically. Session event WebSocket observers are capped with `VISUALCSOUND_SESSION_EVENT_WS_MAX_SUBSCRIPTIONS_TOTAL` and `VISUALCSOUND_SESSION_EVENT_WS_MAX_SUBSCRIPTIONS_PER_SESSION` , while connection attempts are rate-limited with `VISUALCSOUND_SESSION_EVENT_WS_CONNECT_RATE_PER_MINUTE` and `VISUALCSOUND_SESSION_EVENT_WS_CONNECT_RATE_BURST` .

## Practical Tuning Order

1. Start from the defaults shown by the Config page.
2. If the stream crackles or underruns, raise `Steady High Water` and `Startup High Water` first.
3. If latency still feels too high, lower the steady queue targets cautiously.
4. If live note response feels sluggish, reduce `Max Blocks Per Request` or adjust the immediate-note controls in small steps.
5. Apply the settings and test again with a realistic patch load.

Keep in mind that these settings are stored in app state for the current workspace. They are not saved into the patch itself.

They also work together with the engine settings from the patch:

- browser-clock settings tune queueing and request behavior
- patch engine settings such as `sr` , `ksmps` , `-b` , and `-B` tune the backend render block geometry

## Screenshot



### Browser-Clock Latency

Tune the browser PCM queue, render burst size, and request parallelism used by browser-clock audio. These settings only appear when the backend runs in browser-clock mode.

Stored in app state, not in the current patch.

|                                                      |                                                          |                                                                |
|------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------|
| STEADY LOW WATER (MS)<br>120<br>Allowed: 20 - 2000   | STEADY HIGH WATER (MS)<br>200<br>Allowed: 20 - 2000      | STARTUP LOW WATER (MS)<br>180<br>Allowed: 20 - 2000            |
| STARTUP HIGH WATER (MS)<br>320<br>Allowed: 20 - 2000 | UNDERRUN RECOVERY BOOST (MS)<br>120<br>Allowed: 0 - 2000 | MAX UNDERRUN BOOST (MS)<br>500<br>Allowed: 0 - 2000            |
| MAX BLOCKS PER REQUEST<br>96<br>Allowed: 1 - 512     | STEADY PARALLEL REQUESTS<br>2<br>Allowed: 1 - 6          | STARTUP PARALLEL REQUESTS<br>3<br>Allowed: 1 - 6               |
| RECOVERY PARALLEL REQUESTS<br>3<br>Allowed: 1 - 6    | IMMEDIATE NOTE RENDER BLOCKS<br>24<br>Allowed: 1 - 128   | IMMEDIATE NOTE RENDER COOLDOWN (MS)<br>25<br>Allowed: 0 - 1000 |

APPLY BROWSER-CLOCK SETTINGS

The Config page's Browser-Clock Latency section, used when Orchestron runs headless in 'browser\_clock' mode and the browser owns audio playback.



# If, Switch and Drumset

## English — choose a synthesis case at note start

**If** and **Switch**, in the **Control flow** catalog category, are editor constructs that generate Csound `if/elseif/else/endif`. Each instrument instance chooses one case when its note starts. Only that case initializes and performs; selection remains fixed until the voice ends. Different notes, including repetitions of the same note, have independent voices.

1. Add If or Switch. New blocks produce stereo audio. If compares two init-rate inputs using `==`, `!=`, `<`, `<=`, `>` or `>=`. Switch compares its init-rate selector with unique numeric case values. For drums, connect `notnum.inote` to `selector`. Input connections and the existing arithmetic Input Formula Assistant take precedence over literal values in **Configure branches**.
2. Expand the cases on the canvas and drag opcodes from the catalog into a case. A valid drop into **Silence** changes it to **Synthesis** and exposes its result inputs. Ordinary canvas drops create main-graph nodes. Dropping onto a collapsed block expands it; drop again into a specific case.
3. Connect each synthesis case to its managed **Case Result** audio inputs. Several sources can sum at an input, or use an input formula. Connect the block's common output to shared processing or a named Stereo Output. Every non-silent case needs a value for every result channel, even if that case is rarely selected.
4. False and Default initially use **Silence**. Silence assigns zero to every output and contains no synthesis nodes. Changing an existing case to Silence previews the nodes, connections and formulas that will be removed. Switch cases can be renamed, reordered, added and deleted; Default remains last and cannot be deleted. If keeps its two ordered True/False cases.
5. Collapse to see the condition, case summary, shared inputs and common output. Node IDs, selection, positions and the viewport survive collapse, save and reload. Captured input formulas are retained; expand to edit them. Format changes show an impact preview: mono preserves `left` and removes `right` wiring only after confirmation; switching back to stereo leaves `right` unconnected. Block/case deletion removes owned content and associated formulas and configuration together.

## Arrange and transfer nodes

The block stays above and left-aligned with the first case, with room for its complete body and buttons. Case frames measure every visible parameter row at any zoom. They automatically grow to the right and bottom and retain their size when nodes move inward, are removed, or become smaller; **Case Result** stays in a separate row at the bottom-right. Later cases move down or up without changing their internal arrangement. The title band is 40 graph units, padding is 24, and cases have a 32-unit gap. Empty cases keep a 240 × 160 content area.

Drag the **Case Result title or unused body** to resize a branch in both dimensions, including shrinking it. This also works for the Silence marker. The pointer displacement sets the requested size; complete node bounds, padding and the reserved result row limit how small it can become. Other selected nodes stay still and selection is preserved. Sockets, formula editing and help buttons keep their usual actions. Release commits the size and managed positions together; **Escape**, cancellation or focus loss restores the full prior layout. Ordinary dragging retains the largest frame size reached during the gesture. Bright purple borders stay about two screen pixels wide at every zoom. Sizes survive collapse, block movement, reordering, save/reload, copies and export/import.

- Drag main-graph nodes into an expanded case to transfer them. Dragging existing case members normally only rearranges them within that case. Their top and left movement limits keep them clear of the header; selected groups keep their relative spacing. Moving the block moves all contents together.
- Hold **Alt/Option** before starting the drag to transfer case members to another case or the main graph. Keep the desired group selected. **Move selected nodes** in **Configure branches** is the alternative, with identical validation.
- A transfer requires every connection and formula binding touching a moved node to end at another moved node or a node already in the destination scope. A cyan border marks an allowed target; a red border and diagnostic list refused connections and unselected endpoints. Nothing is automatically selected, disconnected or deleted. An isolated ordinary node can always transfer.
- This restriction applies to **membership transfers**. Existing shared main-graph inputs remain valid; you can deliberately connect them after placing the nodes. Case-local values still leave through Case Result only. Main-graph-only constructs, nested blocks and independent Case Result transfers are refused.



- Release to commit positions and membership together. A refused drop, **Escape**, pointer cancellation or loss of focus restores the entire group and clears highlighting. A refused drop leaves Silence unchanged.

## Play the Drumset template

Choose **New from template** → **Drumset**. Notes **36**, **38**, and **42** play kick, snare, and hi-hat; all other notes select Default Silence. The kick uses a falling-pitch sine, the snare combines filtered noise and tone, and the hi-hat uses high-pass-filtered noise. Velocity scales finite amplitude envelopes. Branch-local `xtratism` preserves their decays after short MIDI notes. Audition offers a **Test MIDI note** field; for this template it starts at 36. Save, assign the patch to a MIDI channel in Perform, and route its named Stereo Output to the mixer destination. Send 36, 38 and 42 together to play all three sounds.

Processing after the block runs **once per voice**. Put effects in the performance mixer when they should process the **sum of the whole kit**. The example `examples/velocity_if.patch.json` uses `ampmidi` and `If` to choose sine or saw timbre by velocity; it requires no separate layering subsystem.

Continuous switching, nested blocks, numeric results, raw jumps, crossfades, MIDI learn, ranges and choke groups are outside this release. Keep `outs`, `inleta`, `outleta`, `sfloat` and `maxalloc` in the main graph. GEN nodes and duration controls may belong to cases. Cycles and illegal scope crossings are compile errors, and all cases are validated.

## Patch/API format

Existing patches remain at `schema_version: 1`. Adding a block promotes the patch to **2**; removing the last block does not downgrade it. Create, update and compile use the existing APIs. Unsupported versions and malformed structural records are rejected. Native instrument/performance exports, application state, patch copies, audition and both CSD export modes retain the configuration.

The graph remains flat. Add a typed `graph.control_flow` map keyed by the structural node's ID; each key must reference an `If` or `Switch` node. The example below describes a mono Switch whose ordinary flat node array also contains `kit`, `kick`, `kick_result` and `default_result`:

```
{
  "kit": {
    "kind": "switch",
    "output_format": "mono",
    "operator": "==",
    "cases": [
      {"id": "kick_case", "name": "Kick", "value": 36,
        "node_ids": ["kick", "kick_result"],
        "result_node_id": "kick_result", "silence": false},
      {"id": "default_case", "name": "Default", "value": null,
        "node_ids": ["default_result"],
        "result_node_id": "default_result", "silence": true}
    ]
  }
}
```

- `kind`: `if` or `switch`; `output_format`: `mono` or `stereo`.
- `operator`: one of the six comparison operators; only `If` uses it.
- `cases`: ordered records with stable `id`, editable `name`, numeric `value` for `Switch` cases, and `null` for `Default` or both `If` cases.
- `node_ids`: exclusive membership, including the managed `CaseResult` node identified by `result_node_id`. Nesting and overlapping membership are invalid.
- `silence`: explicit boolean. A silent case contains only its result marker, with no input wiring, parameters or formulas.
- Block inputs: `lhs / rhs` for `If`, `selector` for `Switch`, all `init-rate`. Result and block output ports: `left`, plus `right` for stereo. `CaseResult` has no output ports.
- `graph.ui_layout.editor_state`: optional presentation with `selection` (`nodeIds` and canonical connections) and `viewport` (`x`, `y`, positive zoom `k`).



- `graph.ui_layout.control_flow_blocks[blockId] : true` means collapsed.
- `graph.ui_layout.control_flow_case_sizes[blockId][caseId] : optional { "width": 1200, "height": 900 }` dimensions in graph units. Missing or malformed entries are initialized after all visible bodies are measured. Case/block deletion removes corresponding entries. Positions stay on ordinary node records; measured bounds and drag previews are transient. Input formulas retain their existing `ui_layout.input_formulas["nodeId:portId"]` format. Presentation never defines execution scope.

The compiler orders the main graph with each block as one operation, orders each case separately, emits shared dependencies first and downstream consumers afterwards, and assigns the common audio variables on every path. Score export retains the `notnum` to score-note conversion. CLI transformations of existing graphs must retain `control_flow` and patch version 2; the CLI's high-level layer syntax does not create branching automatically. Use the full drumset fixture as a complete authoring reference.

## Deutsch – Synthesezweig beim Notenstart

**If** und **Switch** sind Editor-Konstrukte in **Kontrollfluss**. Sie erzeugen Csound-Bedingungen. Jede Stimme wählt beim Notenstart genau einen Zweig, der allein initialisiert und ausgeführt wird. Die Auswahl bleibt bis zum Ende dieser Stimme bestehen; auch wiederholte gleiche Noten sind unabhängig.

If vergleicht zwei Eingänge mit i-Rate über `==`, `!=`, `<`, `<=`, `>` oder `>=`. Switch nutzt einen Auswahlwert und eindeutige numerische Zweigwerte sowie einen letzten Standardzweig. Verbindungen und arithmetische Eingangsformeln haben Vorrang vor den Zahlenfeldern. Neue Blöcke liefern Stereo.

Zweige auf dem Canvas ausklappen und Katalogknoten direkt darin ablegen. Ein gültiges Ablegen in **Stille** schaltet auf **Synthese** und zeigt die Ergebniseingänge. Auf freiem Canvas entstehen Hauptgraphknoten. Ein eingeklappter Block wird beim Ablegen zunächst ausgeklappt; danach einen bestimmten Zweig wählen.

Der Block bleibt linksbündig oberhalb des ersten Zweigs, einschließlich seiner Bedienelemente. Rahmen messen alle Parameterzeilen unabhängig vom Zoom und wachsen nach rechts und unten. Sie behalten ihre Größe, wenn Knoten nach innen verschoben, entfernt oder kleiner werden. Das **Zweigergebnis** bleibt unten rechts in einer eigenen Zeile. Nachfolgende Zweige werden mit unveränderter innerer Anordnung verschoben. Titelbereich: 40 Graph-Einheiten, Innenabstand: 24, Zweigabstand: 32, leerer Inhaltsbereich mindestens 240 × 160.

Zum Vergrößern oder Verkleinern eines Zweigs den **Titel oder freien Körper des Zweigergebnisses ziehen**; das funktioniert auch mit dem Stille-Marker. Die Zeigerbewegung bestimmt Breite und Höhe. Vollständige Knotengrenzen, Innenabstände und Ergebniszeile begrenzen die Mindestgröße. Andere ausgewählte Knoten bleiben stehen; die Auswahl bleibt erhalten. Buchsen, Formeleditor und Hilfeschnittflächen funktionieren wie bisher. Loslassen übernimmt Größe und verwaltete Positionen gemeinsam; **Escape**, Abbruch oder Fokusverlust stellt die gesamte vorherige Anordnung wieder her. Normales Ziehen behält die größte während der Geste erreichte Rahmengröße. Helle violette Rahmen bleiben bei jedem Zoom etwa zwei Bildschirmpixel breit. Größen bleiben beim Einklappen, Blockverschieben, Umsortieren, Speichern, Neuladen, Kopieren und Export/Import erhalten.

Hauptgraphknoten können durch normales Ziehen in einen Zweig wechseln. Normales Ziehen vorhandener Zweigknoten ändert nur deren Anordnung; obere und linke Grenzen begrenzen die gesamte Auswahl gemeinsam. Zum Wechsel in einen anderen Zweig oder den Hauptgraphen **Alt/Option bereits beim Beginn des Ziehens halten. Ausgewählte Knoten verschieben** bleibt als Alternative mit derselben Prüfung verfügbar.

Bei einem Zugehörigkeitswechsel müssen alle Verbindungen und Formelbindungen eines verschobenen Knotens zu mitverschobenen Knoten oder Knoten im Zielbereich führen. Türkis markiert erlaubte Ziele; Rot und eine Diagnose nennen unzulässige Verbindungen und nicht ausgewählte Endpunkte. Es wird nichts automatisch ausgewählt, getrennt oder gelöscht. Ein unverbundener gewöhnlicher Knoten kann immer wechseln. Bestehende gemeinsame Eingänge aus dem Hauptgraphen bleiben gültig und können nach der Platzierung bewusst verdrahtet werden. Interne Werte verlassen den Zweig nur über das Zweigergebnis; dort jeden Audiokanal anschließen, bei Bedarf mit Summe oder Eingangsformel. Verschachtelte Blöcke, Hauptgraph-Konstrukte im Zweig und unabhängig verschobene Ergebnisse sind unzulässig.

Loslassen übernimmt Positionen und Zugehörigkeit gemeinsam. Abgewiesenes Ablegen, **Escape**, Abbruch oder Fokusverlust stellt die gesamte Gruppe wieder her und entfernt die Hervorhebung. Stille bleibt bei ungültigem Ablegen unverändert. Der Block nimmt beim Verschieben alle Inhalte mit.



Falsch und Standard beginnen mit **Stille**: alle Ausgänge erhalten null, ohne Syntheseknoten. Wechsel zu Stille, Zweiglöschung und Formatwechsel zeigen betroffene Elemente vor der Bestätigung. Mono behält `left`; Stereo ergänzt einen unverbundenen `right`-Kanal. Standard bleibt zuletzt und kann nicht gelöscht werden. Einklappen, Speichern und Neuladen erhalten IDs, Auswahl, Positionen, Ansichtsfenster und Formeln; für Formeln an ausgeblendeten Eingängen wieder ausklappen.

**Neue Vorlage → Schlagzeug**: 36 Kick, 38 Snare, 42 Hi-Hat, sonst Stille. Anschlagstärke skaliert die Hüllkurven, lokales `xtratin` erhält das Ausklingen nach kurzem Note-off. Die MIDI-Testnote im Vorhören startet bei 36. Speichern, in Perform einen MIDI-Kanal zuweisen und Stereo Output im Mixer verbinden. Effekte nach dem Block wirken pro Stimme; Mixer-Effekte bearbeiten das gesamte summierte Schlagzeug. Das If-Beispiel wählt Sinus oder Sägezahn nach Anschlagstärke.

**Datenformat**: Bestehende Patches bleiben Version 1; Verzweigungen führen dauerhaft zu Version 2. `graph.control_flow` enthält pro Block `kind`, `output_format`, `operator` und geordnete `cases`. Jeder Zweig speichert stabile `id`, `name`, `value`, exklusive `node_ids`, `result_node_id` und `silence`. Die JSON-Feldnamen und das Beispiel oben gelten in allen Sprachen. `ui_layout.control_flow_blocks` speichert den Einklappzustand; `ui_layout.control_flow_case_sizes[blockId][caseId]` speichert `{width, height}` in Graph-Einheiten. Ungültige oder fehlende Größen werden nach vollständiger Messung neu ermittelt, bei Löschung werden die zugehörigen Einträge entfernt. Beides ist reine Darstellung. Speichern, Kopieren, App-Zustand sowie Instrument-, Performance- und CSD-Export behalten die Struktur. Unbekannte Versionen, Zyklen und ungültige Zugehörigkeit werden abgewiesen. Keine Verschachtelung oder laufende Umschaltung; `outs`, `inleta`, `outleta`, `sload`, `maxalloc` bleiben im Hauptgraphen. GEN und Dauersteuerung sind im Zweig erlaubt.

## Français — choisir au début de la note

**If** et **Switch**, dans **Contrôle de flux**, sont des constructions de l'éditeur qui produisent des conditions Csound. Chaque voix choisit un seul cas au début de sa note. Seul ce cas est initialisé et exécuté jusqu'à la fin de la voix. Les notes différentes ou répétées restent indépendantes.

If compare deux entrées de taux `i` avec `==`, `!=`, `<`, `<=`, `>` ou `>=`. Switch compare son sélecteur à des valeurs numériques uniques et garde un cas Par défaut en dernier. Connexions et formules arithmétiques d'entrée sont prioritaires sur les valeurs littérales. Les nouveaux blocs produisent de la stéréo.

Développez les cas et déposez-y directement les nœuds du catalogue. Un dépôt valide dans **Silence** active **Synthèse** et affiche les entrées du résultat. Un dépôt sur le canevas libre crée un nœud dans le graphe principal. Déposer sur un bloc réduit le développe ; déposez ensuite dans un cas précis.

Le bloc reste au-dessus du premier cas, aligné à gauche, avec de la place pour ses boutons. Les cadres mesurent toutes les lignes de paramètres indépendamment du zoom et s'agrandissent vers la droite et le bas. Ils conservent leur taille lorsque les nœuds sont déplacés vers l'intérieur, supprimés ou deviennent plus petits. Le **Résultat du cas** reste en bas à droite dans une ligne réservée. Les cas suivants se décalent sans changer leur disposition interne. Bande de titre : 40 unités du graphe, marge intérieure : 24, espacement entre cas : 32, contenu vide minimal : 240 × 160.

Glissez le **titre ou une zone libre du Résultat du cas** pour agrandir ou réduire la branche dans les deux dimensions, y compris le marqueur Silence. Le déplacement du pointeur définit la taille souhaitée ; les limites complètes des nœuds, les marges et la ligne du résultat fixent le minimum. Les autres nœuds sélectionnés restent immobiles et la sélection est conservée. Prises, formules et boutons d'aide gardent leurs actions habituelles. Relâcher valide ensemble la taille et les positions gérées ; **Échap**, une annulation ou une perte de focus restaure toute la disposition précédente. Un glissement ordinaire conserve la plus grande taille atteinte pendant le geste. Les bordures violet clair gardent environ deux pixels à l'écran quel que soit le zoom. Les tailles survivent à la réduction, au déplacement du bloc, au réordonnancement, à la sauvegarde, au rechargement, aux copies et à l'export/import.

Un glissement normal transfère les nœuds du graphe principal dans un cas. Pour les membres d'un cas, il modifie seulement leur disposition ; les limites supérieure et gauche s'appliquent au groupe entier. Maintenez **Alt/Option** dès le début du **glissement** pour transférer vers un autre cas ou le graphe principal. **Déplacer les nœuds sélectionnés** reste disponible avec la même validation.

Lors d'un transfert, chaque connexion et liaison de formule touchant un nœud déplacé doit aboutir à un autre nœud déplacé ou déjà dans le périmètre cible. Une bordure cyan signale un dépôt autorisé ; une bordure rouge et un diagnostic indiquent les connexions refusées et les extrémités non sélectionnées. Rien n'est sélectionné, déconnecté ou supprimé automatiquement.



Un nœud ordinaire isolé peut toujours être transféré. Les entrées communes du graphe principal restent valides et peuvent être câblées volontairement après placement. Les valeurs internes sortent uniquement par le résultat ; reliez chaque canal, avec somme ou formule si nécessaire. Les blocs imbriqués, les constructions réservées au graphe principal et le transfert indépendant des résultats sont refusés.

Relâcher valide ensemble positions et appartenance. Un refus, **Échap**, une annulation ou une perte de focus restaure tout le groupe et retire la surbrillance. Un dépôt refusé conserve Silence. Déplacer le bloc déplace tout son contenu.

Faux et Par défaut commencent en **Silence**, avec zéro sur chaque sortie et aucun nœud de synthèse. Passer à Silence, supprimer un cas ou changer de format présente les éléments affectés avant confirmation. Mono conserve `left` ; revenir en stéréo ajoute `right` sans connexion. Par défaut reste obligatoire. Réduire, enregistrer et recharger conserve identifiants, sélection, positions, cadrage et formules ; développez-le pour modifier une formule d'entrée masquée.

**Nouveau modèle → Batterie** : 36 grosse caisse, 38 caisse claire, 42 charleston, sinon silence. La vitesse règle les enveloppes, et `xtratim` local conserve leur décroissance après une note courte. La note MIDI de test de l'audition commence à 36. Enregistrez, affectez un canal MIDI dans Perform et routez Stereo Output dans le mixeur. Les effets après le bloc traitent chaque voix ; ceux du mixeur traitent la somme de la batterie. L'exemple If choisit sinus ou dent de scie selon la vitesse.

**Format des données** : les anciens patches restent en version 1 ; les branches font passer durablement à la version 2. `graph.control_flow` stocke `kind`, `output_format`, `operator` et les `cases` ordonnés. Chaque cas contient une `id` stable, `name`, `value`, les `node_ids` exclusifs, `result_node_id` et `silence`. Les champs JSON et l'exemple ci-dessus sont identiques dans toutes les langues. `ui_layout.control_flow_blocks` conserve l'état réduit ; `ui_layout.control_flow_case_sizes[blockId][caseId]` conserve `{width, height}` en unités du graphe. Les tailles absentes ou invalides sont recalculées après mesure complète ; supprimer un cas ou bloc retire ses entrées. Ces champs concernent uniquement la présentation. Sauvegardes, copies, état de l'application et exports instrument, performance et CSD préservent la structure. Versions inconnues, cycles et appartenances invalides sont rejetés. Pas d'imbrication ni de commutation continue ; `outs`, `inleta`, `outleta`, `sfloat`, `maxalloc` restent dans le graphe principal. GEN et les contrôles de durée sont permis dans les cas.

## Español — elegir al inicio de la nota

If y **Switch**, en **Flujo de control**, son construcciones del editor que generan condiciones Csound. Cada voz elige un caso al inicio de su nota. Solo ese caso se inicializa y ejecuta hasta el final de la voz. Las notas distintas o repetidas son independientes.

If compara dos entradas de tasa `i` con `==`, `!=`, `<`, `<=`, `>` o `>=`. Switch compara su selector con valores numéricos únicos y mantiene Por defecto al final. Las conexiones y fórmulas aritméticas de entrada tienen prioridad sobre los números literales. Los bloques nuevos producen estéreo.

Expande los casos y suelta directamente en ellos los nodos del catálogo. Una operación válida en **Silencio** activa **Síntesis** y muestra las entradas del resultado. Soltar sobre el lienzo libre crea nodos en el grafo principal. Soltar sobre un bloque contraído lo expande; después suelta en un caso concreto.

El bloque permanece encima del primer caso, alineado a la izquierda, con espacio para sus botones. Los marcos miden todas las filas de parámetros independientemente del zoom y crecen hacia la derecha y abajo. Conservan su tamaño cuando los nodos se desplazan hacia dentro, se eliminan o se hacen más pequeños. El **Resultado del caso** permanece abajo a la derecha en una fila reservada. Los casos posteriores se desplazan sin alterar su organización interna. Banda de título: 40 unidades del grafo, margen interior: 24, separación entre casos: 32, área vacía mínima: 240 × 160.

Arrastra el **título o una zona libre del Resultado del caso** para ampliar o reducir la rama en ambas dimensiones, también con el marcador Silencio. El desplazamiento del puntero define el tamaño solicitado; los límites completos de los nodos, los márgenes y la fila del resultado fijan el mínimo. Los demás nodos seleccionados permanecen inmóviles y se conserva la selección. Conectores, fórmulas y botones de ayuda mantienen sus acciones habituales. Soltar confirma el tamaño y las posiciones gestionadas juntos; **Escape**, cancelación o pérdida de foco restaura toda la disposición anterior. El arrastre normal conserva el mayor tamaño alcanzado durante el gesto. Los bordes violeta claro mantienen unos dos píxeles de pantalla con cualquier zoom. Los tamaños sobreviven a contraer, mover el bloque, reordenar, guardar, recargar, copiar y exportar/importar.



Arrastrar normalmente transfiere nodos del grafo principal a un caso. Para miembros de un caso, solo cambia su disposición; los límites superior e izquierdo se aplican al grupo completo. Mantén **Alt/Opción desde el inicio del arrastre** para transferir a otro caso o al grafo principal. **Mover nodos seleccionados** sigue disponible con la misma validación.

En una transferencia, cada conexión y vínculo de fórmula que toque un nodo movido debe terminar en otro nodo movido o en uno que ya pertenezca al ámbito de destino. Un borde cian indica un destino válido; un borde rojo y un diagnóstico enumeran conexiones rechazadas y extremos no seleccionados. No se selecciona, desconecta ni borra nada automáticamente. Un nodo ordinario aislado siempre puede transferirse. Las entradas compartidas del grafo principal siguen siendo válidas y pueden conectarse deliberadamente después de colocar los nodos. Los valores internos solo salen por el resultado; conecta cada canal, usando sumas o fórmulas si hace falta. Se rechazan bloques anidados, construcciones exclusivas del grafo principal y transferencias independientes de resultados.

Soltar confirma a la vez posiciones y pertenencia. Un rechazo, **Escape**, una cancelación o la pérdida de foco restaura todo el grupo y elimina el resaltado. Una operación inválida conserva Silencio. Mover el bloque mueve todo su contenido.

Falso y Por defecto empiezan con **Silencio**: cero en todas las salidas y ningún nodo de síntesis. Cambiar a Silencio, borrar un caso o cambiar de formato muestra los elementos afectados antes de confirmar. Mono conserva `left`; volver a estéreo añade `right` sin conectar. Por defecto es obligatorio. Contraer, guardar y recargar conserva IDs, selección, posiciones, encuadre y fórmulas; expande para editar las fórmulas de entradas ocultas.

**Nueva plantilla → Batería:** 36 bombo, 38 caja, 42 charles; el resto produce silencio. La velocidad escala las envolventes, y `xtratism` local conserva el decaimiento tras notas cortas. La nota MIDI de prueba de Audición empieza en 36. Guarda, asigna un canal MIDI en Perform y conecta Stereo Output en el mezclador. Los efectos posteriores al bloque procesan cada voz; los del mezclador procesan la suma de la batería. El ejemplo If elige seno o diente de sierra según la velocidad.

**Formato de datos:** los patches existentes siguen en versión 1; añadir ramas eleva permanentemente la versión a 2. `graph.control_flow` guarda `kind`, `output_format`, `operator` y los casos ordenados. Cada caso contiene una `id` estable, `name`, `value`, `node_ids` exclusivos, `result_node_id` y `silence`. Los campos JSON y el ejemplo anterior son iguales en todos los idiomas. `ui_layout.control_flow_blocks` guarda el estado contraído; `ui_layout.control_flow_case_sizes[blockId][caseId]` guarda `{width, height}` en unidades del grafo. Los tamaños ausentes o inválidos se recalculan tras medir todos los nodos; borrar casos o bloques elimina sus entradas. Ambos campos son solo presentación. Guardado, copias, estado de aplicación y exportaciones de instrumento, performance y CSD conservan la estructura. Se rechazan versiones desconocidas, ciclos y pertenencias inválidas. No se admite anidamiento ni conmutación continua; `outs`, `inleta`, `outleta`, `sfloat`, `maxalloc` permanecen en el grafo principal. GEN y controles de duración sí pueden pertenecer a casos.

## Csound references

- [Csound if](#)
- [Csound xtratism](#)



# Instrument Import / Export and CSD Export

This page documents all user-facing import/export options available from the Instrument Design page.

## Instrument Export Types

### 1. Orchestron Instrument Bundle (Export button)

The `Export` button creates an Orchestron instrument definition bundle containing the patch graph and metadata.

File extensions:

- `.orch.instrument.json` (JSON bundle)
- `.orch.instrument.zip` (ZIP bundle)

### JSON vs ZIP Export Selection

Orchestron chooses the format automatically:

- JSON export when no referenced uploaded GEN01 or `sload` assets are present
- ZIP export when the instrument references uploaded GEN01 audio assets or `sload` SoundFont assets (so audio dependencies can be included)

### 2. Csound CSD Export (Export CSD button)

The `Export CSD` button compiles the patch and downloads a `.csd` file.

Use this when you want to run the generated instrument outside Orchestron in a Csound workflow.

## What Is In An Instrument Bundle

An instrument bundle contains:

- patch name
- patch description
- template flag
- instrument type (Percussion, Melody, Bass, Effects / Noise, or Continuous)
- synchronized always-on flag for compatibility
- schema version
- full patch graph (`nodes`, `connections`, `ui_layout`, `engine_config`)

If a ZIP bundle is required, it also contains referenced uploaded audio/SoundFont assets under an `audio/` folder in the archive.

## Instrument Import (Import button)

The import button accepts:

- standalone instrument bundle JSON
- standalone instrument bundle ZIP
- performance bundle JSON/ZIP (to extract included patch definitions)

This means you can import patches directly from a performance export without manually splitting files. Instrument Type is preserved for both new and overwritten patches, including imports renamed during conflict resolution. Older bundles without a type receive an inferred type; always-on patches become Continuous and unmatched MIDI patches default to Melody.



## Conflict Resolution (Patch Name Already Exists)

When importing one or more patch definitions and a patch name already exists, Orchestron opens a conflict dialog.

Per patch, you can choose:

- Overwrite (update existing patch)
- Rename (disable overwrite and provide a new target name)
- Skip (patch imports only)

The dialog validates name collisions before proceeding.

## Audio Dependency Handling

For ZIP imports that include referenced GEN01 audio or `sload` SoundFont assets:

- Orchestron validates the archive structure
- Imports the audio files into backend storage automatically
- Restores the patch graph with the asset references intact
- Rejects archives that exceed backend bundle limits before unpacking large members

## Advanced Format Notes (for power users)

- Type metadata is exported as `instrumentType`; imports also accept `instrument_type`. Supported values are `percussion`, `melody`, `bass`, `effects_noise`, and `continuous`. An explicit type determines activation even if an older `alwaysOn` / `always_on` flag conflicts. Existing graph schema and bundle version numbers are unchanged.
- Instrument ZIP exports contain exactly one root JSON file plus optional `audio/...` entries
- Missing referenced audio assets in a ZIP import cause import failure (prevents broken GEN01 references)
- Default import limits are 256 MiB compressed request body, 8 MiB root JSON, 512 ZIP entries, and 256 MiB total uncompressed ZIP content
- Deployments can tune these with `VISUALCSOUND_BUNDLE_IMPORT_MAX_BYTES`, `VISUALCSOUND_BUNDLE_IMPORT_JSON_MAX_BYTES`, `VISUALCSOUND_BUNDLE_IMPORT_ZIP_MAX_MEMBERS`, and `VISUALCSOUND_BUNDLE_IMPORT_ZIP_MAX_UNCOMPRESSED_BYTES`
- Individual imported audio/SoundFont assets are still limited by `VISUALCSOUND_GEN_AUDIO_ASSET_MAX_BYTES`

## Instrument Bundle vs Performance Bundle

Use the right export for the right task:

- Instrument bundle: share one patch design
- Performance bundle: share a full performance setup (instrument rack + sequencers + controllers + piano rolls + optional patch definitions)

See [Performance Import / Export](#) for performance bundles.

## Screenshots



## Name Conflicts

Existing items were found. Keep overwrite checked to replace existing entries. Uncheck overwrite to import under a new name. Enable skip to ignore a patch definition.

**Instrument patch: FM Metal**

☒ OVERWRITE ☐ SKIP

NEW NAME

FM Metal Copy

CANCEL

IMPORT

*Patch import conflict dialog with overwrite/rename/skip decisions.*



# Performance Import / Export

Performance import/export lets you move complete live setups between machines or save versioned backups.

## Performance Export (Export)

The Performance page `Export` button creates an Orchestron performance bundle from the current performance workspace.

Use this export when you want to:

- Re-import the performance into Orchestron later
- Move a full performance setup to another machine
- Keep versioned backups of a performance together with its patch references

File extensions:

- `.orch.json`
- `.orch.zip`

## What Gets Exported

A performance export includes:

- Performance metadata ( `name` , `description` )
- Custom names for all six performance device types, including names created through the CLI. Existing names are preserved on import; the GUI applies its name validation only when renaming a device.
- Sequencer/drummer-sequencer/arpeggiator/piano-roll/controller/controller-sequencer configuration snapshot
- Instrument assignments with stable instance IDs
- Explicit audio routes, internal Master settings, insert ownership, strip gain/balance/mute/solo, and send level/pre-post settings
- Referenced patch definitions for the instruments currently assigned in the performance rack
- Patch names (included in the snapshot for easier remapping on import)

If any referenced patch contains uploaded GEN01 audio assets or `sfload` SoundFont assets, export is automatically produced as a ZIP and includes those assets.

## Offline Render Export (Export CSD (MIDI) / Export CSD (SCORE))

The Performance page provides two offline Csound render exports:

- `Export CSD (MIDI)` creates the traditional ZIP with a compiled CSD and a separate MIDI file.
- `Export CSD (SCORE)` creates a ZIP with the performance notes and controller sweeps embedded as Csound score events.
- Both CSD export modes seed enabled manual MIDI Controller lane values at time 0 on their selected MIDI channels (all 16 by default).

Both modes select instruments by device assignment and mixer routing:

- Include rack instances whose MIDI channel is assigned to a melodic sequencer, drummer sequencer, piano-roll keyboard or arpeggiator output. An arpeggiator input alone is not an instrument assignment.
- Device assignments count even when the device is stopped, disabled or has an empty pattern. Selection does not depend on whether a note happens during the exported arrangement.
- Include continuous ( `always on` ) instruments referenced by mixer routes, internal Master settings or insert ownership. Continuous generators with implicit direct audio output are also included.
- Omit other rack instances, routes and mixer controls belonging to those omitted instances, and assets used exclusively by omitted patches. Native `Export` still preserves the complete rack.



Each CSD starts with a comment header containing the performance title, description and creation timestamp, followed by an Orchestron attribution, a short summary of Orchestron's visual instrument design, performance devices, audio routing and portable Csound export features, plus the project GitHub link. Each exported instrument also has CSD comments containing its name, description, rack instance ID and Csound reference. Multiline descriptions remain comments on every line. Routing comments identify source/destination instruments and ports, raw/strip stages, insert ownership, Master processing, direct paths that bypass Master, pre/post-fader taps, and initial gain, balance, mute and solo settings. Mixer instruments are emitted in signal-flow order.

This export is different from the normal `Export` bundle:

- `Export` is for Orchestron import/export workflows
- `Export CSD (MIDI)` and `Export CSD (SCORE)` are for rendering outside Orchestron with standard Csound tools

The MIDI ZIP contains:

- A compiled `.csd` with the selected non-template device instruments and routed continuous instruments
- Offline render settings forced to `sr = 48000` and `ksmps = 1`
- WAV output written as 32-bit float (`-f`) so exported files preserve headroom instead of baking clipping into 16-bit samples
- Always-on effect instruments started with Csound `alwayson`
- A finite `f 0 ...` score duration sized for the exported arranger playback plus a release-tail buffer
- The arranger playback rendered as a `.mid` file from beginning to arrangement end
- Enabled manual MIDI Controller lane values written into the MIDI file at tick 0 on their selected MIDI channels (all 16 by default)
- Uploaded bundled sample audio / SoundFont files used by the exported instruments
- A `README.txt` with the exact Csound command line needed to render the package

The SCORE ZIP contains the same compiled instruments and bundled assets, but omits the `.mid` file. Instead, it embeds note events as `score i` statements, embeds controller sequencer sweeps and enabled manual MIDI Controller lane values as score-controlled CC setter events, and rewrites supported MIDI opcodes such as `cpsmidi`, `ampmidi`, `midi_note`, `notnum`, and `midictrl` for score playback. If export-time approximations are needed, such as best-effort `ampmidi` function-table mapping, the ZIP includes `WARNINGS.txt` and the README lists the warnings.

Only assets stored through Orchestron's upload/import flow are bundled. `GEN01` and `sload` nodes must reference uploaded assets; compile, session start, and offline performance CSD export reject raw filesystem `samplePath` values instead of passing them to Csound.

Offline performance CSD export is bounded before synthesis starts: looping playback is rejected, playback ranges are limited to 65,536 transport steps, a single step can carry at most 16 notes, and the estimated MIDI event budget is limited to 200,000 events. MIDI generation also stops if it exceeds the event budget or takes more than 5 seconds.

ZIP layout:

- The ZIP contains a single top-level directory
- That directory has the same basename as the exported performance `.csd`
- Inside the MIDI export directory you will find the `.csd`, `.mid`, `README.txt`, and `assets/` subdirectory
- Inside the SCORE export directory you will find the `.csd`, `README.txt`, optional `WARNINGS.txt`, and `assets/` subdirectory

Typical extracted structure:

```
Offline_Export/  
  Offline_Export.csd  
  Offline_Export.mid  
  README.txt  
  assets/
```

Typical SCORE structure:



```
Offline_Export/  
  Offline_Export.csd  
  README.txt  
  WARNINGS.txt  
  assets/
```

Typical use:

- Share a performance as a portable offline render package
- Render the arrangement outside Orchestron with stock Csound
- Archive a self-contained .csd + .mid + assets bundle, or a single inline-score .csd + assets bundle, for later mastering or batch rendering

Rendering workflow:

1. Click Export CSD (MIDI) or Export CSD (SCORE) on the Performance page.
2. Extract the downloaded ZIP.
3. Change into the bundled directory inside the extracted archive.
4. Run the command from README.txt, for example the MIDI export command:

```
csound -d -W -f -o Offline_Export.wav -F Offline_Export.mid Offline_Export.csd
```

For SCORE exports the command omits -F :

```
csound -d -W -f -o Offline_Export.wav Offline_Export.csd
```

The exported .csd already includes matching CsOptions, so `csound Offline_Export.csd` also works after you change into that directory.

## Performance Import (Import)

The Performance page Import button supports two categories of files:

Bundle imports are bounded by backend request and ZIP limits before large archive members are decompressed. Defaults are 256 MiB compressed request body, 8 MiB import JSON, 512 ZIP entries, and 256 MiB total uncompressed ZIP content. Tune these with `VISUALCSOUND_BUNDLE_IMPORT_MAX_BYTES`, `VISUALCSOUND_BUNDLE_IMPORT_JSON_MAX_BYTES`, `VISUALCSOUND_BUNDLE_IMPORT_ZIP_MAX_MEMBERS`, and `VISUALCSOUND_BUNDLE_IMPORT_ZIP_MAX_UNCOMPRESSED_BYTES`; individual bundled audio/SoundFont assets still use `VISUALCSOUND_GEN_AUDIO_ASSET_MAX_BYTES`. Persistent generated-asset storage is capped by `VISUALCSOUND_GEN_AUDIO_ASSETS_MAX_TOTAL_BYTES` and `VISUALCSOUND_GEN_AUDIO_ASSETS_MAX_COUNT`; unreferenced generated assets older than `VISUALCSOUND_GEN_AUDIO_ASSET_GC_MIN_AGE_SECONDS` are eligible for garbage collection.

### 1. Full Orchestron Performance Export (recommended)

If the file matches the performance export format, Orchestron opens an import options dialog.

### 2. Raw/legacy sequencer snapshot JSON

If the file is not a full performance export bundle, Orchestron attempts to apply it directly as a sequencer configuration snapshot.

This is useful for advanced workflows and backward compatibility.

## Import Options Dialog (Performance Bundle)

When importing a full performance bundle, you can choose whether to import:



- performance
- patch definitions (if present in the bundle)

You may import either or both.

Typical uses:

- Import both (full environment restore)
- Import only patch definitions (extract instruments from a performance file)
- Import only performance (if the destination machine already has matching patches)

## Conflict Resolution Dialog

If names already exist, Orchestron opens a conflict dialog.

### For Patches

Per patch definition you can choose:

- Overwrite existing patch
- Rename and create a new patch
- Skip importing that patch

### For Performance

For the performance itself you can choose:

- Overwrite existing performance
- Rename and create a new performance

(Performance entries do not use `Skip` in the conflict dialog; skipping is handled via the import options step.)

## Patch ID Remapping During Performance Import

Performance bundles store instrument patch references. On import, Orchestron remaps patch IDs using:

1. Imported patch definitions (source ID -> newly created/updated destination patch ID)
2. Existing patch name matches (when patch definitions are not imported but patch names match)

This makes performance imports more portable across machines and patch databases.

## Import Validation (Resolvable Instruments Required)

A performance import fails if no instrument assignments can be resolved to available patches.

In that case:

- import patch definitions from the bundle, or
- create/match patches locally by name first

## Related Workflows

- Instrument bundle import/export is documented in [Instrument Import / Export](#) and [CSD Export](#).
- You can also import patch definitions from a performance bundle on the Instrument Design page.

## Mixer snapshot

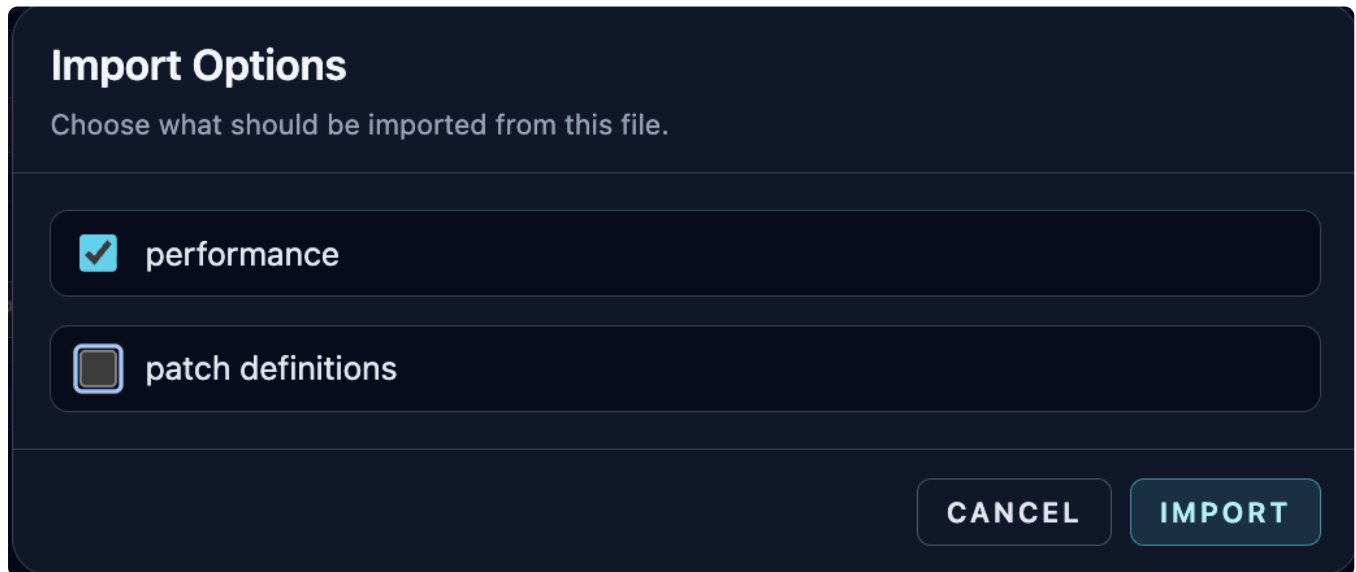
Native bundles and both performance CSD modes capture version 11 routing and mixer settings: gain, pan/balance,



mute/solo, sends including pre/post, Master and inserts. Export waits for outstanding live mixer acknowledgments and reports synchronization failures. Mixer values are initialized inside each CSD and require no control client. MIDI and SCORE notes retain their authored velocities; SCORE instrument references come from the compiler manifest.

Offline rendering remains 48 kHz with ksmps=1, float WAV output, packaged assets and release tails. Equivalent routing and settings are guaranteed, not sample-identical live/offline waveforms. Finite exports without MIDI notes are supported when a routed continuous source can produce audio. Standalone patch exports remain independent of Perform settings. See [Audio mixer and routing](#) for migration and signal flow.

## Screenshots



*Performance import options dialog for importing performance and/or patch definitions.*



### Name Conflicts

Existing items were found. Keep overwrite checked to replace existing entries. Uncheck overwrite to import under a new name. Enable skip to ignore a patch definition.

|                                            |                                               |                                          |
|--------------------------------------------|-----------------------------------------------|------------------------------------------|
| Instrument patch: Bass                     | <input checked="" type="checkbox"/> OVERWRITE | <input checked="" type="checkbox"/> SKIP |
| Instrument patch: Moog                     | <input checked="" type="checkbox"/> OVERWRITE | <input checked="" type="checkbox"/> SKIP |
| Instrument patch: FM Organ                 | <input checked="" type="checkbox"/> OVERWRITE | <input type="checkbox"/> SKIP            |
| Instrument patch: Bass Drum                | <input type="checkbox"/> OVERWRITE            | <input type="checkbox"/> SKIP            |
| NEW NAME                                   |                                               |                                          |
| <input type="text" value="Bass Drum New"/> |                                               |                                          |
| Instrument patch: FM Metal                 | <input checked="" type="checkbox"/> OVERWRITE | <input checked="" type="checkbox"/> SKIP |
| Instrument patch: TB303                    | <input checked="" type="checkbox"/> OVERWRITE | <input type="checkbox"/> SKIP            |
| Performance: Demo Performance              | <input checked="" type="checkbox"/> OVERWRITE |                                          |

CANCELIMPORT

Conflict dialog for imported patch/performance name collisions.

## Performance controller settings (version 12)

Native JSON/ZIP bundles and both CSD export modes preserve each rack instance's `performanceControllerValues`, keyed by the `perf_controller` node ID. Unchanged controls use patch defaults; reset removes the override. Version 12 adds these values while preserving version 11 routing and mixer data; versions 1–11 still load. Patch-ID remapping during import preserves node identities and their settings.

CSD (MIDI) and CSD (SCORE) initialize values in the Csound orchestra before notes and continuous instruments start. No additional MIDI CC messages or external control client are required. Export waits for outstanding live controller updates and reports synchronization failures. Standalone patch exports use defaults. See [Performance controllers](#).

## MIDI controller channel selections (version 13)

Version 13 stores each MIDI Controller and Controller Sequencer's `targetChannels` selection as a sorted list of MIDI channels 1–16. At least one channel is required. Selections survive Save/Load Performance, browser reload, and native JSON/ZIP export/import. Versions 1–12 still load; missing channel selections default to all 16 channels (OMNI).

Both CSD modes route manual initial values and automated curves to the selected channels. Manual lanes sharing a CC number keep independent values on different channels; at render start, the later enabled lane wins only where both channel and CC number overlap. Channel selection does not change which instruments are included in the CSD. The native bundle envelope remains version 1, and browser app state remains version 2.



## Internal Master (version 14)

The fixed Master is stored as `$master` in `audioGraph.masterId`, route endpoints, mixer strips and insert ownership. It has no instrument assignment or patch definition. Native imports normalize designated neutral legacy Masters before importing patches, preventing duplicate library entries. Both CSD modes generate the internal output and initialize its saved controls and inserts. Direct Audio Output paths retain their own controls and continue to bypass Master.

Versions 1–13 remain readable. App state remains version 2 and native bundle envelopes remain version 1. See [Master migration and repair](#) for backups, customized processors and missing references.

## Arpeggiator Patterns (version 15)

Version 15 introduced arpeggiator patterns. App-state remains version 2; the native envelope remains version 1. Arpeggiators store eight musical pads, rhythm steps, seeds, processing/hold/playback settings, and Pad Looper arrangements. User presets store musical pad settings only.

On load/import, old musical settings migrate to #1 and accent cycles become step velocities. The remaining pads use defaults. Old performances adopt Arranger synchronization, Hold Off, Continue, corrected swing and full-range ordering; there is no legacy playback engine. Migration stays in memory until normal Save/export. See [Arpeggiator migration](#).

MIDI and SCORE CSD exports use the same arpeggiator scheduler as live audio, including source timestamps, pad boundaries, rests, ties, ratchets, overlap and seeded variation. Export event estimates account for chord expansion, Hold and ratchets; the existing hard event and time limits still apply. Export contains sequenced inputs; live keyboard gestures must be represented as sequence data to appear offline.

## Per-note timing (version 16)

Current saves and exports write performance config **16** and accept versions **1–16**. Melodic steps and drummer cells store `timingOffsetPercent`, an integer from –50 to +50; missing values mean zero. Browser app state remains version 2 and the native bundle envelope remains version 1. CLI runtime conversion maps this value to `timing_offset_percent`. MIDI and SCORE exports use the same shifted attacks and releases as live playback, within each format's timing resolution.

## Arrangement compatibility and temporary audition

The arrangement remains performance format v16, with imports of v1–16. Root sequences, group/supergroup references and pause tokens retain their meaning. Unused and empty draft definitions survive round trips. An imported enabled empty arrangement becomes an explicit occurrence of its previous active pad, retaining its repeat behavior. New empty lanes use Manual pads.

Save, autosave, native bundles and both CSD modes read authored configuration. Temporary audition selection, queued commands, playback position and active-pad/enablement overrides are excluded; edits to musical content still save normally.

Optional `padLoopPattern.definitionColors` stores per-definition display colours in performance v16. Keys are `pad:0 – pad:7`, `group:<ID>` and `super:<ID>`; values are six-digit hexadecimal colours. Invalid/missing entries use defaults. Unused definitions retain their colours. Save/load, autosave, native bundles and CLI handling preserve valid metadata; compiled sequences and both CSD modes ignore it. Momentary speaker previews and their restoration state remain excluded from every persistence path.



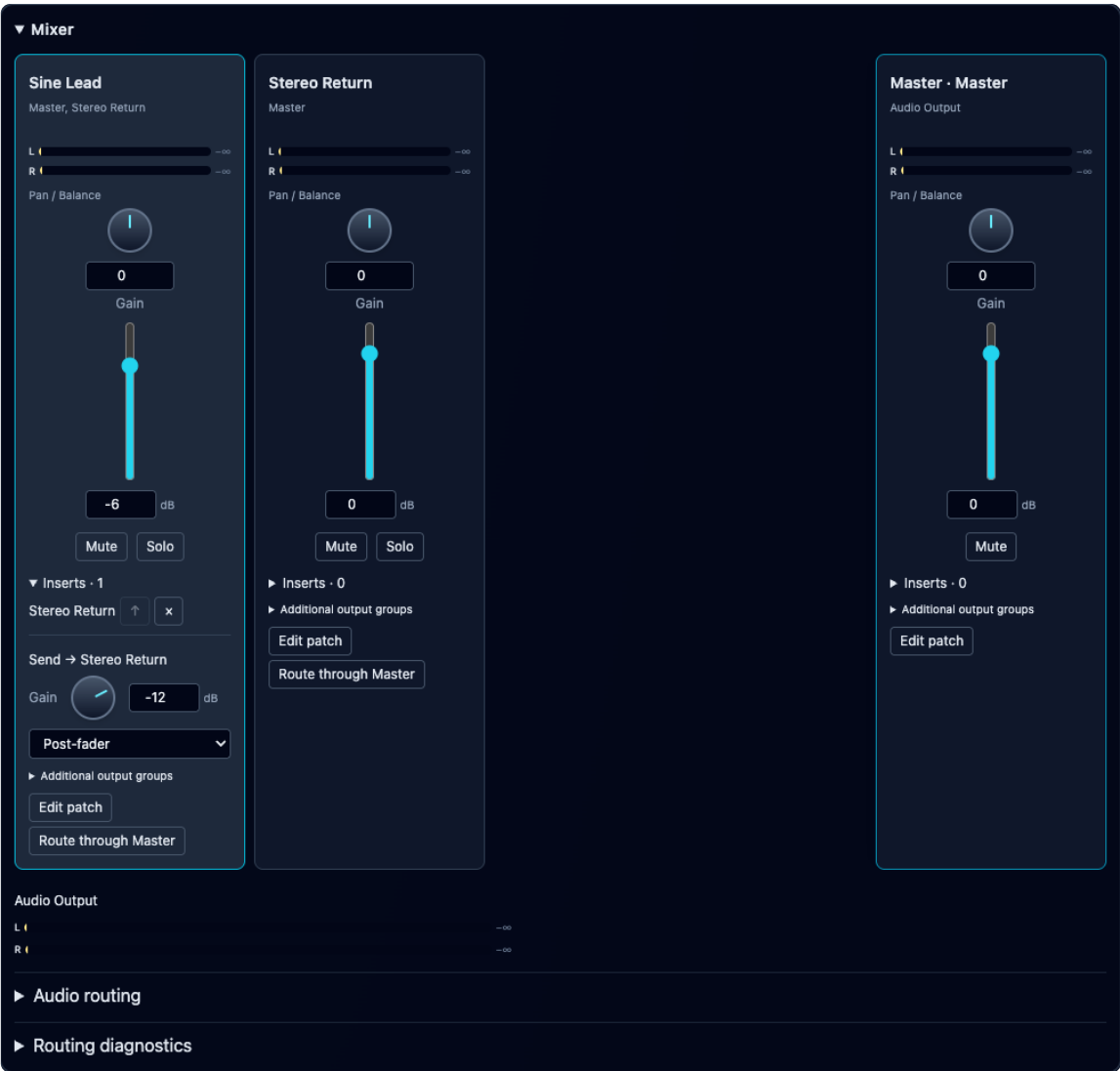
## Audio mixer and routing

The collapsible mixer sits below the instrument rack. Select a strip to edit its outputs and sends. The pinned Master controls audio routed into it. A **Direct Audio Output** badge identifies paths that bypass Master; these still respond to their own strip's fader, balance, mute and solo.

Collapsing Mixer suspends its visual contents and meter subscriptions while audio controls and meter collection keep running. Nested sections suspend independently and remember their expansion state when their parent closes. Reopening restores routing selections, editor feedback and strip scroll position, and meters show the latest snapshot. See [Collapsible Panels](#) for state lifetime and gesture cleanup.

### Build your first mix

1. In Instrument Design, choose **New → Playable instrument**, name the draft and **Save** it. Create and save an **Audio effect** when you need a processor. The built-in effect is a stereo pass-through starter; add your processing inside its graph.
2. In Perform, use **Add Instrument** and select the saved instrument. Set its MIDI channel. Guided instruments connect to the fixed internal Master automatically. For existing patches, inspect their destination labels and use **Route through Master** if needed.
3. Expand **Mixer** below the rack. Each strip header selects its source for the routing editor. The Master strip stays pinned at the right; scroll horizontally when the rack has many strips.
4. Start with 0 dB gain and centered Pan / Balance. Build routes and insert chains while stopped, then open **Routing diagnostics → Check routing**.
5. Click **Start Instruments** and play via a piano roll, sequencer or MIDI input. Change gain, balance, mute/solo and existing send settings while listening. Watch strip meters and the final Audio Output meter.
6. Use **Save Performance** to retain this mix. Stop the engine before adding/removing routes, replacing assignments or reordering inserts.



A stopped example mix: Sine Lead at -6 dB, a -12 dB post-fader send to Stereo Return, and pinned Master. The insert uses a separate instance of the same pass-through patch; no effect processing is implied by its name.

## Signal flow

| Path            | Processing order                                                                        |
|-----------------|-----------------------------------------------------------------------------------------|
| Instrument      | Summed voices → Perform inserts → pan/balance → fader → destination                     |
| Pre-fader send  | After inserts → send amount → destination                                               |
| Post-fader send | After pan/balance and fader → send amount → destination                                 |
| Return          | Summed inputs → effect patch → inserts → balance → return fader                         |
| Master          | Incoming routes → internal stereo bus → inserts → balance → Master fader → Audio Output |

Effects drawn inside an instrument graph run per voice. Perform inserts receive summed voices. Each inserted effect is a dedicated rack instance. Reordering an insert changes explicit connections; if custom connections no longer form a simple chain, **Custom routing** preserves those connections.

Additional output groups retain their own port names and mappings. They share the owning strip's gain and mute; inserts process the designated main output. Custom routes can express other processing arrangements.



## Inserts versus shared returns

An **insert** processes one strip in series after its voices are summed. Select that strip, choose a saved continuous patch with audio inputs and outputs in the **Inserts...** selector below the routing controls, then click the adjacent **Add**. Open **Inserts** on the strip to inspect its chain. Click a processor name to edit its patch, **↑** to move it earlier, or **x** to remove that insert instance. Reusing one saved effect creates separate processing instances; an insert does not appear as a separate mixer strip.

A **return** is a continuous rack instance that can receive sends from several sources and has its own mixer strip. Add it with **Add Instrument**, connect its output to Master, then add sends from the source strips. For a parallel reverb/delay arrangement, design the return patch to output the desired wet signal; the starter pass-through would merely add another copy of the dry signal.

## Add a stereo send

1. Stop the engine. Add the saved effect as a rack instance and use its **Route through Master** button to connect its main output.
2. In **Audio routing**, choose the instrument under **Source** and the return under **Destination**.
3. Leave **Stereo** checked and verify **Exact channel mapping**, for example `left / right → left / right`. Choose **Send** in the Route selector and click the routing row's **Add**.
4. On the source strip, find **Send → [return name]**. New sends start at silence; enter a value such as -12 dB to hear the return.
5. Choose **Post-fader** when the send should follow the source's fader/balance, or **Pre-fader** for an independent send amount. The source's Mute still silences either kind of send.

Adding a send does not replace the main output route. Avoid adding duplicate connections: parallel paths sum and can increase output level. Up to 1,024 individual channel connections and 64 rack instances (including processors, excluding internal Master) are supported; a stereo pair uses two connections.

## Controls

- Faders span -60 to +12 dB, with an exact-silence position below -60. Clearing the numeric value selects silence. Double-click resets to 0 dB.
- Sends span silence to +6 dB. New sends start silent and post-fader. Pre/post can change during playback.
- Rotary controls support vertical dragging, arrow keys, Page Up/Down, Home/End, exact numeric editing and double-click reset. Shift-drag provides finer adjustment.
- Stereo balance preserves both channels at unity in the center and attenuates the opposite channel toward either side. Explicit mono-to-stereo routing uses equal-power pan. Custom mono mappings stay unchanged.
- Mute blocks all outgoing paths, including pre-fader sends. Processors keep running, allowing downstream tails to decay.
- Multiple solos are allowed. Solo preserves required upstream inputs and downstream processors. Soling a return excludes its sources' unrelated dry branches. Mute takes precedence. Master has no Solo control.
- Gain, balance, send and mute/solo changes use finite 20 ms ramps. Playback starts at the saved values, without an initial fade.

Meters show stereo peak and RMS. CLIP indicates a peak at or above 0 dBFS; no automatic limiting, normalization or clipping is applied. The final **Audio Output** meter includes Master and direct paths. Connectivity diagnostics describe structure; meters measure actual audio activity.

## Routing and repair

Choose a source, destination and explicit channel mapping before adding a main route, send or custom connection. Stereo groups expand into their exact member ports. Uncheck **Stereo** to select one exact source port and destination inlet at a time, including custom or mono mappings. Group display names are not inlet names; legacy label suggestions never rename ports.

Open **Destination matrix** to inspect incoming routes grouped by destination. Expand a route row to change its source port, destination instance or destination port; **Remove** deletes that grouped connection. This is also where broken references can be repaired instead of discarding the performance.



Open **Routing diagram** to follow the same connections as source → destination cards. Expand **Exact channel mapping** on a card to inspect port names and processing stages. Click source/destination names to select routing endpoints; use the strip's **Edit patch** action to open its instrument graph.

**Audio routing**

Master Master

Source Sine Lead · f7f3 Destination Stereo Return · 2d9e ☒ Stereo Send Add

Exact channel mapping: left / right → left / right

Stereo Return Add

**Destination matrix**

Sine Lead

- ▶ Stereo Return → Sine Lead · insert · 2 ch Remove

Master

- ▶ Sine Lead → Master · main · 2 ch Remove
- ▶ Stereo Return → Master · main · 2 ch Remove

Stereo Return

- ▼ Sine Lead → Stereo Return · send · 2 ch Remove

|       |   |               |       |
|-------|---|---------------|-------|
| left  | → | Stereo Return | left  |
| right | → | Stereo Return | right |

Stereo Return

- ▶ Sine Lead → Stereo Return · insert · 2 ch Remove

**Routing diagram**

- Sine Lead → Master  
▶ Exact channel mapping
- Stereo Return → Master  
▶ Exact channel mapping
- Sine Lead → Stereo Return  
▶ Exact channel mapping
- Sine Lead → Stereo Return  
▶ Exact channel mapping
- Stereo Return → Sine Lead  
▶ Exact channel mapping

The send's left and right mappings expanded in Destination matrix. Matrix and diagram show the same routes, including the dedicated insert's input and return-to-strip connections.

**Route through Master** replaces the selected strip's main routes with stereo connections to the designated Master in this performance. Sends and custom routes are retained. Master is a fixed internal stereo output with volume, balance, mute, meters and effect inserts. It is always available, has no editable patch and occupies no library entry or rack slot. Its controls and inserts belong to the performance. **Audio Output** remains available as an editable starter for custom output processors.

New guided instruments connect to Master automatically. Loading an older performance preserves its authored direct/custom paths. Master gain or mute affects only paths routed into Master: a direct strip can still be audible. Check the **Direct Audio Output** badge and the final Audio Output meter when tracking down sound that remains after lowering Master.

Topology changes require stopping the engine. **Stop to edit routing** makes that action explicit. Existing mixer controls remain live. Invalid references and genuine feedback cycles block playback and CSD export. Broken routes remain available for repair. Unused ports, incomplete stereo pairs and parallel direct/processed paths are warnings.

## Troubleshoot routing and silence

Expand **Routing diagnostics** and click **Check routing** after editing connections. Diagnostics describe connectivity, while meters report measured sound. Clicking an instance-related diagnostic can open its patch for inspection. A **Muted or silent path** entry selects the corresponding strip.



| Symptom or diagnostic                            | What to check                                                                                                                                |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Repair route / Missing reference                 | Restore the referenced patch or repair the destination and exact port names in Destination matrix.                                           |
| Feedback cycle                                   | Remove a connection that feeds a processing path back into itself. Playback and CSD export remain blocked until genuine cycles are resolved. |
| Connect an audio output / Select an audio source | Follow the source to an output sink; route audio into effects that require inputs.                                                           |
| Complete stereo mapping / Unused ports           | Check both channel mappings and the declared audio groups. These structural warnings do not prove a patch is silent.                         |
| Review parallel output paths                     | Check whether both direct and processed copies intentionally reach the output.                                                               |
| Muted or silent path                             | Clear Mute or raise a fader/send from silence; also check active solos and whether the source is producing notes/audio.                      |
| CLIP                                             | Reduce gain along the contributing paths. No limiter is applied automatically.                                                               |

#### Routing diagnostics

Connectivity is structural; meters show measured audio.

Check routing

Sine Lead: Muted or silent path

*A valid route can still be silent: here the Sine Lead strip is muted.*

## Persistence and compatibility

Performance configuration version 15 retains the internal Master introduced in version 14, with stable instance IDs, explicit routes, mixer strips, sends and insert ownership. Versions 1–14 remain readable. App state version 2 saves this same model. Save/Load, Clone and native JSON/ZIP bundles preserve the current settings. Mixer automation recording is not included.

Versions 1–10 convert old Level values using  $20 * \log_{10}(\text{level} / 10)$ , including continuous effects. Missing Level means 0 dB. Legacy inlet selection is resolved once and saved explicitly. Level no longer scales MIDI velocity; velocity-sensitive patches can therefore change timbre after migration. Authored note velocities remain unchanged.

Both performance CSD modes initialize the same routing, gain, balance, mute, solo and send settings without a browser. Offline rendering retains 48 kHz, `ksmps=1`, float WAV output and release tails. Live/offline settings match; different engine rates need not produce identical waveforms. A routed continuous source can render a finite range with no MIDI notes. Standalone instrument exports contain patch design and interface metadata, without Perform mixer settings.

## Control API

Session creation and validation accept `audio_graph` and `mixer`. `GET /api/sessions/{id}/mixer` returns desired values and revision; `PUT` accepts batched partial scalar updates and an optional expected revision. Stale revisions return 409. The active browser-clock controller can send the equivalent `mixer_update` message and receive `mixer_ack`/`mixer_error` replies. Updates are queued and applied at render-block boundaries. Meter frames carry engine sample timestamps and are delivered at most 15 times per second.

`POST /api/sessions/preview` accepts a session plus inline draft patch definitions. It creates a transient runtime without saving drafts into the patch library. Normal stop/delete session operations clean it up.

The performance CLI writes version 16 and preserves routing, mixer data, instance settings and existing insert chains. `--level` is deprecated and converts to audio gain. New CLI routes require exact destination inlet selection when names differ; `-inlet` makes that mapping explicit.



## Author a mix with the performance CLI

The performance creator skill can stage main/send routes, strip and Master controls, and atomic stereo send updates. From its directory, with a staged performance and the named effect patches available:

```
uv run orchestron_cli --json edit add-standard-effects --send-gain-db -12
uv run orchestron_cli --json edit routes list
uv run orchestron_cli --json edit mixer list
uv run orchestron_cli --json edit mixer strip set \
  --binding '$master' --gain-db -3
```

The optional preset routes dry instrument outputs and a shared reverb return through a compressor into Master. No speaker instrument is required. It redirects existing direct outputs without cloning patches; new sends default to silence unless an initial amount is supplied. Repeating the preset preserves matching route IDs and saved mixer settings.

For explicit routing, edit routes add --kind main|send|custom selects the path type. Quote \$master and \$direct.left/right in shell commands. Main routes capture direct output ports; send/custom routes alone leave their direct-output bypass active. Remove an exact connection with edit routes remove --id ROUTE\_ID .

Use edit mixer strip set --binding ID for gain, balance, mute/solo (Master has no solo), and edit mixer send set --route LEFT\_ID --route RIGHT\_ID for a stereo send's gain and pre/post tap. Add --gain-db silence for exact silence. Omitted controls retain their saved values. Use the app for insert creation/reordering; the CLI preserves existing insert chains.

Validate and commit to save the mix. edit push-runtime applies mixer values to a matching live runtime; use edit rebuild-runtime after routing changes. Continuous-effect parameter overrides require rack restart, whereas ordinary mixer controls remain live. Full CLI examples are in the performance creator skill.

## Upgrading existing Masters

On backend startup, the SQLite library is migrated before it is served. The migration creates a sibling .before-internal-master-<timestamp>.bak database backup before changing data. Repeated starts make no further changes after migration.

Known neutral Master graphs are replaced by the internal output while preserving each performance's routes, gain, balance, mute and inserts. Customized Masters retain their processing as ordinary rack instruments feeding the internal Master. Unreferenced neutral Master copies are removed from the library only after references in saved performances and app state have been migrated. Modified patches, edited drafts and patches still used elsewhere are retained.

A genuinely missing legacy patch stays visible for repair. While stopped, **Replace missing Master with internal output** transfers compatible stereo connections and saved Master controls; save the performance afterward. Unknown port mappings must first be repaired in the routing matrix.

To preview the migration manually with the backend stopped, run `uv run python -m backend.tools.migrate_internal_master --database backend/data/visualcsound.db`. Add --apply to create a backup and apply it. Use the actual database path for Docker or custom installations. To roll back, stop the backend and restore the backup together with the previous application version.



# Performance

This chapter covers the Perform page (labeled Perform / Performance depending on language), where you build a playable multi-instrument setup and perform it live.

The performance description field displays three lines with automatic text wrapping and preserves manual line breaks. Longer descriptions scroll vertically within the field. Hover over the description to see its full text in a tooltip.

## Device Names

Melodic sequencers, drummer sequencers, controller sequencers, arpeggiators, piano rolls, and MIDI controllers all support custom names. Click the pen immediately to the right of a device name to edit it. The input receives focus with its current name selected.

- Save or Enter applies the name. Cancel or Escape discards the draft. Clicking elsewhere does not save; switching performances or removing the device discards the draft.
- Names must be nonempty after trimming surrounding whitespace and contain at most 65 Unicode characters. HTML and angle brackets ( < and > ) are rejected.
- Names must be unique across all six device types within the performance, ignoring case and surrounding whitespace. Validation messages appear while typing, and Save is disabled until the name is valid.
- Names retain their capitalization. Melodic synchronization choices and arranger track titles show the stored names. New devices receive an unused numbered default name.
- Renaming preserves device IDs, synchronization, routing, patterns, and playback state. Names survive performance save/load, browser persistence, raw snapshot import, and native JSON/ZIP export/import.
- Existing/imported names remain intact under the existing loading rules, even if they violate the new editing rules. They display as plain text and need a valid replacement only when renamed. CLI/API acceptance is unchanged.

## Collapsible Panels

Click a panel title or focus it and press Enter or Space to collapse or expand its contents. Header actions, status and help remain available. Panels collapse independently, start expanded and remember your choices across view switches until browser reload. Adding a device opens its group. Collapsing preserves edits and does not stop playback. The shared tempo stays outside the sequencer groups.

Instrument Rack, Melodic Sequencers, Drummer Sequencers, Controller Sequencers, Arpeggiators, Piano Rolls (keyboards), MIDI Controllers, and Multitrack Arranger each have their own toggle. Empty drummer/controller groups keep their Add buttons. The arranger appears when at least one melodic, drummer, controller sequencer, or arpeggiator exists. These layout choices are not saved into performances or exports.

Collapsed visual bodies are suspended: hidden grids and timelines are not built, animation loops stop, and hidden meters unsubscribe. This includes the Mixer and each nested routing, matrix, diagram, channel-mapping, diagnostic and insert section. Playback, pad switching, MIDI processing, audio controls and meter collection continue. Visible headers, rack summary, shared tempo and footer transport status keep updating. Expanding displays the current playback position and latest meters immediately, without replaying missed animation frames.

Draft names and controller values, validation feedback, arpeggiator preset drafts, pad-loop selections, open containers, arranger selections, zoom and scroll positions survive collapse/expand while the Perform page remains open. Nested Mixer sections remember their open state when a parent collapses. New, Load and Import clear retained editor state after successfully replacing the workspace, even when reloading the same performance. Saving does not clear it. Removed devices and invalid selections are discarded; restored scroll positions are limited to the current content. This temporary editor state is not stored in performance files or browser storage and does not survive a page reload or leaving Perform.

Collapsing closes temporary menus and pickers, cancels active drag previews and releases manually held piano notes. Already-applied edits remain, and existing input save/commit rules still apply. Sequenced notes continue playing. Offscreen panels that remain expanded still render normally. Collapsing reduces visual work; any reduction in audio dropouts depends on the workload and audio system.



The collapsed rack shows channel and patch name in one horizontally scrolling row. See [Compact Rack](#) for continuous instruments, long names, and keyboard scrolling.

## What You Can Do Here

- Assemble an instrument rack from saved patches and assign MIDI channels
- Mix audio with dB faders, pan/balance, mute/solo, meters, inserts and pre/post sends
- Route instruments and returns through Master or explicit direct paths
- Start/stop the instrument engine session
- Create multiple melodic sequencers, drummer sequencers, and controller sequencers
- Add backend-run arpeggiators that turn held notes into routed arpeggiated instrument output
- Edit reusable pads and phrases, assemble song order in the arranger, and audition phrases without changing the song
- Arrange multiple track timelines in the multitrack arranger with shared transport and loop-range playback
- Perform live with piano rolls and manual MIDI controller knobs
- Save/load/clone/delete performances
- Import/export performance bundles (with optional patch definitions)
- Export offline Csound render ZIPs either with .csd + .mid playback or with inline Csound score playback

## Chapter Contents

- [Instrument Rack and Engine Transport](#)
- [Audio Mixer and Routing](#)
- [Melodic Sequencers and Step Editing](#)
- [Drummer Sequencers](#)
- [Pattern Pads and Reusable Phrases](#)
- [Multitrack Arranger](#)
- [Controller Sequencers](#)
- [Arpeggiators](#)
- [Piano Rolls](#)
- [MIDI Controllers](#)
- [Performance Import / Export](#)
- [Live Status and Safety Controls](#)

## Important Concept: Patch vs Performance

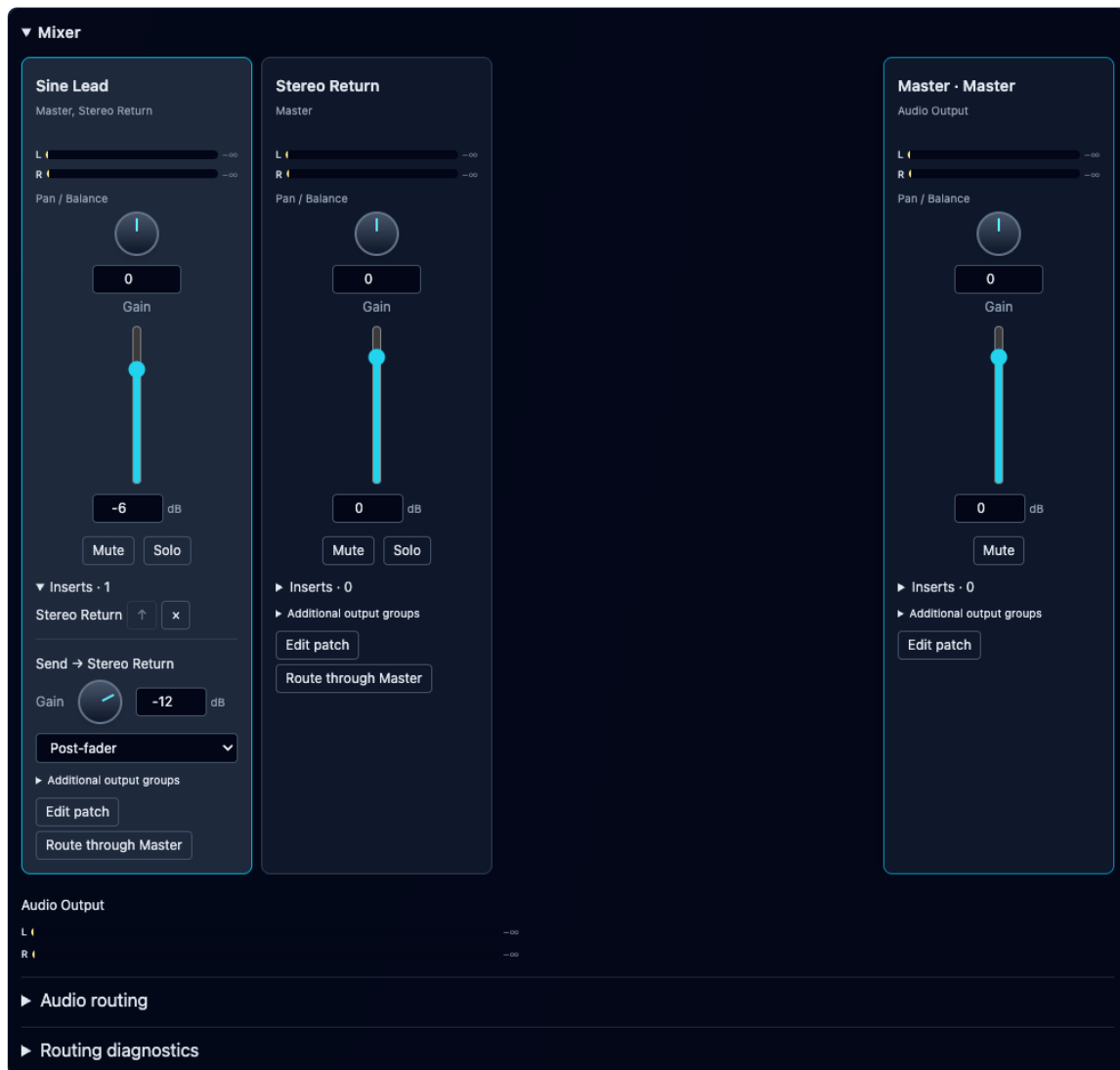
- **Patch (Instrument Design page):** one instrument graph
- **Performance (Perform page):** a full live setup that references one or more patches and stores sequencer/piano-roll/controller state plus explicit audio routes and mixer settings

## Practical Workflow

1. Save at least one patch in Instrument Design.
2. Add instruments to the performance rack and assign channels.
3. Open Mixer, check the destination of each strip, and connect effects/returns. Guided instruments get a neutral Master automatically; existing direct routes stay direct.
4. Use Routing diagnostics → Check routing, then Start Instruments. Adjust existing mixer controls during playback.
5. Build melodic sequencers / drummer sequencers / controller sequencers / piano rolls.
6. Save the performance to retain the rack, routes and current mix.
7. Export the performance bundle for backup/sharing, or use Export CSD (MIDI) / Export CSD (SCORE) when you need an offline Csound render package.



## Screenshots



Perform mixer with a post-fader send, dedicated insert and pinned Master. See the following chapters for rack and sequencer controls.

Arranger lane palettes and Playback settings have separate disclosures that start collapsed. Melodic and drummer sequencers have an always-visible upper-right pattern workspace for assembling reusable phrases; controller and arpeggiator phrase panels remain collapsible. The arranger uses context menus for grouping and ungrouping. Lane Mute/Solo is temporary and independent of the instrument mixer; see [compact tracks and lane controls](#).



# Instrument Rack and Engine Transport

The Instrument Rack is the top section of the Perform page and controls the live session, instrument assignments, and performance metadata.

## Compact Rack

When collapsed, the rack keeps Add Instrument and Start/Stop available. A single horizontally scrolling row shows MIDI channel and patch name for every instance, including duplicates; continuous instruments show Continuous instead of a channel. Long names have full-name tooltips. Focus the row and use the arrow keys to scroll. Metadata, file actions and detailed controls return when expanded.

The header contains rack status, Add Instrument, Start/Stop, and help in both views. Collapse choices survive view switches until browser reload.

## Performance Metadata and Library Actions

The rack includes fields and actions for the current performance:

- Performance Name
- Description
- Load Performance dropdown
- Save Performance
- Clone
- Delete
- Export
- Export CSD (MIDI)
- Export CSD (SCORE)
- Import

These actions operate on the performance configuration (instrument rack + audio routing/mixer + sequencers + controllers + piano rolls), not on individual patch definitions.

- Export writes an Orchestron `.orch.json` / `.orch.zip` performance bundle for backup, sharing, and re-import.
- Export CSD (MIDI) writes an offline-render ZIP with a compiled `.csd`, the arranger performance as `.mid`, bundled uploaded sample/SF assets, and a `README.txt` with the render command. Export CSD (SCORE) embeds notes and controller sweeps directly in the Csound score, omits the `.mid`, rewrites supported MIDI opcodes for score playback, and writes a matching `no- -f` render command. Both modes seed enabled manual MIDI Controller lane values at time 0 on each assigned instrument channel and use 32-bit float WAV output ( `-f` ) to preserve headroom. GEN01 and `sfload` sample files must be uploaded/imported assets; raw local `samplePath` values are rejected before compilation.

## Instrument Assignments (Rack Slots)

Each rack entry selects a saved patch and a MIDI channel (1–16), or displays Continuous activation. Up to 64 user patch instances are supported, including processors. Generated mixer stages do not occupy rack slots or MIDI channels. Use distinct MIDI channels for note-triggered instances.

Patches containing ``perf_controller`` show five compact tuning knobs per standard rack-card row; extras wrap below. They remain editable while playing. New notes use updated values; continuous instruments adopt them after restarting the rack. Save Performance stores explicit overrides.

The separate [audio mixer](#) below the rack provides audio faders, pan/balance knobs, mute/solo, pre/post sends, inserts and meters. The old numeric Level field has been removed. Mixer gain affects held notes and external MIDI audio without changing note velocity.



While instruments run, adding/removing assignments, changing patch/channel assignments, and changing routing topology are locked. Existing mixer controls stay live. Use **Stop to edit routing** before changing connections or insert order.

## Add Instrument

- Use `Add Instrument` to create another rack slot.
- Open the slot's patch picker to browse instruments in collapsed type groups: Percussion, Melody, Bass, Effects / Noise, and Continuous (always-on). Expand a group to see its instruments alphabetically.
- Search names and descriptions across all types by entering at least four characters; results appear after a 500 ms typing pause. Clear the search to return to browsing. Template patches are excluded from the rack picker.
- This enables multi-instrument performances driven by different MIDI channels.
- Continuous effects run as explicit rack instances. Adding an insert creates a dedicated instance automatically; simply saving an effect in the library does not start it.
- In the `orchestron-performance-creator` CLI, use `edit instruments list` to discover stable rack binding IDs and audio ports. Build main/send paths with `edit routes add/remove/clear/list`, then set strip and send controls with `edit mixer`. The optional `edit add-standard-effects --send-gain-db -12` preset routes dry instruments and a shared reverb through a compressor into the built-in Master; no speaker instance is needed.
- The button is unavailable while the engine is running; stop instruments before changing rack assignments.

The CLI validates its staged rack through the same backend route resolver used by session creation and compilation. Run `edit validate`, then `edit create-runtime --start` to create a CLI-owned engine session. Because instrument assignments and Csound audio connections are fixed at compile time, use `edit rebuild-runtime` after changing the rack or its routes; `edit push-runtime` updates mixer, sequencer and arpeggiator settings when the compiled rack still matches.

## Rack Transport (Instrument Engine Control)

The rack transport controls the instrument runtime session itself.

Buttons:

- `Start Instruments`
- `Stop Instruments`

### Start Instruments / Stop Instruments

These start/stop the underlying instrument engine session.

Global arrangement transport now lives in the multitrack arranger section. There, cassette-style `Rewind`, `Stop`, `Play`, and `Fast forward` buttons drive sequencers that have `Arrangement` selected as playback source and move the shared playhead in 1-beat transport blocks. Arranger `Play` stops sequencers whose `Manual pads` selected, while arranger `Stop` stops only the Arrangement sequencers; manually started `Manual pads` sequencers can keep running. The arranger `Stop` button does not stop the instrument engine; `Stop Instruments` in the rack does that. Double-clicking arranger `Stop` resets the playhead to the selected loop start or to step 0 when no manually running sequencer keeps transport active.

## Session State Badge

The rack shows current session state (localized label), for example:

- `running`
- `stopped / idle`

This is the state of the instrument engine session, not just the sequencer transport.

## Error Banner

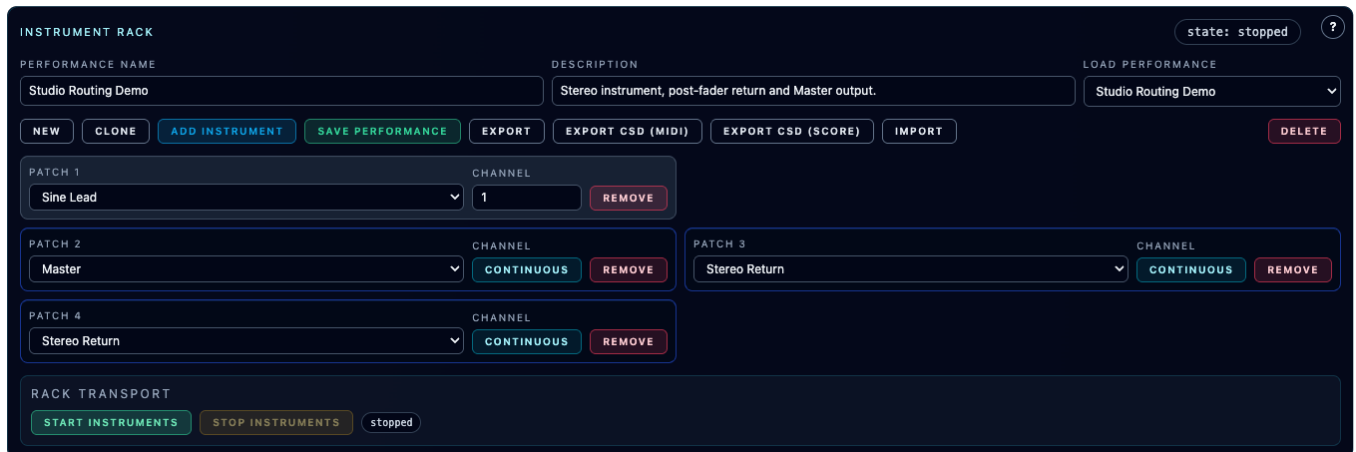
If engine start/stop or transport actions fail, the Perform page shows an error banner below the rack transport section.



## Tips

- Save the performance after significant changes (rack assignments, sequencers, piano rolls, controller mappings).
- Use distinct MIDI channels for note-triggered rack instances.
- Use `sname` labels on source `outlet_a` nodes to name the available matrix channels. The label can be stored directly on the node or supplied by a direct `const_s` connection.

## Screenshots



*Instrument rack detail with performance metadata, assignments, and transport controls.*



# Performance Controllers

Use `perf_controller` to adapt a patch to a particular performance: envelope times, sustain, distortion amount, or another parameter accepting l-rate. Each rack instance owns its settings, even when several instances use the same patch.

## Define a Controller

Add `perf_controller` from Constants in Instrument Design. Click its gear button and configure:

| Field   | Initial value | Meaning                             |
|---------|---------------|-------------------------------------|
| Minimum | 0             | Lowest setting                      |
| Maximum | 1             | Highest setting                     |
| Default | 0.5           | Setting before a rack override      |
| Scale   | Linear        | Linear or logarithmic knob movement |
| Label   | Parameter     | Caption displayed in the rack       |

The fields are fixed values, without input sockets or formulas. Numbers must be finite, minimum smaller than maximum, and default within the range. Logarithmic minimum must be positive. For a logarithmic attack-time control, try minimum `0.001`, maximum `5`, default `0.01`, and label `Attack (s)`. Connect `iout` to the envelope's l-rate attack input, then save the patch.

## Tune a Rack Instrument

Controllers appear in patch-node order beneath patch and channel. Their compact rack layout fits five controls across a standard instrument card before wrapping onto another row. Each shows a caption, scale dashes, endpoints, LIN/LOG indicator, and current value. Linear ticks are equally spaced; logarithmic ticks follow ratios and mark decade boundaries.

- Drag vertically, or use arrow keys. Hold Shift for fine adjustment.
- Enter an exact number below the knob; Enter or leaving the field applies it. Escape cancels.
- Home and End choose the endpoints. Double-click, Delete, or Backspace on the focused knob resets to the patch default.
- Changes affect newly started notes. Sounding notes keep their initialized value. Continuous instruments require a rack restart.

## Save, Reload, and Export

Untouched controls follow the patch default. After a value changes, an explicit instance override remains until reset, including when it is subsequently set to the current default. App state retains edits immediately through the existing persistence workflow; Save Performance is required to update a saved performance. Native JSON/ZIP bundles and both CSD (MIDI) and CSD (SCORE) exports preserve instance values. The CSD contains its initial settings and runs without a controller client. These controls do not send MIDI CC and do not become MIDI automation.

Node IDs identify controls; duplicate labels are allowed. Changing the selected patch clears its overrides. Editing a patch range clamps saved values; deleting a controller removes its override, with a notice in the rack. Standalone patch export and isolated audition use defaults; audition within a performance uses the selected instance's values.

`perf_controller` is an Orchestron editor construct lowered to `l-rate chnget` and channel initialization. It is not a native Csound opcode.

## Agent CLI

Patch specs accept an ordered `performance_controllers` list with stable node IDs, target inputs, ranges, defaults, scales, and labels. The performance CLI exposes `edit performance-controllers list/set/reset --binding ID ; set`



additionally takes `--node NODE_ID --value NUMBER`, and `reset` takes `--node NODE_ID`. Use `edit validate` and `edit commit` to save, or `edit push-runtime` to update an attached session with unchanged rack and patch definitions. Always-on instruments still need restart. See the skill references for patch authoring and instance settings.

## Deutsch

Die Agent-CLI unterstützt `performance_controllers` in Patch-Spezifikationen und `edit performance-controllers list/set/reset --binding ID` für Instanzwerte. `edit commit` speichert die Performance; `edit push-runtime` aktualisiert die Laufzeit. Kontinuierliche Instrumente benötigen weiterhin einen Neustart.

Virtueller Orchestron-Opcode für instrumentspezifische Performance-Einstellungen wie ADSR-Zeiten oder Verzerrung. Das Zahnrad öffnet feste Felder für Minimum, Maximum, Standardwert, Skala (linear oder logarithmisch) und Beschriftung; es gibt keine Eingangsanschlüsse. Vorgaben: 0, 1, 0.5, linear und Parameter. Zahlen müssen endlich sein, das Minimum kleiner als das Maximum und der Standardwert im Bereich liegen. Das logarithmische Minimum muss positiv sein. Jede Rack-Instanz zeigt eigene beschriftete Regler unter Patch und Kanal; weitere Regler umbrechen in neue Zeilen. Vertikal ziehen, Pfeiltasten verwenden (Umschalt für Feineinstellung) oder einen exakten Wert eingeben. Doppelklick stellt den Patch-Standard wieder her. Der I-rate-Ausgang wird beim Notenstart gelesen: gehaltene Noten behalten ihre Einstellung, neue Noten nutzen den geänderten Wert. Kontinuierliche Instrumente übernehmen Änderungen nach einem Rack-Neustart. Unberührte Regler folgen dem Patch-Standard; geänderte Werte bleiben bis zum Zurücksetzen gespeichert. Der App-Zustand behält Änderungen; Performance speichern und native JSON/ZIP-Exporte sichern sie für späteres Laden. CSD (MIDI) und CSD (SCORE) initialisieren die Einstellungen im Orchester ohne MIDI-CC oder externen Client. Einzelpatch-Export und isoliertes Vorhören nutzen Standardwerte; Vorhören einer Performance-Instanz nutzt deren Einstellungen. Beschriftungen dürfen sich wiederholen; Knoten-IDs identifizieren die Regler. Ein Patch-Wechsel löscht die Einstellungen. Geänderte Bereiche begrenzen gespeicherte Werte, entfernte Knoten löschen ihre Werte; ein Hinweis erscheint.

## Français

La CLI des agents accepte `performance_controllers` dans les spécifications de patch et `edit performance-controllers list/set/reset --binding ID` pour les valeurs par instance. `edit commit` enregistre la performance; `edit push-runtime` actualise la session. Les instruments continus nécessitent toujours un redémarrage.

Opcode virtuel Orchestron pour les réglages propres à un instrument dans une performance, comme les temps ADSR ou la distorsion. La roue dentée ouvre les champs fixes Minimum, Maximum, Valeur par défaut, Échelle (linéaire ou logarithmique) et Libellé; aucun port d'entrée. Valeurs initiales: 0, 1, 0.5, linéaire et Parameter. Les nombres doivent être finis, le minimum inférieur au maximum et la valeur par défaut dans la plage. Le minimum logarithmique doit être positif. Chaque instance du rack affiche ses propres boutons sous le patch et le canal, sur plusieurs lignes si nécessaire. Glissez verticalement, utilisez les flèches (Maj pour un réglage fin) ou saisissez une valeur exacte. Double-cliquez pour rétablir la valeur du patch. La sortie I-rate est lue au début de chaque note: les notes tenues conservent leur réglage, les nouvelles utilisent la valeur modifiée. Les instruments continus adoptent les changements après le redémarrage du rack. Les contrôles intacts suivent les valeurs du patch; les valeurs modifiées persistent jusqu'à réinitialisation. L'état de l'application conserve les modifications; Enregistrer la performance et les exports JSON/ZIP les préservent pour un chargement ultérieur. CSD (MIDI) et CSD (SCORE) initialisent les réglages dans l'orchestre sans CC MIDI ni client externe. L'export du patch seul et l'écoute isolée utilisent les valeurs par défaut; l'écoute en performance utilise les réglages de l'instance. Les libellés peuvent se répéter; les identifiants des nœuds identifient les contrôles. Changer de patch efface les réglages. Les plages modifiées limitent les valeurs enregistrées et les nœuds supprimés retirent leurs valeurs, avec un message.

## Español

La CLI de agentes admite `performance_controllers` en las especificaciones de patch y `edit performance-controllers list/set/reset --binding ID` para valores por instancia. `edit commit` guarda la interpretación; `edit push-runtime` actualiza la sesión. Los instrumentos continuos siguen necesitando un reinicio.

Opcode virtual de Orchestron para ajustes de un instrumento específicos de una interpretación, como tiempos ADSR o distorsión. El engranaje abre campos fijos de Mínimo, Máximo, Valor predeterminado, Escala (lineal o logarítmica) y Etiqueta; no hay puertos de entrada. Valores iniciales: 0, 1, 0.5, lineal y Parameter. Los números deben ser finitos, el mínimo menor que el máximo y el valor predeterminado dentro del rango. El mínimo logarítmico debe ser positivo. Cada instancia del rack muestra sus propios controles debajo del patch y canal, en filas adicionales si hace falta. Arrastre verticalmente, use las



flechas (Mayús para ajuste fino) o introduzca un valor exacto. Haga doble clic para restaurar el valor del patch. La salida l-rate se lee al comenzar la nota: las notas sostenidas conservan su ajuste y las nuevas usan el valor modificado. Los instrumentos continuos adoptan cambios tras reiniciar el rack. Los controles intactos siguen los valores del patch; los valores modificados persisten hasta restablecerlos. El estado de la aplicación conserva las modificaciones; Guardar interpretación y las exportaciones JSON/ZIP las preservan para cargarlas después. CSD (MIDI) y CSD (SCORE) inicializan los ajustes en la orquesta sin CC MIDI ni cliente externo. La exportación del patch y la escucha aislada usan valores predeterminados; la escucha en interpretación usa los ajustes de la instancia. Las etiquetas pueden repetirse; los identificadores de nodo identifican los controles. Cambiar de patch borra los ajustes. Los rangos modificados limitan los valores guardados y los nodos eliminados retiran sus valores, con un aviso.



# Melodic Sequencers and Step Editing

Melodic sequencers are step-based pattern sequencers used for note playback.

This page covers the **melodic sequencer** editor. Drum-machine style programming is documented in [Drummer Sequencers](#).

Use the pen beside the device name to rename it. See [Device Names](#) for editing controls, validation, and import/export behavior.

## Collapse the Panel

Melodic, drummer, and controller groups collapse independently. The shared tempo remains above the groups. Each group has its own Add button, which expands it. Panels start expanded and remember your choice across view switches until browser reload. Collapsing does not stop playback or discard edits.

## Global Sequencer Clock

The sequencer section contains a global clock with:

- Running/stopped state badge
- BPM field (range 30..300)

The backend runs the step timing (native runtime clock), which reduces browser timing jitter during playback.

## Adding / Removing Melodic Sequencers

- Add Melodic Sequencer creates a new melodic sequencer card
- Add Drummer Sequencer creates a drummer sequencer card (documented separately in [Drummer Sequencers](#))
- A performance can contain up to 16 melodic sequencers
- Each melodic sequencer has its own Remove button

## Per-Sequencer Controls

Each melodic sequencer card provides:

- sequencer state badge (running/stopped or queued start/stop state)
- Start / Stop (sequencer enable state; can be used manually while the multitrack arranger is stopped)
- Remove
- Clear Steps
- MIDI Channel (1..16)
- Scale (root + scale type)
- Mode (Ionian, Dorian, Phrygian, Lydian, Mixolydian, Aeolian, Locrian)
- Meter (2..7 over 4 or 8)
- Grid (2, 4, or 8, steps per beat)
- Beat Ratio (1:1, 2:1, 3:2, 4:3, 3:4, 5:4, 4:5, 7:4)
- Beats (1..8, with the current meter numerator exposed directly)

The step editor width is derived from the sequencer's own timing and length:

- steps per beat = the sequencer's selected Grid
- pad steps = beats \* steps per beat
- default timing (4/4, grid 4) gives 16 steps for a 4-beat pad
- odd meters can still use matching one-bar pad lengths such as 3 beats in 3/4 or 5 beats in 5/4

Beat Ratio changes how quickly the sequencer advances relative to the shared transport beat without changing its stored pad length, meter, or grid:



- 1:1 keeps normal speed
- ratios above 1:1 make the melodic sequencer run faster
- ratios below 1:1 make the melodic sequencer run slower

This is separate from **Meter** and **Grid**, so you can combine polymeter and true polyrhythm on the same performance page.

## Queued Start/Stop State Labels

When changes are queued to take effect at the next cycle boundary, the sequencer state label can show queued states such as:

- starting at step 1
- stopping at step 1

## Scale, Scale Type, and Mode

Orchestron separates:

- Scale root (for example C, Db, F#)
- Scale type (major, neutral, minor)
- Mode (7 diatonic modes)

This supports guided note entry and better readability in the sequencer step editor and piano roll highlighting.

## Pattern Workspace

The upper-right **pattern workspace** assembles pads and reusable phrases with drag-and-drop. Cmd/Ctrl-click selects items to group; right-click splits groups. Apply commits shared phrase edits explicitly. Play loops the whole workspace; hold a speaker for 250 ms to preview an individual pad or phrase until release. Only pads with a note above zero velocity are green; rests, holds without notes and zero-velocity patterns stay dark. Selection outlines remain separate.

## Step Editor (Per-Step Note Programming)

Each step cell supports:

- Rest or note selection
- Chord selection, including the standard 5 power chord (root + perfect fifth)
- Hold toggle (HOLD)
- Octave field (separate from pitch class)
- In-scale/out-of-scale status display
- Playhead highlighting during playback

## Step Note Entry Workflow

Each step has a pitch-class dropdown grouped into:

- Rest
- In-scale notes (for the current sequencer scale/mode)
- Out-of-scale notes

The octave is edited separately via the **OCT** field.

This split note/octave design makes pattern entry faster than selecting full MIDI note numbers.

## Theory Aids In Steps

- In-scale notes are highlighted
- In-scale notes show degree labels (1..7)



- Out-of-scale notes remain selectable (for chromatic writing)
- Enharmonic note labels are available where relevant (for example sharps/flats)
- Chord choices are grouped as none, diatonic, or chromatic for the selected step root and mode. A 5 power chord is diatonic only when both the root and perfect fifth are in the current mode.

## Hold Toggle

Each step includes a `HOLD` toggle.

Use `HOLD` to sustain behavior across steps according to the sequencer runtime logic and the note/gate pattern.

## Step Cell Visual States

Step cells visually indicate:

- Active playhead step (when the melodic sequencer + transport are running)
- In-scale programmed note
- Out-of-scale programmed note
- Rest / hold combinations

## Early and Late Notes

Timing moves an individual attack by **–50% to +50% of one local step**, in 1% increments. Zero means **On grid**. The millisecond readout follows the current tempo, grid and beat ratio: at 120 BPM with Grid 4 and ratio 1:1, –20% is 25 ms early. Timing belongs to each pad and does not change its length, meter or playback speed.

Moving a note also moves its release, preserving its length and any `HOLD` extension. A following attack can shorten the preceding note to avoid overlap. Chords move together. If neighboring attacks land at exactly the same instant, the later logical step wins.

An early first step plays before the boundary when the same pad repeats. On a fresh start or a different-pad launch it plays at the boundary instead. Queued stops, pauses and finite arrangement ends suppress repeat anticipation. A command received after an anticipated attack has already sounded cannot undo that attack.

Copying steps or pads preserves timing. Clear Steps resets it. Inactive drum cells retain their timing for reactivation. Old performances load on grid; Save/Load, browser restoration, native bundles and both CSD export modes preserve offsets. Timing edits during playback use the normal coalesced live-edit workflow.

Use the **Timing** slider or signed percentage field in each step to adjust its position. The reset arrow restores zero. The slider supports arrow keys; the note/chord copy handle also copies timing.

The performance CLI can discover tracks with `edit sequencers list` and inspect or change offsets with `edit step-timing list|set|reset`. Pads use 1–8; steps are zero-based. YAML/JSON scores also support per-note timing. See the skill timing reference for the full workflow and examples.

## Clear Steps

`Clear Steps` resets the current melodic sequencer step contents (for the active pad pattern context) so you can quickly reprogram it.

## Related Features

- Pattern pads (#1..#8), queued pad switching, pad copying, transposition, and phrase library and arranger controls are documented in [Pattern Pads and Reusable Phrases](#).

## Screenshots



The screenshot displays the 'MELODIC SEQUENCER 1' interface. At the top, there are buttons for 'running', 'STOP', and 'CLEAR STEPS'. Below this, the 'MIDI CHANNEL' is set to 2, and 'SYNC TO' is set to 'none'. The 'SCALE' is 'C minor' and the 'MODE' is 'Aeolian'. The 'METER' is 4/4, 'GRID' is 4, and 'BEAT RATIO' is 1:1. The 'BEATS' section shows a sequence of 1, 2, 4, and 8. The 'PATTERN PADS' section contains 12 pads, each with a 'HOLD' button, a 'CHD' (chord) button, an 'OCT' (octave) button, and a 'VEL' (velocity) button. A context menu is open over the first pad (P1), showing a list of notes: Rest, C, D, D#, E, F, G, G#, A#, B, C#, D, E, F#, A. The 'In scale: C minor / Aeolian' option is selected. The 'PAD SEQUENCE' section shows 'PAD LOOPER: ON', 'REPEAT: ON', 'compiled: 0', and 'steps: 0'. The 'MAIN SEQUENCE' section shows a list of pads (P1, P2, P4, P8, P16) and a note: 'No groups yet. Select pads/groups, right-click, Group. Click here, press 1-8, or drop pads'.

Melodic sequencer detail showing step editing, note selection, octave, and hold controls.



# Drummer Sequencers

Drummer sequencers are drum-machine style step sequencers for fixed MIDI drum keys.

They are optimized for per-step drum hit programming instead of melodic note/chord entry.

Use the pen beside the device name to rename it. See [Device Names](#) for editing controls, validation, and import/export behavior.

## Collapse the Panel

The drummer group collapses independently of melodic and controller sequencers. Its header and Add button remain available even when empty; adding a drummer sequencer expands the group. Panels start expanded and remember your choice across view switches until browser reload. Collapsing does not stop playback or discard edits.

## What Makes Them Different

Compared to melodic sequencers, drummer sequencers:

- use MIDI key numbers ( 0 . . 127 ) per row (drum instrument selection)
- do not use scale / mode / chord controls
- do not use pad transpose edge buttons
- let you program multiple drum rows per step
- store velocity per active drum hit (per row + step)

## Adding / Removing Drummer Sequencers

- Use the Add Drummer Sequencer button in the sequencer section header to add a drummer sequencer card
- A performance can contain up to 16 drummer sequencers
- Each drummer sequencer card has its own Remove button

## Per-Drummer-Sequencer Controls

Each drummer sequencer card provides:

- sequencer state badge (running/stopped or queued start/stop state)
- Start / Stop (sequencer enable state; can be used manually while the multitrack arranger is stopped)
- Remove
- Clear Steps
- + Key (add another drum row)
- MIDI Channel ( 1 . . 16 )
- Meter ( 2 . . 7 over 4 or 8 )
- Grid ( 2 , 4 , or 8 , steps per beat)
- Beat Ratio ( 1:1 , 2:1 , 3:2 , 4:3 , 3:4 , 5:4 , 4:5 , 7:4 )
- Beats ( 1 . . 8 , with the current meter numerator exposed directly)
- Playback source and upper-right pattern workspace (same as melodic sequencers)

## Drum Rows (Keys)

The left side of the drummer grid contains vertically stacked drum rows.

Each row has:

- row index label
- MIDI key input ( 0 . . 127 )



- row remove button ( x )

## Auditioning Drum Keys While Editing

When the instrument engine session is running, changing a row key sends a short one-shot MIDI note preview on the drummer sequencer's MIDI channel.

This helps identify which drum sound is mapped to a given MIDI key number.

## Drum Step Grid (LED Matrix)

The grid is row-based:

- rows = selected drum keys
- columns = beats \* steps per beat

This keeps the `Keys` column horizontally aligned with the LED rows.

### LED States

- inactive step: dark LED with visible border
- active step: red LED
- current playing column: highlighted across every drum row
- active step in the current playing column: flashing green LED

### Velocity (Per Hit)

Velocity is shown by LED color saturation (stronger color = higher velocity).

The LED border color stays visible even at low velocities, so low-velocity active hits remain distinguishable from inactive steps.

While dragging vertically on a hit, the UI also shows a live numeric velocity: `xx` readout next to the edited step.

### Editing Workflow

- Click an inactive LED to activate a hit
- Click an active LED (without dragging) to deactivate it
- Click and drag vertically on an LED to change velocity (0..127)
- While dragging, watch the live `velocity` readout for the exact value
- Keyboard:
  - Enter / Space toggles the hit
  - Arrow Up / Arrow Down adjusts velocity

## Early and Late Notes

Timing moves an individual attack by **-50% to +50% of one local step**, in 1% increments. Zero means **On grid**. The millisecond readout follows the current tempo, grid and beat ratio: at 120 BPM with Grid 4 and ratio 1:1, -20% is 25 ms early. Timing belongs to each pad and does not change its length, meter or playback speed.

Moving a note also moves its release, preserving its length and any HOLD extension. A following attack can shorten the preceding note to avoid overlap. Chords move together. If neighboring attacks land at exactly the same instant, the later logical step wins.

An early first step plays before the boundary when the same pad repeats. On a fresh start or a different-pad launch it plays at the boundary instead. Queued stops, pauses and finite arrangement ends suppress repeat anticipation. A command received after an anticipated attack has already sounded cannot undo that attack.

Copying steps or pads preserves timing. Clear Steps resets it. Inactive drum cells retain their timing for reactivation. Old



performances load on grid; Save/Load, browser restoration, native bundles and both CSD export modes preserve offsets. Timing edits during playback use the normal coalesced live-edit workflow.

Drag a hit **left/right** to adjust timing, or **up/down** to adjust velocity. The first drag direction locks the property until release; clicking without dragging still toggles the hit. Changed hits show a signed percentage and a displaced indicator.

Use **Left/Right** arrow keys for timing and **Up/Down** for velocity. Right-click a hit, or press **Shift+F10**, to open its Timing controls with a slider, numeric field and reset. The controls affect that hit in the displayed pad. Closing or collapsing the editor ends the gesture.

The performance CLI can discover tracks with `edit sequencers list` and inspect or change offsets with `edit step-timing list|set|reset`. Pads use 1–8; steps are zero-based. YAML/JSON scores also support per-note timing. See the skill timing reference for the full workflow and examples.

## Pattern Pads and Reusable Phrases

Drummer sequencers support the same #1..#8 pattern-pad workflow as melodic sequencers.

If a drummer sequencer is running, pad changes are queued to the next loop boundary. If it is stopped, pad changes apply immediately so you can edit another pad while the shared transport keeps running elsewhere.

Drummer pad length is stored in beats, while meter, grid, and beat ratio are configured per drummer sequencer.

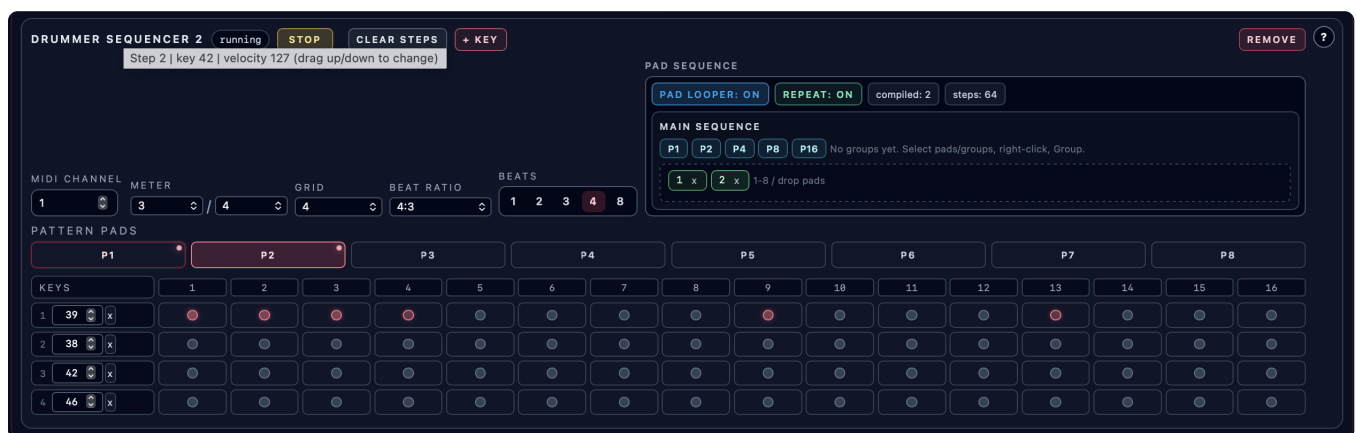
This keeps drummer pads beat-based while still allowing one-bar patterns in odd meters such as 3/4, 5/4, or 7/8 by matching the pad beat count to the meter, while `Beat Ratio` changes how fast the row pattern cycles against the shared transport without changing the stored beat length.

Edit pad contents here and assemble reusable groups/supergroups in the upper-right [pattern workspace](#). Drop and reorder pads, Cmd/Ctrl-click to group selections, and right-click to ungroup. Apply commits edits to existing phrases; loose workspace items stay temporary. Play loops the assembly, and holding a speaker for 250 ms previews one item until release. Place saved phrases in the multitrack arranger. Playback source chooses Arrangement or Manual pads; repetition is set in the arranger lane. Click selects a pad; Launch pad starts it explicitly.

Differences vs melodic sequencer pads:

- no transpose edge buttons (– / +)
- pad content is drum-row hit data + per-hit velocities (not note/chord/theory state)

## Screenshots



Drummer sequencer showing multiple drum rows, per-step hits, and velocity editing on the LED grid.

## Typical Uses

- Kick / snare / hat step programming
- Percussion layers with multiple rows



- Switching between groove variations using pattern pads
- Automating drum pattern changes with the arranger while performing on piano roll/controllers



# Controller Sequencers

Controller Sequencers automate MIDI CC values using editable curves.

Use the pen beside the device name to rename it. See [Device Names](#) for editing controls, validation, and import/export behavior.

## Collapse the Panel

The controller group collapses independently of melodic and drummer sequencers. Its header and Add button remain available even when empty; adding a controller sequencer expands the group. Panels start expanded and remember your choice across view switches until browser reload. Collapsing does not stop playback or discard edits.

## What A Controller Sequencer Does

A controller sequencer sends its programmed CC curve to the backend sequencer, which samples that curve over a repeating length and emits timed MIDI Control Change messages during playback.

Typical uses:

- filter sweeps
- morph controls
- modulation depth animation
- macro movement synchronized to transport

## Adding / Removing Controller Sequencers

- `Add Controller Sequencer` creates a new controller sequencer card
- A performance can contain up to 16 controller sequencers
- Each controller sequencer has a `Remove` button

## Per-Controller-Sequencer Controls

Each controller sequencer provides:

- Running/stopped state badge
- `Start / Stop` enable toggle; can be used manually while the multitrack arranger is stopped
- `Controller #` (0..127)
- **MIDI Channels** checkboxes, 1–16 in one row beside `Clear Steps`
- `Meter` (2..7 over 4 or 8)
- `Grid` (2, 4, or 8, steps per beat)
- `Beat Ratio` (1:1, 2:1, 3:2, 4:3, 3:4, 5:4, 4:5, 7:4)
- `Curve length in beats` (1..8, plus 16)
- `CC label preview` (CC N)
- `Curve editor`

## MIDI Channels

Check the channels that receive the curve's CC messages. All 16 are checked by default (OMNI); at least one must remain checked. The row scrolls horizontally on narrow screens. This is a device setting shared by all its pads.

Channel edits use the same 80 ms live-edit synchronization as curve edits and take effect at the next engine block without restarting transport or losing queued pads. The current curve value is sent to the new selection. Removed channels retain their last received value.

Selections survive Save/Load Performance, browser reload, native JSON/ZIP export/import, and both CSD export modes.



Older performances without the setting load with all 16 channels checked.

## Curve Length ( 1 . 8 , plus 16 , beats)

This defines the repeat length for the curve sampling relative to that controller sequencer's own timing. Keypoints stay normalized across the full pad duration, so the same curve shape stretches automatically when you choose a longer beat length.

Beat Ratio changes how quickly the controller sequencer moves through that curve relative to the shared transport. Faster ratios create repeating automation polyrhythms without changing the stored keypoint positions.

## Pattern Workspace

The always-visible workspace sits beside the musical controls, or below them on narrow screens. Drag pads into it, select several with Cmd/Ctrl-click, then Group or Supergroup. Reorder by dragging, split through the context menu, and Apply shared phrase changes explicitly. See [Pattern Workspace](#) for editing, deletion and retained drafts.

Play loops the displayed assembly and outlines its current pad, phrase or rest. Hold a speaker for 250 ms to temporarily override it; release resumes the workspace at its current position. Stop restores prior playback. Controller curves and channel routing remain effective. Two zero-valued endpoints alone look empty; additional points or nonzero values highlight the pad. This is a content indicator: zero still sends a valid CC value.

## Curve Editor Interactions

The curve editor is an interactive graph view (spline-based curve display and sampling).

When all keypoints have the same value, the curve appears as a horizontal line across the editor. This also applies to the minimum ( 0 ) and maximum ( 127 ) CC values.

Supported interactions:

- Click background to add a point (interior points only)
- Drag a point to change position/value
- Double-click an interior point to remove it
- Endpoints are boundary anchors (positions remain at start/end of the curve)

## Visual Playback Indicators (When Running)

When the main transport is running and the controller sequencer is enabled, the editor shows:

- a vertical playback position line
- a marker showing the currently sampled CC value on the curve

This makes it easy to understand exactly what value is being sent at each transport moment.

## Live Use Notes

- Controller sequencers run alongside melodic sequencers on the same backend transport clock.
- Curve edits are combined after an 80 ms editing pause and prepared while the previous curve keeps playing. They take effect at the next engine block without resetting transport. The editor modifies the displayed pad even when the arranger selected it automatically.
- See [Editing During Playback](#) for failure handling and transport behavior.
- Pad clicks select the editing pad without changing playback. Launch pad in Manual pads mode queues the selected pad at the next boundary; the separate speaker previews it after a 250 ms hold.
- You can combine automated controller sequencers with manual MIDI controller knob lanes on the same performance page.
- If you automate the same CC number from multiple sources, the last-sent value wins at the MIDI receiver side (plan mappings accordingly).



Screenshots



Controller sequencer with curve editor, CC configuration, and playback indicator.



# Arpeggiators

An arpeggiator processes MIDI notes and drives an existing rack instrument. Send chords from a melodic sequencer, an on-screen keyboard, or external MIDI to its **Input Channel**. Choose the named instrument in **Target Channel**. Both controls are always visible in the top row, with **Input Channel → Target Channel** showing the MIDI direction. Inputs must be unique; an arpeggiator cannot target another arpeggiator input.

## Start and Synchronize

**Arranger** is the default playback mode. Arranger Play starts arpeggiators with Arrangement as playback source; Stop silences them while independent Live devices and manually started sequencers can continue. Incoming chords join the next subdivision of the shared beat. Chord changes preserve the phrase position. Seeks and selected loop wraps reconstruct the phrase at the destination; finite arrangements stop at their end.

**Live** runs independently while the instrument engine is active. Its Start button can start the engine. The first incoming note establishes the Live clock; subsequent chords retain its pulse. Both modes use global BPM, including tempo changes.

**Restart** controls phrase position without moving the subdivision clock:

- **Continue:** keep advancing through chord changes (default).
- **First held note:** restart the phrase on a fresh key gesture.
- **Every beat / Every bar:** restart at the next subdivision of each master beat/bar.

Straight, triplet and dotted rates use musical beat positions. Swing alternates long and short intervals with an unchanged total duration for each pair.

## Hold and Processing

**Hold Off** is the default. Releasing all input notes silences the output, preserving rests in a source sequence. **Hold and replace** remembers the last chord after release; the next fresh gesture replaces that chord. **Add/remove notes** toggles pitches explicitly. **Clear held notes** releases the arpeggiator's notes and clears pending input and launches.

**Active** generates the pattern; **Bypass** forwards incoming notes to the target; **Mute** consumes notes silently. Stop, removal, routing changes and processing/playback-mode changes release notes owned by this arpeggiator.

## Eight Musical Pads

Each pad stores a complete musical variation: order, rate, octave range, transpose, expression, harmony, rhythm, rotation and random seed. Click **#1–#8** to choose the editing pad. Hold its separate speaker for 250 ms to preview it. Use Launch pad for an explicit launch in Live/manual mode. The playing pad has a dot; a queued pad has a clock mark. Automatic pad changes leave your editing pad and unfinished preset name in place.

Each pad has two independent lengths:

- **Rhythm steps:** 1–32 steps (default 16 sixteenths).
- **Pad duration:** 1–8 or 16 master beats (default four beats), used by the arranger.

For example, a five-step rhythm continues across successive four-beat occurrences of #1. Changing to #2 restarts the musical cursors at that launch boundary. Arrangement pauses silence output while still tracking incoming pitches.

Drag one pad onto another to copy its musical variation. The shared phrase library defines groups and supergroups; song placement and repeat controls live in the multitrack arranger. New arpeggiators play #1 once. Their named arranger lanes contribute to Fit, arrangement bounds, seeking and both CSD exports, including performances containing only arpeggiators.

In Live mode, pad launches use **Next cycle** by default, or **Next master bar**. Stopped launches select immediately and Cancel launch removes a queued launch. In Arranger mode, held speakers start at the next pad/rest boundary while the arranger runs, or at an audio block while stopped. Releasing restores the previous playback.



## Pattern Workspace in Arranger and Live

The upper-right workspace offers the same drag assembly, Group/Supergroup, local Ungroup and explicit Apply/Save as new/Discard controls as the other sequencers. It stacks beneath routing controls on narrow screens; pads and the rhythm grid remain full-width. See [Pattern Workspace](#) for draft retention and definition deletion.

Play loops the entire displayed assembly, highlighting its current pad, phrase or rest. Its phrase timing uses pad duration, independently of rhythm length. A speaker held for 250 ms overrides the workspace until release; the workspace resumes at its current clock position. Stop restores the prior arrangement, manual or stopped state. Edits enter at the next workspace cycle boundary.

Live starts follow Next cycle or Next master bar when a Live clock is established, otherwise an audio block. Live hides arrangement-only controls but keeps phrase editing and audition available. Input or held notes are required; no test chord is generated. Hold and Active/Bypass/Mute still apply. Chord changes during audition remain current when the underlying Live pulse resumes. A temporarily auditioned stopped device accepts notes without changing its saved Start/Stop setting.

Green pads contain a note-producing step with positive velocity and probability. Rest-only, tie-only, zero-velocity and fully suppressed pads remain dark, regardless of held notes. Selection and queued outlines remain separate. Zero-velocity steps are silent.

## Edit the Rhythm Grid

Select a step, then choose its action:

| Action              | Result                                                                                          |
|---------------------|-------------------------------------------------------------------------------------------------|
| Next note           | Play the next pitch in the selected order                                                       |
| Chord-note position | Play a relative position in the ascending, octave-expanded input pool; oversized positions wrap |
| Rest                | Advance time silently                                                                           |
| Tie                 | Extend the preceding note/chord without retriggering; silent without a preceding note           |
| Chord               | Play the entire expanded input pool                                                             |

Velocity bars scale incoming accents in percent. Each step also has probability, a gate override and **1–4 ratchets**. Gate supports **5–200%**; values above 100% overlap different pitches. **Use pad gate** clears the step override. Rests and probability misses keep the automatic note cursor stationary unless **Advance through rests** is enabled. **Note repeats** repeats an automatic pitch before moving on. **Rotation** changes the rhythm's starting step.

Keyboard editing: Left/Right selects adjacent steps; Space toggles a rest; Delete/Backspace inserts a rest; T inserts a tie; C plays the chord; N restores Next note. Double-click also toggles a rest. Controls have accessible text labels.

Note previews use large 18 px labels. Narrow grids scroll horizontally; long chord labels stay on one line with an ellipsis. Hover over a step for the complete note list and scale degrees, also available to screen readers.

With **Follow source** or **Custom** scale handling, step borders use the piano keyboard's degree colours: **1 red, 2 orange, 3 yellow, 4 green, 5 teal, 6 blue, 7 violet**. Degrees describe the generated pitches after transposition and quantization. Follow source uses each note's incoming scale context, with the pad's scale as fallback. Octave repetitions keep the same colour. Chords show equal, distinct border segments for their degrees, ordered 1–7. **Scale Off**, missing previews, rests and ties keep neutral borders. Previews and colours appear only when editing the playing pad. A cyan outer outline marks the selected step independently of its degree colour; the playback background and keyboard focus remain separate.

## Expression and Harmony

Input velocity preserves accents; Fixed and Random velocity remain available. Timing and velocity humanization change individual strikes. Timing variation stays inside each strike's scheduling window and never moves the next clock position.



**Repeat variation** repeats seeded random choices; **Evolve** varies them by cycle. **New variation** changes the saved seed. Each pad has its own seed, independent of other devices; identical input and transport positions produce reproducible events in playback and exports.

Octaves expand the input pool before applying note order. Down descends through the complete range. Up/Down reverses without duplicating endpoints. **Octave by octave** retains an alternative traversal in Advanced timing.

Scale handling defaults to **Off**. **Follow source** uses the source's root/mode and falls back to the pad's configured scale when absent. **Custom** always uses the pad's scale. The effective scale and resolved pitch preview help explain the output. Playback highlights follow audible PCM status.

## Presets, Live Edits and Collapse

Presets apply only to the editing pad. **Modified** marks changes from its preset. **Update preset** overwrites a user preset; **Save as preset** creates a new one. Built-in presets remain available. Routing, playback settings and other pads are preserved.

Live edits coalesce for 80 ms and apply at audio render boundaries, preserving transport and pending launches. Invalid edits retain the working configuration. Status updates never resubmit musical settings. Collapse hides visual work while preserving playback, pad selection, details and drafts.

The screenshot displays the 'Arpeggiator 1' interface with the following sections:

- Header:** 'Arpeggiator 1' with 'Playing' and 'Stop' buttons, and a 'Remove' button.
- Routing:** 'Input Channel' (6) and 'Target Channel' (3 - Furnace Techno Stab).
- Playback:** 'Arranger' mode, 'Active' status, and 'Hold' setting (Hold and replace).
- Held Notes:** C4, E=4, G4, B=4, Active Note: —, with a 'Clear held notes' button.
- Pad Row:** Eight pads (P1-P8) with play buttons.
- Instructions:** 'Select a pad to edit; use its play button to launch. Drag pads to copy, Space toggles a step; Delete inserts a rest; T ties; C plays the chord.'
- Settings:**
  - Preset: 'outch', Rate: '1/16', Pattern: 'Up', Octaves: '2'.
  - Gate (%): '72', Swing (%): '75', Rhythm steps: '16', Pad duration (beats): '4'.
- Pad Grid:** 16 pads with notes: C5, D5, G5, A5, C6, D6, G6, A6, C5, D5, G5, A5, C6, D6, G6, A6.
- Step 1:** 'Next note', Velocity (%): '109', Probability (%): '100', Ratchets: '1', Gate (%): '72', and 'Use pad gate' button.
- Expression:** Collapsed section.
- Harmony:**
  - Transpose (st): '12', Scale: 'Custom', Mode: 'Ionian'.
  - Effective scale: G Ionian.
- Advanced timing:** Collapsed section.
- Save as preset:** 'Preset name' field, 'Modified' status, 'Update preset' button, and 'Save as preset' button.
- PAD SEQUENCE:** 'PAD LOOPER: ON', 'REPEAT: ON', 'compiled: 1', 'steps: 32'.
- MAIN SEQUENCE:** 'P1 P2 P4 P8 P16' with a note 'No groups yet. Select pads/groups, right-click, Group.' and a sequence '1 x 1-8 / drop pads'.

Arpeggiator showing its input and target routing, eight musical pads, per-step rhythm controls, custom harmony settings, and pattern workspace sequence.

## Migration to Performance Version 15

Versions 1–14 load in memory into the new model. Existing settings become #1; the other seven pads use defaults. Identities, routes, rate, order, octaves, expression and user presets are retained; old accent cycles become grid velocities.

Existing performances adopt Arranger mode, Hold Off, Continue, corrected swing and full-range ordering, so they can sound different. Save/export writes version 16; app-state version 2 and native bundle-envelope version 1 stay unchanged. Runtime positions and held notes are not saved.



# Piano Rolls

Piano Rolls provide manual performance input with scale-aware keyboard highlighting.

Use the pen beside the device name to rename it. See [Device Names](#) for editing controls, validation, and import/export behavior.

## Collapse the Panel

Collapse the Piano Rolls header to hide the keyboards and their settings. Add and help remain available; adding a keyboard expands the group. Panels start expanded and remember your choice across view switches until browser reload. Collapsing does not stop playback or discard edits.

## Purpose

Piano Rolls are for live/manual playing on an on-screen keyboard while the instrument engine is running.

They coexist with melodic sequencers and controller sequencers, which means you can jam manually on top of sequenced playback.

Starting a piano roll is independent from the multitrack arranger transport. If the instrument engine is stopped, **Start** brings up the engine so the keyboard can play without starting arrangement playback.

## Add / Remove / Enable

Each piano roll supports:

- Add Piano Roll
- Remove
- Start / Stop (enable/disable that piano roll)

A piano roll only sends notes when:

- that piano roll is enabled
- instruments are running

## Piano Roll Controls

Each piano roll has:

- MIDI Channel (1..16)
- Scale
- Mode

## Scale/Mode Following Behavior During Live Playback

When the instrument engine is running and a piano roll is enabled, the piano roll can follow the currently running melodic sequencers for harmonic guidance.

### Shared Scale/Mode Case

If running melodic sequencers agree on the same scale+mode:

- the piano roll shows that shared scale/mode
- keys are highlighted according to that theory



## Mixed Scale/Mode Case

If running melodic sequencers disagree:

- piano roll controls show `mixed`
- highlighting is computed from the intersection of the active theories
- only notes common to the active scales/modes are highlighted

This is especially helpful when multiple tracks create potentially conflicting harmonic contexts.

## Keyboard Range and Layout

The piano keyboard spans:

- C1 .. B7 (84 notes)

UI behavior:

- Full-width horizontal keyboard section
- Scroll arrows appear when the keyboard overflows the viewport
- Smooth horizontal scrolling controls

## Key Highlighting and Degree Labels

Highlighted keys show:

- in-scale status
- scale degree labels ( 1 . 7 )
- rainbow degree coloring for quick visual recognition

In mixed-theory situations, a key can show combined degree labels (for example 2/5 ) when it belongs to multiple active track theories.

## Playing Notes (Pointer Interaction)

- Press/hold on a key sends `note_on`
- Release sends `note_off`
- Works for both white and black keys

The UI tracks held notes and clears them if the piano roll becomes inactive or the session stops.

## Practical Use Cases

- Live melody or bass improvisation on top of sequencer patterns
- Testing an instrument patch in musical context without external MIDI hardware
- Performing harmonically guided lines while melodic sequencers define the key center

## Screenshots



PIANO ROLLS

ADD PIANO ROLL

PIANO ROLL 1

running

STOP

REMOVE

MIDI CHANNEL

2

SCALE

mixed

MODE

mixed

In-scale notes for mixed / mixed are highlighted with degrees.

7 octaves keyboard (C1..B7).

C#1

D#1

E1

F1

G1

A1

B1

C2

D2

E2

F2

G2

A2

B2

C3

D3

E3

F3

G3

A3

B3

C4

D4

E4

F4

G4

A4

B4

C1

D1

E1

F1

G1

A1

B1

C2

D2

E2

F2

G2

A2

B2

C3

D3

E3

F3

G3

A3

B3

C4

D4

E4

F4

G4

A4

B4

Piano roll with scale-aware key highlighting and mixed-theory display behavior.



# MIDI Controllers

The MIDI Controllers panel provides manual CC control lanes with interactive knobs.

Use the pen beside the device name to rename it. See [Device Names](#) for editing controls, validation, and import/export behavior.

## Collapse the Panel

Collapse the MIDI Controllers header to hide the knobs and their settings. The count, Add button and help remain available; adding a controller expands the group. Panels start expanded and remember your choice across view switches until browser reload. Collapsing does not stop playback or discard edits.

## Overview

This panel is for manual MIDI Control Change performance (hands-on CC control), separate from controller sequencer automation.

## Capacity

- Up to **6** controller lanes can be added

The `Add Controller` button is disabled after 6 lanes are present.

## Per-Controller Lane Controls

Each controller lane includes:

- Controller name label ( `Controller N` if unnamed)
- Running/stopped state badge
- `Controller #` field ( `0..127` )
- **MIDI Channels** checkboxes at the top right: 1–8 above 9–16
- `Start / Stop` enable toggle
- `Remove` button
- Knob control (value `0..127` )
- Numeric value display

## MIDI Channels

Check the channels that should receive this lane's CC messages. All 16 are checked by default (OMNI); at least one must remain checked. The last checked box cannot be unchecked until another channel is selected.

When the lane and instrument session are running, changing channels immediately sends the current value to the new selection. Unchecked channels retain their last received value. Starting a lane or session, changing the CC number, and adjusting the knob all use the selected channels.

The selection survives Save/Load Performance, browser reload, native JSON/ZIP export/import, and both CSD exports. Older performances without a selection load with all 16 channels checked. Controllers using the same CC on different channels can have independent values; overlapping mappings receive the latest transmitted value.

## Knob Interaction

The knob uses pointer drag interaction:

- Click and drag vertically to change the value



- Up increases the CC value
- Down decreases the CC value

The UI also shows a numeric value readout next to the knob for exact control.

## Live Sending Behavior

When a controller lane is enabled and the instrument session is running:

- changing the knob sends MIDI `control_change` messages immediately

This is ideal for expressive live performance adjustments.

## Manual Controllers vs Controller Sequencers

Use manual controller lanes when you want:

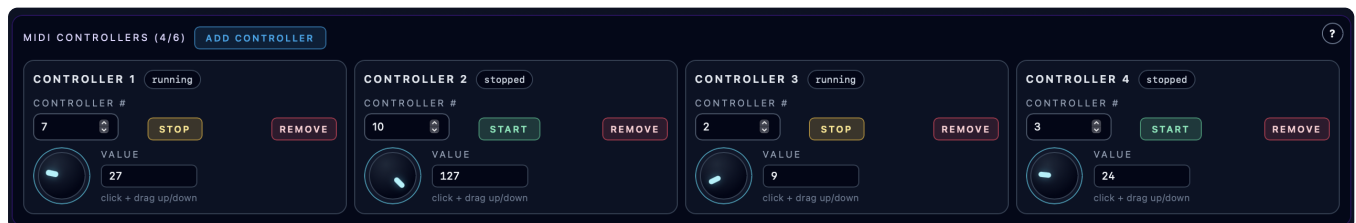
- direct, improvised control
- tactile-feeling UI knobs
- on-the-fly sweeps and tweaks

Use controller sequencers when you want:

- repeatable automation
- transport-synced movement
- curve-based modulation patterns

Both can exist in the same performance.

## Screenshots



Manual MIDI controller lanes with knobs, CC numbers, and values.



# Opcode Catalog and Integrated Documentation

The opcode catalog is the source list of all supported building blocks you can place in the graph editor.

## What The Catalog Supports

- Search by opcode name
- Search by category
- Search by description text
- Search by tags
- Click to add an opcode to the graph
- Drag-and-drop an opcode into the graph canvas at a chosen position

## Search Behavior

The search field filters on these opcode fields:

- name
- category
- description
- tags

This makes it easy to find opcodes by function (for example `filter`, `mid`, `reverb`, `soundfont`) even if you do not remember the exact Csound opcode name.

## Adding Opcodes

### Click To Add

- Clicking an opcode row inserts the node into the graph using the app's add-node behavior.
- This is fastest when exact placement is not important yet.

### Drag-and-Drop To Place

- Drag an opcode from the catalog and drop it into the graph canvas.
- The graph editor computes the drop location and places the node there.
- While dragging over the graph, the canvas highlights to confirm drop targeting.

## Visual Information In The Catalog

Each catalog entry shows:

- Opcode icon (backend-served icon asset)
- Opcode name
- Category label

These same categories influence node coloring in the graph editor for quick visual grouping.

## Integrated Opcode Documentation ( ? on node)

After placing an opcode in the graph, use the node's ? button to open the opcode documentation modal.

The opcode documentation modal provides:

- Localized description (English/German/French/Spanish)
- Category and syntax/template summary



- Input/output port descriptions (including optional/default/accepted types)
- Tags
- Direct link to the Csound reference page (Open Csound Reference)

## Atone High-Pass Filters

Search for `atone` or `highpass` in the `filter` category:

- `atone` filters an audio signal (`asig`) and produces audio (`aout`). Its `khp` cutoff accepts control-rate or init-rate values and starts at 200 Hz.
- `atonek` filters a control signal (`ksig`) and produces control output (`kout`) for modulation. Its `khp` cutoff starts at 10 Hz. Both inputs accept control-rate or init-rate values.
- `atonex` cascades multiple `atone` stages for a sharper cutoff. Its audio input/output are `asig` / `aout`; `xhp` accepts audio-rate, control-rate, or init-rate cutoff values and starts at 200 Hz. The optional init-rate `inumlayer` sets the stage count and defaults to 4.

Each filter exposes optional init-rate `iskip`: 0 clears its internal state at initialization, while a nonzero value retains previous state. The default is 0. Setting only `iskip` on `atonex` retains the default four layers. Cutoff defaults are editor starting values; Csound requires a cutoff argument.

Connect the audio outputs of `atone` and `atonex` to downstream audio nodes or both `outs` inputs. Connect `atonek` to a control input such as oscillator amplitude or frequency. The node's `? help` documents every port in English, German, French, and Spanish.

## STK Instruments

Search for `stk` to find the 27 Synthesis Toolkit instruments in the `physical_modeling`, `fm`, and `oscillator` categories. They include strings, winds, percussion, voices, organs, and electric pianos. Each node produces one mono audio signal; connect it to both `outs` inputs for centered stereo output.

- `ifrequency` and `iampitude` are init-rate inputs: frequency in Hz and note amplitude from 0 to 1. Use `cpsmidi` for note pitch; apply envelopes or gain after the audio output for continuous level changes.
- `STKDrummer` uses the frequency to select a drum sample. Connect `cpsmidi` and play MIDI drum notes. `STKDrummer`, `STKPlucked`, and `STKSitar` have no documented controller inputs.
- The other nodes expose the manual's controller number/value pairs in order. The controller number is prefilled with the documented number. Set or connect its paired `kv` value to enable that control. Both ports accept init-rate or control-rate sources, so controller values can change during a note.
- Unset controller values omit the entire pair and retain STK's instrument defaults. You can enable a later pair without filling earlier pairs. For example, set only `kv7` to 3 on `STKBandedWG` to send instrument preset controller 16 with value 3.
- Most values range from 0 to 127, but preset selectors use model-specific ranges. `STKShakers` uses controller **1071**, not a MIDI CC in the 0–127 range, for its instrument selection. Check the node's `? help` for each controller's meaning.

Runtime requires the Csound STK plugin (`stkopd`) and its rawwave data on the machine running the Csound backend. Set `RAWWAVE_PATH` to the installed rawwaves directory before starting the backend; some Csound builds do not register any STK opcodes without it. For example, an Apple Silicon Homebrew installation of STK uses `RAWWAVE_PATH=/opt/homebrew/opt/stk/share/stk/rawwaves/`. Catalog availability and successful graph compilation do not install this runtime dependency.

## Context Help (? in page sections)

In addition to opcode-level docs, the app provides context help buttons for page sections (toolbar, catalog, graph, runtime, sequencer panels, config panels). These open integrated markdown help content in the currently selected GUI language.

For the opcode catalog specifically, the integrated help now focuses on search/add workflow and on the distinction between catalog help and the node-level opcode documentation modal.



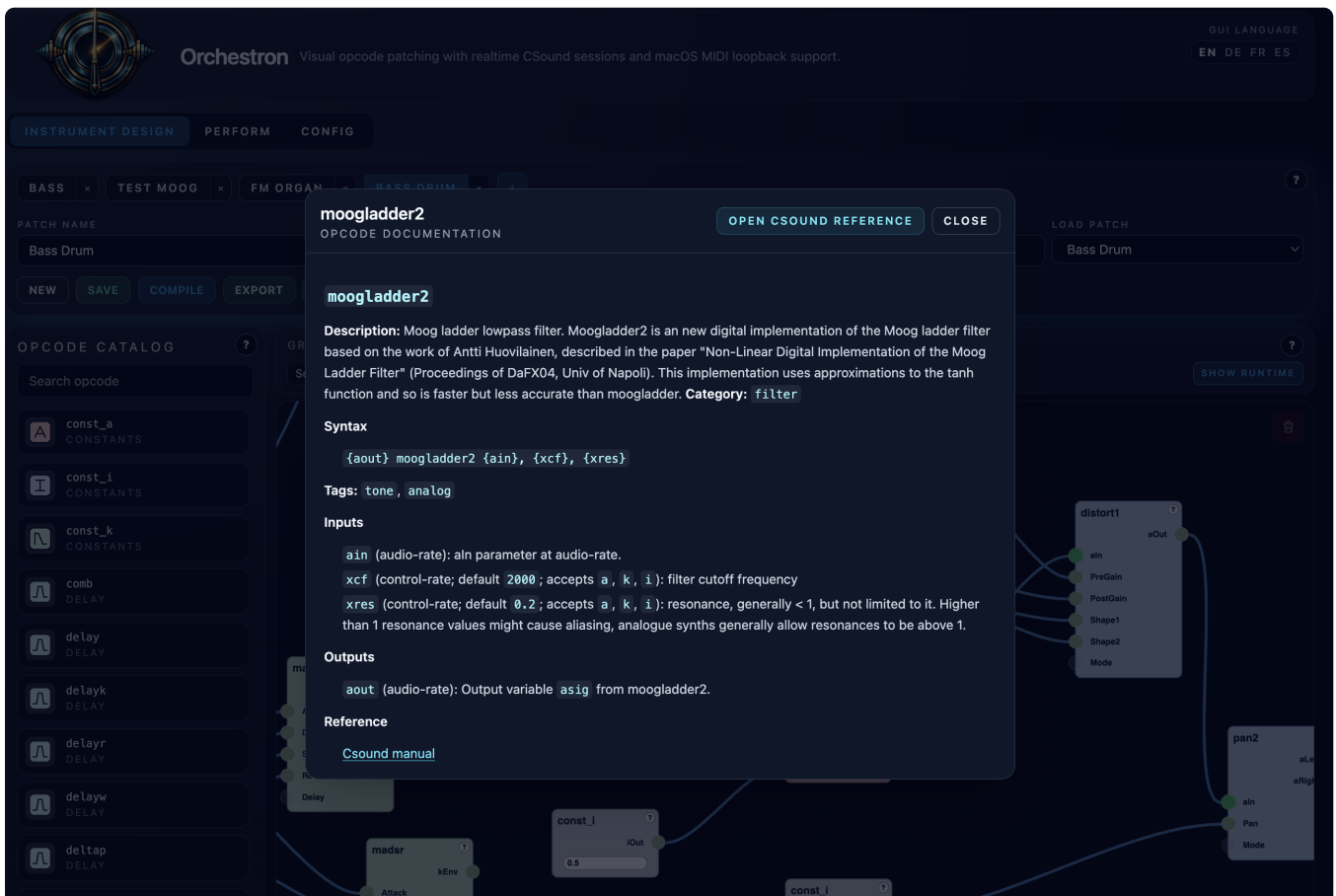
See also:

- [GUI Language and Integrated Help](#)
- [Supported Opcodes](#) for the full opcode index table

## Screenshots



Opcode catalog filtered by search text.



Integrated opcode documentation modal opened from a graph node.



# GUI Language and Integrated Help

Orchestron includes multilingual UI labels and integrated documentation so you can stay inside the app while learning and working.

## GUI Language Selector

The language selector is in the top-right area of the main app header.

Supported languages:

- English
- German
- French
- Spanish

The selector is presented as quick language links (short labels) for fast switching.

## What Changes When You Switch Language

Language switching affects user-facing text such as:

- Page labels ( Instrument Design , Perform , Config )
- Toolbar/button labels
- Context help pages (integrated help modals)
- Opcode documentation modal content (localized generated markdown)
- Many labels inside advanced editors (including the GEN editor)

## Integrated Help ( ? Buttons)

Orchestron provides context help buttons ( ? ) across the UI.

Examples include:

- Instrument patch toolbar
- Opcode catalog
- Graph editor header/area
- Runtime panel
- Sequencer sections (rack, tracks, multitrack arranger, piano rolls, MIDI controllers)
- Config page sections

Clicking a context ? opens a markdown help modal tailored to that UI area.

On the Instrument Design, Perform, and Config pages, these help modals focus on the current tool, device, transport section, or engine setting panel instead of repeating generic UI conventions.

## Opcode Documentation ( ? On Opcode Node)

Placed opcode nodes provide a node-level ? button that opens opcode documentation.

The modal includes:

- Localized description
- Inputs and outputs (with detailed port descriptions)
- Syntax/template summary
- Tags



- Open [Csound Reference](#) link to the official Csound documentation page for that opcode

## Modal Behavior

Both help modals and opcode documentation modals support:

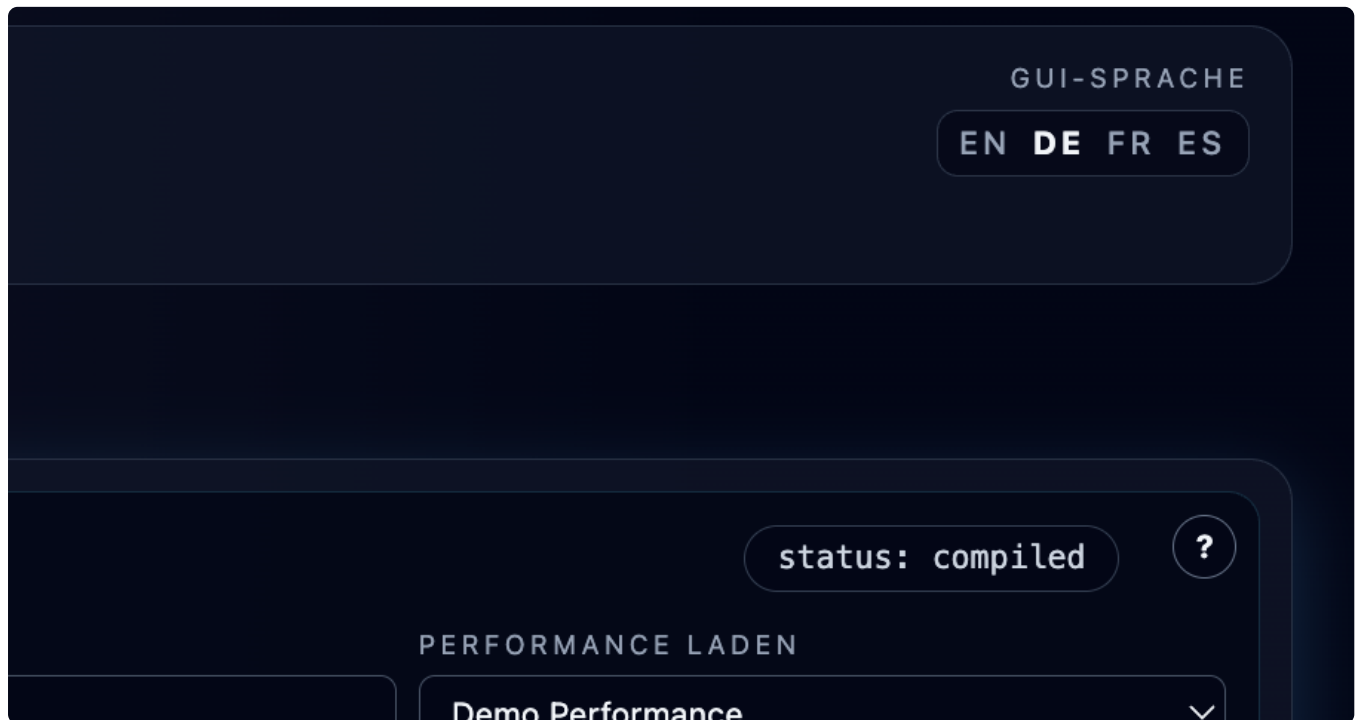
- Close button
- Click outside modal to close
- Esc key to close

## Best Practice

Use context help for workflow guidance and opcode docs for technical/operator details.

This combination is faster than switching repeatedly between the app and external documentation while designing patches.

## Screenshots



*Top-right header language selector links.*





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# Supported Opcodes

This appendix is generated from `backend/app/data/opcodes.json` and currently lists **177** user-selectable opcodes and editor constructs in the Instrument Design opcode catalog.

## How To Use This Appendix

- Use the table for quick discovery (category, signal I/O, one-line purpose).
- In the app, click the `?` on an opcode node to open the integrated, localized documentation with port-level details and a Csound reference link.
- The `Inputs` / `Outputs` columns show port count and signal-rate shorthand.
- Signal legend: `a` = audio-rate, `k` = control-rate, `i` = init-rate, `S` = string, `f` = phase-vocoder frame signal.

## Category Counts

| Category          | Count |
|-------------------|-------|
| analysis          | 2     |
| constants         | 5     |
| control_flow      | 2     |
| delay             | 11    |
| distortion        | 4     |
| dynamics          | 2     |
| envelope          | 12    |
| filter            | 19    |
| fm                | 17    |
| math              | 3     |
| midi              | 7     |
| mixer             | 3     |
| modulation        | 10    |
| noise             | 3     |
| oscillator        | 19    |
| output            | 1     |
| physical_modeling | 29    |
| reverb            | 4     |
| routing           | 4     |
| soundfont         | 3     |
| spectral          | 8     |
| tables            | 4     |
| utility           | 5     |

## Opcode Index

### analysis



| Opcode  | Inputs      | Outputs | Short Description                                               |
|---------|-------------|---------|-----------------------------------------------------------------|
| follow2 | 3 (a, k, k) | 1 (a)   | Envelope follower with independent attack and release controls. |
| rms     | 3 (a, i, i) | 1 (k)   | RMS envelope follower for an audio signal.                      |

## constants

| Opcode          | Inputs | Outputs | Short Description                                                  |
|-----------------|--------|---------|--------------------------------------------------------------------|
| const_a         | -      | 1 (a)   | Audio-rate constant value.                                         |
| const_i         | -      | 1 (i)   | Init-rate constant value.                                          |
| const_k         | -      | 1 (k)   | Control-rate constant value.                                       |
| const_s         | -      | 1 (S)   | String constant value.                                             |
| perf_controller | -      | 1 (i)   | Virtual per-instance performance setting with a labeled rack knob. |

## control\_flow — editor constructs

These nodes generate structured Csound conditionals; they are not ordinary opcode calls. [If](#), [Switch](#) and [Drumset](#) (EN/DE/FR/ES) documents note-start selection and the managed Case Result boundary.

| Construct | Inputs   | Outputs    | Short Description                                                |
|-----------|----------|------------|------------------------------------------------------------------|
| If        | 2 (i, i) | 1 or 2 (a) | Select True/False by an init-rate comparison for each voice.     |
| Switch    | 1 (i)    | 1 or 2 (a) | Select a named numeric case or mandatory Default for each voice. |

## delay



| Opcode   | Inputs               | Outputs  | Short Description                                           |
|----------|----------------------|----------|-------------------------------------------------------------|
| comb     | 4 (a, k, i, i)       | 1 (a)    | Comb filter / feedback delay.                               |
| delay    | 3 (a, i, i)          | 1 (a)    | Simple non-interpolating audio delay line.                  |
| delayk   | 3 (k, i, i)          | 1 (k)    | Control-rate delay line.                                    |
| delayr   | 2 (i, i)             | 1 (a)    | Read tap from a classic delay-line memory buffer.           |
| delayw   | 1 (a)                | -        | Write into the delay-line memory buffer.                    |
| deltap   | 1 (k)                | 1 (a)    | Read a delay tap using linear interpolation.                |
| deltap3  | 1 (k)                | 1 (a)    | Read a delay tap using cubic interpolation.                 |
| flanger  | 4 (a, k, k, i)       | 1 (a)    | Flanger effect with delay modulation and feedback.          |
| vcomb    | 6 (a, k, k, i, i, i) | 1 (a)    | Variable-time comb reverb with controllable loop time.      |
| vdelay3  | 4 (a, k, i, i)       | 1 (a)    | Variable delay line with cubic interpolation.               |
| vdelayxs | 6 (a, a, a, i, i, i) | 2 (a, a) | Stereo variable delay with high-quality sinc interpolation. |

## distortion

| Opcode     | Inputs               | Outputs | Short Description                                           |
|------------|----------------------|---------|-------------------------------------------------------------|
| clip       | 4 (a, i, i, i)       | 1 (a)   | Signal clipper with selectable transfer curves.             |
| distort1   | 6 (a, k, k, k, k, i) | 1 (a)   | Waveshaping distortion with configurable transfer curve.    |
| fold       | 2 (a, k)             | 1 (a)   | Artificial foldover effect for audio signals.               |
| powershape | 3 (a, k, i)          | 1 (a)   | Power-law waveshaper for controllable nonlinear distortion. |

## dynamics

| Opcode | Inputs               | Outputs | Short Description                                                   |
|--------|----------------------|---------|---------------------------------------------------------------------|
| dam    | 6 (a, k, i, i, i, i) | 1 (a)   | Dynamic amplitude processor (downward compressor/noise suppressor). |
| limit  | 3 (a, k, k)          | 1 (a)   | Hard clamp limiter.                                                 |

## envelope



| Opcode   | Inputs                        | Outputs | Short Description                                                      |
|----------|-------------------------------|---------|------------------------------------------------------------------------|
| adsr     | 4 (i, i, i, i)                | 1 (k)   | Control-rate ADSR envelope.                                            |
| envlpxr  | 8 (k, i, i, i, i, i, i, i)    | 1 (k)   | Control-rate envelope with optional release scaling controls.          |
| expon    | 3 (i, i, i)                   | 1 (k)   | Control-rate exponential segment between two points.                   |
| expseg   | 9 (i, i, i, i, i, i, i, i, i) | 1 (k)   | Control-rate exponential breakpoint envelope generator.                |
| expsega  | 9 (i, i, i, i, i, i, i, i, i) | 1 (a)   | Audio-rate exponential breakpoint envelope generator.                  |
| expsegr  | 5 (i, i, i, i, i)             | 1 (k)   | Control-rate exponential envelope with release segment.                |
| linenr   | 4 (k, i, i, i)                | 1 (k)   | Control-rate envelope with note-release stage.                         |
| linseg   | 7 (i, i, i, i, i, i, i)       | 1 (k)   | Control-rate linear breakpoint envelope generator.                     |
| linsegr  | 5 (i, i, i, i, i)             | 1 (k)   | Control-rate linear breakpoint envelope with release segment.          |
| madsr    | 6 (i, i, i, i, i, i)          | 1 (k)   | MIDI release-sensitive ADSR envelope.                                  |
| mxadsr   | 6 (i, i, i, i, i, i)          | 1 (k)   | Extended MIDI release-sensitive ADSR envelope.                         |
| transegr | 7 (i, i, i, i, i, i, i)       | 1 (k)   | Control-rate transition envelope with release-triggered final segment. |

## filter



| Opcode       | Inputs               | Outputs        | Short Description                                                      |
|--------------|----------------------|----------------|------------------------------------------------------------------------|
| atone        | 3 (a, k, i)          | 1 (a)          | Audio-rate high-pass filter complementary to tone.                     |
| atonek       | 3 (k, k, i)          | 1 (k)          | Control-rate high-pass filter complementary to tonek.                  |
| atonex       | 4 (a, k, i, i)       | 1 (a)          | Cascaded atone high-pass filters with a sharper cutoff.                |
| butterbp     | 4 (a, k, k, i)       | 1 (a)          | Second-order Butterworth band-pass filter.                             |
| butterbr     | 4 (a, k, k, i)       | 1 (a)          | Second-order Butterworth band-reject filter.                           |
| butterhp     | 3 (a, k, i)          | 1 (a)          | Second-order Butterworth high-pass filter.                             |
| butterlp     | 3 (a, k, i)          | 1 (a)          | Second-order Butterworth low-pass filter.                              |
| diode_ladder | 6 (a, k, k, i, i, i) | 1 (a)          | Diode ladder low-pass filter model.                                    |
| exciter      | 5 (a, k, k, k, k)    | 1 (a)          | Harmonic exciter that adds controlled upper partials.                  |
| fofilter     | 5 (a, k, k, k, i)    | 1 (a)          | Formant filter.                                                        |
| moogladder   | 3 (a, k, k)          | 1 (a)          | Moog ladder low-pass filter.                                           |
| moogladder2  | 3 (a, k, k)          | 1 (a)          | Nonlinear Moog-style ladder filter with audio-rate modulation support. |
| moogvcf      | 5 (a, k, k, i, i)    | 1 (a)          | Moog ladder voltage-controlled filter emulation.                       |
| resonx       | 6 (a, k, k, i, i, i) | 1 (a)          | Multi-layer resonant band-pass filter.                                 |
| rezzy        | 5 (a, k, k, i, i)    | 1 (a)          | Resonant low-pass or high-pass filter.                                 |
| skf          | 5 (a, k, k, i, i)    | 1 (a)          | Sallen-Key low-pass or high-pass filter.                               |
| statevar     | 5 (a, k, k, i, i)    | 4 (a, a, a, a) | State-variable filter with simultaneous HP/LP/BP/BR outputs.           |
| tbvcf        | 6 (a, k, k, k, k, i) | 1 (a)          | TB-303 style voltage-controlled filter model.                          |
| vc1pf        | 4 (a, k, k, i)       | 1 (a)          | Virtual-analog low-pass filter.                                        |

**fm**



| Opcode       | Inputs                                  | Outputs  | Short Description                                                      |
|--------------|-----------------------------------------|----------|------------------------------------------------------------------------|
| STKBeeThree  | 12 (i, i, k, k, k, k, k, k, k, k, k, k) | 1 (a)    | STK four-operator FM organ inspired by Hammond organs.                 |
| STKFMVoices  | 12 (i, i, k, k, k, k, k, k, k, k, k, k) | 1 (a)    | STK FM singing-voice instrument with vowel and spectral-tilt controls. |
| STKHeavyMetl | 12 (i, i, k, k, k, k, k, k, k, k, k, k) | 1 (a)    | STK four-operator FM instrument for metallic timbres.                  |
| STKPercFlut  | 12 (i, i, k, k, k, k, k, k, k, k, k, k) | 1 (a)    | STK four-operator FM percussive flute instrument.                      |
| STKRhodey    | 12 (i, i, k, k, k, k, k, k, k, k, k, k) | 1 (a)    | STK four-operator FM electric piano inspired by Rhodes pianos.         |
| STKTubeBell  | 12 (i, i, k, k, k, k, k, k, k, k, k, k) | 1 (a)    | STK four-operator FM tubular-bell instrument.                          |
| STKWurley    | 12 (i, i, k, k, k, k, k, k, k, k, k, k) | 1 (a)    | STK four-operator FM electric piano inspired by Wurlitzer pianos.      |
| crossfmi     | 9 (k, k, k, k, k, i, i, i, i)           | 2 (a, a) | Interpolating crossed frequency-modulation oscillator pair.            |
| crossfmpmi   | 9 (k, k, k, k, k, i, i, i, i)           | 2 (a, a) | Interpolating crossed frequency/phase-modulation oscillator pair.      |
| crosspmi     | 9 (k, k, k, k, k, i, i, i, i)           | 2 (a, a) | Interpolating crossed phase-modulation oscillator pair.                |
| fmb3         | 10 (k, k, k, k, k, i, i, i, i, i)       | 1 (a)    | B3 organ FM model.                                                     |
| fmbell       | 12 (k, k, k, k, k, k, i, i, i, i, i, i) | 1 (a)    | Bell FM model.                                                         |
| fmmetal      | 11 (k, k, k, k, k, k, i, i, i, i, i)    | 1 (a)    | Metallic FM model.                                                     |
| fmpercfl     | 11 (k, k, k, k, k, k, i, i, i, i, i)    | 1 (a)    | Percussive flute FM model.                                             |
| fmrhode      | 11 (k, k, k, k, k, k, i, i, i, i, i)    | 1 (a)    | Rhodes electric piano FM model.                                        |
| fmvoice      | 11 (k, k, k, k, k, k, i, i, i, i, i)    | 1 (a)    | Voice-like FM model.                                                   |
| fmwurlie     | 11 (k, k, k, k, k, k, i, i, i, i, i)    | 1 (a)    | Wurlitzer electric piano FM model.                                     |

## math

| Opcode | Inputs   | Outputs | Short Description                                               |
|--------|----------|---------|-----------------------------------------------------------------|
| a_mul  | 2 (a, a) | 1 (a)   | Multiply two audio signals.                                     |
| k_mul  | 2 (k, k) | 1 (k)   | Multiply two control signals.                                   |
| tanh   | 1 (a)    | 1 (a)   | Hyperbolic tangent function for audio waveshaping and limiting. |

## midi



| Opcode       | Inputs      | Outputs  | Short Description                                                 |
|--------------|-------------|----------|-------------------------------------------------------------------|
| ampmidi      | 2 (i, i)    | 1 (i)    | Generate init-rate amplitude from MIDI note velocity.             |
| ampmidicurve | 3 (k, k, k) | 1 (k)    | Map MIDI velocity to gain using dynamic range and curve exponent. |
| ampmidid     | 2 (k, i)    | 1 (k)    | Map MIDI velocity to amplitude using a decibel range.             |
| cpsmidi      | -           | 1 (i)    | Read active MIDI note pitch as cycles-per-second.                 |
| midi_note    | 1 (i)       | 2 (k, k) | Extract MIDI note frequency and velocity amplitude.               |
| midictrl     | 3 (i, i, i) | 1 (k)    | Read a MIDI controller value with optional scaling.               |
| notnum       | -           | 1 (i)    | Read the MIDI note number for the current note event.             |

## mixer

| Opcode | Inputs      | Outputs  | Short Description                           |
|--------|-------------|----------|---------------------------------------------|
| mix2   | 2 (a, a)    | 1 (a)    | Mix two audio signals.                      |
| ntrpol | 3 (a, a, k) | 1 (a)    | Linear crossfade between two audio signals. |
| pan2   | 3 (a, k, i) | 2 (a, a) | Stereo panner.                              |

## modulation



| Opcode   | Inputs                        | Outputs | Short Description                                                            |
|----------|-------------------------------|---------|------------------------------------------------------------------------------|
| jitter   | 3 (k, k, k)                   | 1 (k)   | Control-rate segmented random line generator.                                |
| lfo      | 3 (k, k, i)                   | 1 (k)   | Low-frequency oscillator for control-rate modulation.                        |
| portk    | 3 (k, k, i)                   | 1 (k)   | Control-rate portamento/slew limiter with optional init value.               |
| random   | 2 (k, k)                      | 1 (k)   | Control-rate random value generator between a minimum and maximum.           |
| randomh  | 5 (k, k, k, i, i)             | 1 (k)   | Control-rate random sample-and-hold generator with update frequency control. |
| randomi  | 5 (k, k, k, i, i)             | 1 (k)   | Control-rate interpolated random generator with update frequency control.    |
| release  | -                             | 1 (k)   | Return 1 during note release phase and 0 otherwise.                          |
| samphold | 4 (k, k, i, i)                | 1 (k)   | Control-rate sample-and-hold processor.                                      |
| vibr     | 4 (k, k, i, i)                | 1 (k)   | Simple vibrato control oscillator with table lookup.                         |
| vibrato  | 9 (k, k, k, k, k, k, k, k, i) | 1 (k)   | Randomized vibrato generator.                                                |

## noise

| Opcode  | Inputs         | Outputs | Short Description                                                |
|---------|----------------|---------|------------------------------------------------------------------|
| noise   | 4 (k, k, i, i) | 1 (a)   | Variable-color random audio noise.                               |
| pinker  | -              | 1 (a)   | Pink noise generator.                                            |
| pinkish | 2 (a, i)       | 1 (a)   | Pinkening filter for shaping white noise toward a pink spectrum. |

## oscillator



| Opcode     | Inputs                                                                | Outputs  | Short Description                                                             |
|------------|-----------------------------------------------------------------------|----------|-------------------------------------------------------------------------------|
| STKMoog    | 12 (i, i, k, k, k, k, k, k, k, k, k, k)                               | 1 (a)    | STK sampled-wave synthesizer with swept formant filters.                      |
| STKSimple  | 10 (i, i, k, k, k, k, k, k, k, k)                                     | 1 (a)    | STK wavetable and noise synthesizer with filters and an ADSR envelope.        |
| fof        | 15 (k, k, k, k, k, k, k, k, i, i, i, i, i, i, i)                      | 1 (a)    | Formant/granular source using sinusoid bursts.                                |
| fof2       | 15 (k, k, k, k, k, k, k, k, i, i, i, i, k, k, i)                      | 1 (a)    | FOF source with per-grain phase indexing and glissando.                       |
| foscili    | 7 (k, k, k, k, k, i, i)                                               | 1 (a)    | Audio-rate FM oscillator with harmonic ratios.                                |
| gbuzz      | 7 (k, k, k, k, k, i, i)                                               | 1 (a)    | Generalized buzz oscillator with controllable harmonics.                      |
| grain      | 10 (k, k, k, k, k, k, i, i, i, i)                                     | 1 (a)    | Classic granular synthesis oscillator with table-based grains.                |
| grain2     | 9 (k, k, k, i, k, i, i, i, i)                                         | 1 (a)    | Easy-to-use granular synthesis texture generator.                             |
| grain3     | 13 (k, k, k, k, k, k, i, k, i, k, k, i, i)                            | 1 (a)    | Granular oscillator with independent frequency and phase modulation controls. |
| granule    | 22 (k, i, i, i, i, i, i, i, i, i, k, i, k, i, i, i, i, i, i, i, i, i) | 1 (a)    | Multi-voice granular processor with independent gap and grain-size controls.  |
| moog       | 9 (k, k, k, k, k, k, i, i, i)                                         | 1 (a)    | Mini-Moog style synthesizer model source.                                     |
| oscil3     | 4 (k, k, i, i)                                                        | 1 (a)    | Cubic-interpolating oscillator with low distortion.                           |
| oscili     | 3 (k, k, i)                                                           | 1 (a)    | Classic interpolating oscillator.                                             |
| poscil3    | 4 (k, k, i, i)                                                        | 1 (a)    | High-precision cubic interpolating oscillator.                                |
| syncphasor | 3 (k, a, i)                                                           | 2 (a, a) | Audio-rate normalized phase generator with sync trigger I/O.                  |
| vco        | 5 (k, k, i, k, i)                                                     | 1 (a)    | Band-limited voltage-controlled oscillator.                                   |
| vco2       | 6 (k, k, i, k, k, i)                                                  | 1 (a)    | Improved anti-aliased analog-style oscillator.                                |
| voice      | 8 (k, k, k, k, k, k, i, i)                                            | 1 (a)    | Formant-based vocal source synthesizer.                                       |
| vosim      | 8 (k, k, k, k, k, k, i, i)                                            | 1 (a)    | Vocal formant synthesis using overlapping sinusoidal bursts.                  |

## output

| Opcode | Inputs   | Outputs | Short Description   |
|--------|----------|---------|---------------------|
| outs   | 2 (a, a) | -       | Stereo output sink. |

## physical\_modeling



| Opcode       | Inputs                                              | Outputs | Short Description                                                       |
|--------------|-----------------------------------------------------|---------|-------------------------------------------------------------------------|
| STKBandedWG  | 16 (i, i, k, k, k, k, k, k, k, k, k, k, k, k, k, k) | 1 (a)   | STK banded waveguide model of bowed bars, glasses, and bowls.           |
| STKBlowBotl  | 10 (i, i, k, k, k, k, k, k, k, k)                   | 1 (a)   | STK blown-bottle model with a Helmholtz resonator and jet excitation.   |
| STKBlowHole  | 12 (i, i, k, k, k, k, k, k, k, k, k, k)             | 1 (a)   | STK clarinet model with register and tone holes.                        |
| STKBowed     | 12 (i, i, k, k, k, k, k, k, k, k, k, k)             | 1 (a)   | STK bowed-string waveguide instrument.                                  |
| STKBrass     | 12 (i, i, k, k, k, k, k, k, k, k, k, k)             | 1 (a)   | STK brass waveguide instrument with lip and slide controls.             |
| STKClarinet  | 12 (i, i, k, k, k, k, k, k, k, k, k, k)             | 1 (a)   | STK single-reed clarinet waveguide instrument.                          |
| STKDrummer   | 2 (i, i)                                            | 1 (a)   | STK drum synthesizer using sampled raw waves and one-pole filters.      |
| STKFlute     | 12 (i, i, k, k, k, k, k, k, k, k, k, k)             | 1 (a)   | STK flute waveguide instrument with jet and breath controls.            |
| STKMandolin  | 12 (i, i, k, k, k, k, k, k, k, k, k, k)             | 1 (a)   | STK mandolin model with paired strings and sampled body responses.      |
| STKModalBar  | 16 (i, i, k, k, k, k, k, k, k, k, k, k, k, k, k, k) | 1 (a)   | STK modal percussion instrument with bar and bell presets.              |
| STKPlucked   | 2 (i, i)                                            | 1 (a)   | STK Karplus-Strong plucked-string instrument.                           |
| STKResonate  | 12 (i, i, k, k, k, k, k, k, k, k, k, k)             | 1 (a)   | STK noise-driven resonator with pole and zero controls.                 |
| STKSaxofony  | 16 (i, i, k, k, k, k, k, k, k, k, k, k, k, k, k, k) | 1 (a)   | STK saxophone-like reed waveguide instrument.                           |
| STKShakers   | 14 (i, i, k, k, k, k, k, k, k, k, k, k, k, k)       | 1 (a)   | STK stochastic percussion and environmental-sound models.               |
| STKSitar     | 2 (i, i)                                            | 1 (a)   | STK plucked-string sitar model.                                         |
| STKStiffKarp | 8 (i, i, k, k, k, k, k, k)                          | 1 (a)   | STK stiff-string model with pickup, sustain, and stretch controls.      |
| STKVoicForm  | 12 (i, i, k, k, k, k, k, k, k, k, k, k)             | 1 (a)   | STK four-formant voice synthesizer with voiced and unvoiced excitation. |
| STKWhistle   | 12 (i, i, k, k, k, k, k, k, k, k, k, k)             | 1 (a)   | STK whistle model with a pea and fipple modulation.                     |
| dripwater    | 8 (k, i, i, i, i, i, i, i)                          | 1 (a)   | Stochastic dripping-water physical model source.                        |
| marimba      | 11 (k, k, i, i, i, k, k, i, i, i, i)                | 1 (a)   | Physical model of a marimba bar and resonator.                          |
| pluck        | 6 (k, k, i, i, i, i)                                | 1 (a)   | Karplus-Strong plucked-string model.                                    |
| sekere       | 5 (i, i, i, i, i)                                   | 1 (a)   | PhISEM sekere shaker model.                                             |
| sleighbells  | 8 (k, i, i, i, i, i, i, i)                          | 1 (a)   | PhISEM sleigh bells model.                                              |



| Opcode     | Inputs                            | Outputs | Short Description                                                 |
|------------|-----------------------------------|---------|-------------------------------------------------------------------|
| wgbow      | 8 (k, k, k, k, k, k, i, i)        | 1 (a)   | Waveguide bowed-string model.                                     |
| wgbowedbar | 9 (k, k, k, k, k, i, i, i, i)     | 1 (a)   | Waveguide bowed-bar model.                                        |
| wgclar     | 10 (k, k, k, i, i, k, k, k, i, i) | 1 (a)   | Waveguide clarinet model.                                         |
| wgflute    | 10 (k, k, k, i, i, k, k, k, i, i) | 1 (a)   | Waveguide flute model.                                            |
| wgpluck2   | 5 (i, k, i, k, k)                 | 1 (a)   | Waveguide plucked string model with pick and reflection controls. |
| wguide2    | 4 (a, k, k, k)                    | 1 (a)   | Two-point waveguide resonator.                                    |

## reverb

| Opcode   | Inputs                        | Outputs  | Short Description                                                             |
|----------|-------------------------------|----------|-------------------------------------------------------------------------------|
| freeverb | 6 (a, a, k, k, i, i)          | 2 (a, a) | Stereo Freeverb processor with room size and high-frequency damping controls. |
| platerev | 9 (i, i, k, i, i, i, i, a, a) | 2 (a, a) | Physical-model plate reverb with stereo output taps.                          |
| reverb2  | 4 (a, k, k, i)                | 1 (a)    | Schroeder reverb processor.                                                   |
| reverb3c | 7 (a, a, k, k, i, i, i)       | 2 (a, a) | 8-delay-line stereo FDN reverb based on Sean Costello's design.               |

## routing

| Opcode  | Inputs   | Outputs | Short Description                                                 |
|---------|----------|---------|-------------------------------------------------------------------|
| inleta  | 1 (S)    | 1 (a)   | Receive an audio-rate signal from a named instrument inlet port.  |
| inletk  | 1 (S)    | 1 (k)   | Receive a control-rate signal from a named instrument inlet port. |
| outleta | 2 (S, a) | -       | Send an audio-rate signal to a named instrument outlet port.      |
| outletk | 2 (S, k) | -       | Send a control-rate signal to a named instrument outlet port.     |

## soundfont

| Opcode   | Inputs                     | Outputs  | Short Description                                                     |
|----------|----------------------------|----------|-----------------------------------------------------------------------|
| sfinstr3 | 8 (i, i, k, k, i, i, i, i) | 2 (a, a) | Play a SoundFont instrument as stereo audio with cubic interpolation. |
| sfloat   | -                          | 1 (i)    | Load a SoundFont2 file and return a file handle.                      |
| sfplay3  | 8 (i, i, k, k, i, i, i, i) | 2 (a, a) | Play a SoundFont preset as stereo audio with cubic interpolation.     |

## spectral



| Opcode    | Inputs                     | Outputs | Short Description                                                        |
|-----------|----------------------------|---------|--------------------------------------------------------------------------|
| pvsanal   | 7 (a, i, i, i, i, i, i)    | 1 (f)   | Phase-vocoder analysis from an audio input to an fsig stream.            |
| pvshift   | 6 (f, k, k, k, k, k)       | 1 (f)   | Shift fsig partial frequencies by a fixed amount in Hz.                  |
| pvssmooth | 3 (f, k, k)                | 1 (f)   | Smooth fsig amplitude and frequency trajectories with lowpass filtering. |
| pvsmorph  | 4 (f, f, k, k)             | 1 (f)   | Morph between two fsig streams by amplitude and frequency interpolation. |
| pvsosc    | 8 (k, k, k, i, i, i, i, i) | 1 (f)   | Generate oscillator spectra directly as an fsig stream.                  |
| pvsvoc    | 5 (f, f, k, k, k)          | 1 (f)   | Cross-synthesize fsig amplitudes and excitation frequencies.             |
| pvs warp  | 7 (f, k, k, k, k, k, k)    | 1 (f)   | Warp and shift the spectral envelope of an fsig stream.                  |
| pvsynth   | 2 (f, i)                   | 1 (a)   | Resynthesize audio from an fsig stream via overlap-add.                  |

## tables

| Opcode    | Inputs                                  | Outputs | Short Description                                                                                            |
|-----------|-----------------------------------------|---------|--------------------------------------------------------------------------------------------------------------|
| GEN       | -                                       | 1 (i)   | Routine-aware function table generator (meta-opcode) that renders ftgen/ftgenonce from a specialized editor. |
| ftgen     | 12 (i, i, i, i, i, i, i, i, i, i, i, i) | 1 (i)   | Create a function table at init time using a GEN routine.                                                    |
| ftgenonce | 12 (i, i, i, i, i, i, i, i, i, i, i, i) | 1 (i)   | Generate a function table once and reuse it across instances.                                                |
| vco2init  | 6 (i, i, i, i, i, i)                    | 1 (i)   | Precompute band-limited wavetable sets for vco2 oscillators.                                                 |

## utility

| Opcode   | Inputs   | Outputs | Short Description                                                      |
|----------|----------|---------|------------------------------------------------------------------------|
| downsamp | 2 (a, i) | 1 (k)   | Convert an audio signal to control-rate by downsampling.               |
| k_to_a   | 1 (k)    | 1 (a)   | Interpolate control signal to audio-rate.                              |
| maxalloc | 1 (i)    | -       | Limit simultaneous instrument allocations for the compiled instrument. |
| upsamp   | 1 (k)    | 1 (a)   | Convert a control signal to audio-rate by repeating k-values.          |
| xtratim  | 1 (i)    | -       | Extend current note duration by an additional init-time amount.        |



# Graph Editor

The graph editor is the central visual patching surface for instrument design.

## Core Graph Editing Features

- Typed input/output ports (audio, control, init, string, function-table)
- Node placement and repositioning (drag nodes on the canvas)
- Connection creation between compatible ports
- Connection and node selection highlighting
- Deletion of selected nodes/connections via explicit delete action
- Zoom controls and fit-to-graph navigation
- Node category color coding
- Inline constant value editing for `const_a`, `const_i`, `const_k`, `const_s`

## Canvas Navigation

The graph editor supports pan/zoom navigation and includes a zoom HUD in the lower-right corner:

- Zoom percentage display
- – zoom out
- + zoom in
- `Fit` to fit the current graph into view

If the graph is empty, `Fit` resets toward a centered default view.

Horizontal and vertical scrollbars along the canvas edges let you reach every part of a large patch, including expanded If/Switch case frames. They remain available when **Patch controls** is collapsed and adjust when Runtime or the browser window changes the canvas size. Drag a scrollbar, or focus it with **Tab** and use the arrow keys for that direction. Scrolling moves the view without moving nodes or clearing selection. The scrollbars stay synchronized with background panning, zoom, and `Fit`.

## Node Appearance and Visual Cues

### Category Coloring

Nodes are color-coded by category (for example generator, filter, MIDI, constants, etc.) to help visually scan signal flow.

### Optional Inputs

Optional inputs are visually distinct from required inputs.

### Formula Highlight On Sockets

If an input socket has a saved Input Formula (combine formula), the socket is highlighted so you can immediately see custom combine logic exists.

## Adding Nodes

You can add nodes in two ways:

- Click from the opcode catalog
- Drag an opcode from the catalog and drop it into the graph canvas

Drag-and-drop placement is preferable for larger patches because it preserves your layout intent.



## Connections

### Creating Connections

- Connect outputs to compatible inputs.
- The backend compiler remains the source of truth for final type/compile validation.

### Selecting Connections

- Connections can be selected directly (not just nodes).
- Selected connections receive a visual highlight.
- Ctrl-based accumulation allows multi-select of connections and nodes.

### Connection Deletion Is Intentional

Orchestron prevents accidental edge removal during socket interaction.

- Connections are not removed by casual socket clicks/drag.
- To remove a connection, select it explicitly and use the delete action.

### Selection Behavior

- Click nodes to select.
- Ctrl-click accumulates node selections.
- Click connections to select them.
- Ctrl-click accumulates/deselects connection selections.
- Background click clears selection (when Ctrl is not held).

### Delete Selected Elements

The graph editor overlay includes a trash/delete button (top-right of the graph canvas area).

- Disabled when nothing is selected
- Enabled when at least one node or connection is selected

Deletion flow:

- If one item is selected, deletion can happen directly.
- If multiple items are selected, Orchestron opens a confirmation dialog listing all affected items (nodes and connections) before deletion.

## Node-Specific Controls

### Constant Nodes

The constant opcodes (`const_a`, `const_i`, `const_k`, `const_s`) provide inline editable values directly on the node. `const_s` accepts lowercase letters, digits, and underscores, must begin with a lowercase letter, and is limited to 50 characters.

### Documentation Button ( ? )

Nodes with documentation expose a ? button to open the opcode documentation modal.

### GEN Button (for GEN node)

The GEN meta-opcode node shows a GEN button that opens the specialized GEN table editor.



## Layout Persistence

- Node positions are stored in the patch graph.
- The graph structure (nodes + connections) is part of the patch data.
- The visual layout and related UI metadata (for example input formulas, GEN node configuration) are stored in `graph.ui_layout`.

## Compile Implications

- Compilation order is derived from graph dependencies.
- Invalid wiring, missing required inputs, or invalid formulas surface compile diagnostics.
- Graph compile status is reflected in the instrument page header (`compiled`, `pending changes`, `errors`).

## Integrated Help Focus

The graph editor ? help now concentrates on graph-specific behavior: persisted layout, deliberate deletion, constant-node inline editing, and the Input Formula Assistant workflow for combining multiple signals on one input.

## If and Switch at note start

Add **If** or **Switch** from **Control flow**. Drag catalog nodes directly into expanded cases, then patch each case into its managed **Case Result**. Frames measure the complete node bodies, automatically grow to the right and bottom and retain their dimensions, and keep the result in a separate bottom-right row. Drag the **Case Result title or unused body** to resize or shrink the frame; complete member bounds set its minimum. Other selected nodes stay still. This also works for Silence. Escape, cancellation or focus loss restores the previous dimensions and positions. Sizes persist through reload and export/import; bright purple outlines remain about two screen pixels thick at every zoom. The block remains above the first case; top and left movement limits keep members clear of the header.

Ordinary main-graph drags into a case transfer the selection. Ordinary case-member drags rearrange the current case; hold **Alt/Option at drag start** to transfer to another case or the main graph. The cyan/red border shows whether a drop is allowed. Every connection and formula binding must lead to another moved node or an endpoint already in the destination scope. Invalid drops list the connections and restore the whole group; Escape cancels. Existing shared main-graph inputs remain valid and can be wired after placement. A valid drop into Silence activates Synthesis. Dropping onto a collapsed block expands it without choosing a case.

The common output carries only the selected case. Collapse preserves selection, positions, viewport, shared input wiring and formulas. **Configure branches** retains **Move selected nodes** with the same transfer validation, plus case editing, Silence and reviewed mono/stereo changes. The built-in **Drumset** template demonstrates overlapping voices.

See [If, Switch and Drumset](#) for the complete EN/DE/FR/ES workflow, scope rules and version-2 patch/API format.

## Stereo blocks and audio interfaces

Stereo Input and Stereo Output are collapsible views of ordinary paired inlet/outlet nodes. Expand them to edit the underlying opcodes and connections. Exact port names and input formulas survive collapsing, expansion, saving and export. Direct Audio Output (outs) stays distinct.

The Orchestron Patch Creator CLI generates a collapsed Stereo Output block with named `left` / `right` channels and a main stereo mapping. Route this group through the performance mixer to Master for playback. In patch specs, output formulas target `output_left.asignal` and `output_right.asignal`.

The interface panel describes musical role, stable group IDs, display names, exact member ports, purpose and designated main input/output. Stereo blocks and their mappings share the same creation and deletion operations. Advanced mono/custom groups remain metadata views; changing a stereo group to another layout exposes the ordinary nodes without deleting them.



## Create, rename and remove a stereo interface

1. Click **Stereo Input** or **Stereo Output** in the opcode catalog, drag either onto the canvas, or expand **Stereo Input / Stereo Output · Exact channel mapping** and use its matching add button. Each action creates two channels and their mapping. Connect the new audio ports as needed.
2. Set Role and review the group's display name, purpose and designated Main input/Main output. New pairs use `left / right` or an available `busN.left / busN.right` pair. Channel names are unique within each direction.
3. Edit **Left** and **Right**, then choose **Apply** to review and confirm the rename. **Cancel** discards the draft. Renaming changes the mapping and underlying opcode names together. A string constant supplying an old name remains available to its other connections.
4. **Show underlying nodes** exposes the pair; **Collapse** returns to its grouped view. Delete a collapsed block or choose **Remove** on its mapping to remove both channels, attached connections and affected input formulas. The confirmation lists all affected elements. Deleting only one expanded channel removes the mapping and leaves the other as an ordinary opcode. Deleting a connection alone retains the mapping.
5. Deleting the designated main group clears its selection without promoting another group. Select a remaining group explicitly when needed.
6. Compile and save before assigning the patch in Perform. Check [Audio Mixer and Routing](#) after any external port changes.

## Upgrade or repair an existing instrument

- **Group existing channels:** choose Stereo Input or Stereo Output, select the existing Left/Right nodes and review the change. This preserves the channels' exact names, IDs, positions, audio wiring and formulas. Channel names become part of the stereo block's configuration: values from connected `const_s` nodes are stored directly on the underlying channels, and their naming connections are removed. Dedicated naming constants disappear from the canvas. Constants still used by other nodes or formulas remain available to those consumers; editing or deleting them no longer changes the grouped names. Edit the block's **Left/Right** fields to rename its channels. Channels already used by another mapping, duplicate names and unresolved expressions must be repaired first.
- **Convert Direct Audio Output:** select one `outs` node and review its replacement with a named stereo outlet pair. Each side retains its incoming wires, literal value and input formula. Missing inputs remain unconnected. Other direct outputs are left intact. This changes the destination model: connect the named ports in the performance mixer to hear them.
- Ungrouped old patches are not converted automatically. Previously saved valid stereo groups receive the same channel-name cleanup when opened, including groups that are currently expanded. Stale mappings show a reason and **Repair / Remove mapping**. Repair can bind a chosen existing pair or explicitly **Create missing channels**. Removing an unresolved mapping removes only metadata, preserving existing nodes.
- Conversion and renaming preserve role and activation. Previews list affected current-performance routes for explicit repair in the mixer after saving; other saved performances remain unchanged.



OPCODE CATALOG

Search opcode

follow2

ANALYSIS

rms

ANALYSIS

const\_a

CONSTANTS

const\_i

CONSTANTS

const\_k

CONSTANTS

const\_s

CONSTANTS

comb

DELAY

delay

DELAY

delayk

DELAY

delayr

DELAY

delayw

DELAY

Audition

GRAPH EDITOR (4 NODES, 2 CONNECTIONS) (compiled)

Selected: 0 opcode(s), 0 connection(s)

SHOW RUNTIME

Stereo Input / Stereo Output - Exact channel mapping

+ Stereo Input - inlets

+ Stereo Output - outlets

Role 

Audio effect

Stereo Input

Stereo

left, right

Main output

Main input

Show underlying nodes

Stereo Output

Stereo

left, right

Main output

Main output

Show underlying nodes

Stereo Return

Stereo Input - Stereo Input (left / right)

left

right

Stereo Output - Stereo Output (left / right)

left

right

100%

-

+

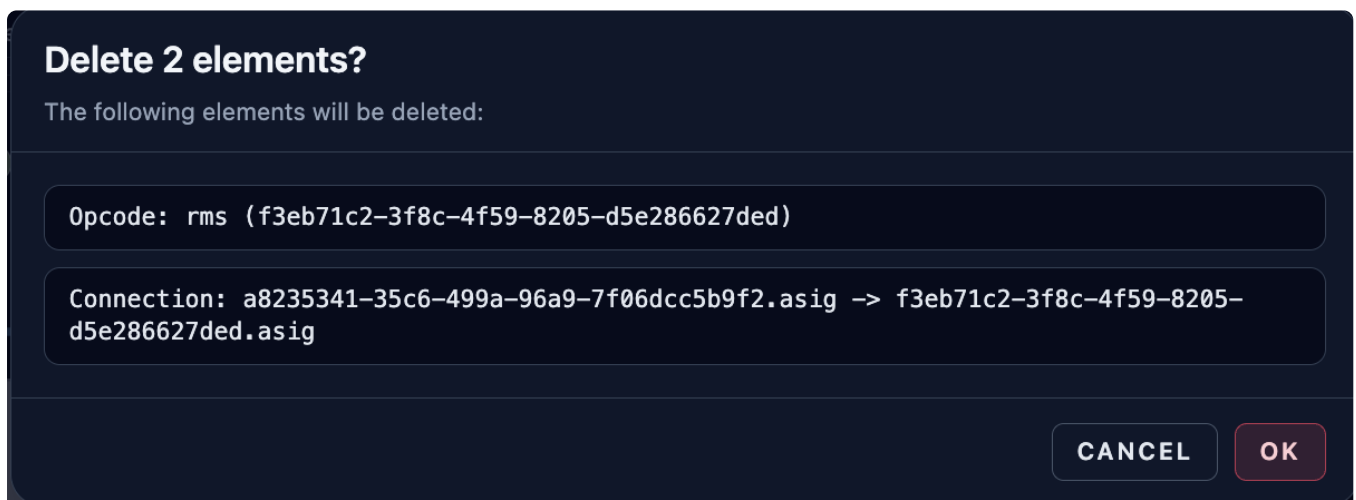
FIT

A continuous pass-through effect with explicit left/right groups and collapsed Stereo Input and Stereo Output blocks.

## Screenshots



Graph editor detail showing selection state, delete button, and zoom controls.



Delete-selection confirmation dialog listing selected graph items.



# Input Formula Assistant

The Input Formula Assistant lets you define exactly how incoming signals are combined on a single target input.

## When To Use It

Most commonly, you use it when **multiple outputs are connected to the same input**.

Examples:

- Mix two modulation sources before feeding a filter cutoff input
- Scale one source more than another ( $\text{in1} * 0.8 + \text{in2} * 0.2$ )
- Apply utility transforms ( $\text{abs}(\text{in1})$ ,  $\text{dbamp}(\text{in1})$ ) before an input

It is also useful for advanced cases with only one or even zero connected inputs:

- One input: reshape or scale the signal ( $\text{ceil}(\text{in1})$ ,  $\text{in1} * 0.5$ )
- Zero inputs: provide a constant expression ( $0.25$ ,  $\text{ampdb}(6)$ ) as a formula-driven value

## How To Open It

- Double-click an **input socket** in the graph editor.
- The assistant opens for that exact target node + target input port.

## What The Assistant Shows

### Connected Inputs (Token Mapping)

The left panel lists inbound connections to the target input and assigns tokens such as:

- `in1`
- `in2`
- `in3`

Each token shows source information (node/opcode/port) so you can see what signal you are referencing.

### Operator and Function Buttons

The assistant provides insertion buttons for:

- Operators: `+`, `-`, `*`, `/`
- Grouping: `(`, `)`
- Unary functions: `abs()`, `ceil()`, `floor()`, `ampdb()`, `dbamp()`
- Built-in literals: `sr` (configured audio sample rate)
- Custom numeric literals via the number entry field

### Formula Text Area

You can type directly into the formula field.

### Token Selection / Delete Helper

The assistant tokenizes the current expression and displays clickable tokens.

- Click a token to select that exact range in the text area.
- Use `Delete Selection` to remove the selected token/range quickly.



## Validation Rules

The formula must be syntactically valid before it can be saved.

Validation includes:

- Non-empty expression
- Valid numeric literals
- Supported characters/operators only
- Balanced parentheses
- Valid token ordering
- Known identifiers only ( `inN` tokens, supported built-in literals like `sr` , or supported unary function names)

If validation fails:

- The formula text area border changes to an error style
- A list of validation errors is shown
- `Save Formula` is disabled

## Save / Clear / Cancel

- `Save Formula` : stores the expression and token bindings for that target input
- `Clear Formula` : removes the saved formula for the target input
- `Cancel / Close` : closes the dialog without changing the saved formula

## What Happens During Compile

- If multiple connections target one input and **no** formula is defined, Orchestron falls back to summing the inbound signals.
- If a formula exists, the compiler resolves the token bindings and compiles the expression.
- Formula validation also happens backend-side, so invalid formulas fail compile with diagnostics.

## Formula Examples

### Weighted Mix

```
in1 * 0.7 + in2 * 0.3
```

### Gain Conversion

```
dbamp(in1)
```

### Rectified Modulation

```
abs(in1)
```

### Constant Override

```
0.5
```

### Sample-Rate-Aware Expression



sr \* 0.5

## Screenshots

Input Formula Assistant

2FDBDCAC-0F7E-4DE8-855B-37CC5B81537E.KCF (MOOGLADDER / CUTOFF)

CONNECTED INPUTS

in1const\_k (kout)

in2lfo (kout)

in3madsr (kenv)

in4midictrl (kval)

INSERT OPERATOR

+

-

\*

/

(

)

abs()

ceil()

floor()

ampdb()

dbamp()

INSERT NUMBER

1

ADD

FORMULA

((in3+0.2) \* (in1 \* (1+in2)))\*in4|

TOKEN SELECTION (CLICK TO SELECT RANGE)

(

(

in3

+

0.2

)

\*

(

in1

\*

(

1

+

in2

)

)

)

\*

in4

DELETE SELECTION

CLEAR FORMULA

CANCEL

SAVE FORMULA

Input Formula Assistant with connected input tokens and editable formula.

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# GEN Table Editor (GEN Meta-Opcode)

The GEN meta-opcode is a specialized table generator node that opens a dedicated editor for building `ftgen` / `ftgenonce` lines visually.

## Why The GEN Meta-Opcode Exists

Csound function-table generation is powerful but argument-heavy. The GEN editor provides a structured UI for common GEN routines and a preview of the generated `ftgen` / `ftgenonce` line.

## Opening The GEN Editor

- Add the GEN opcode to the graph.
- Click the GEN button on the node card.

## Editor Overview

The editor has two main areas:

- Left side: generation mode, routine selection, parameters
- Right side: preview (Effective GEN, flattened args, rendered line) and notes

## Table Generation Mode (Top Section)

### Opcode Mode

Choose which Csound table opcode style to generate:

- `ftgen`
- `ftgenonce`

### Routine Selection

You can choose from structured editors for common routines and a named routine option:

- GEN01 - Audio File
- GEN02 - Value List
- GEN07 - Segments
- GEN08 - Cubic Spline Segments
- GEN10 - Harmonic Sine Partial
- GEN11 - Harmonic Cosine Partial
- GEN17 - Step Table From x/y Pairs
- GEN20 - Window / Distribution Function
- GENpadsynth (named routine)
- Custom routine number / raw arguments

### Core Table Fields

- Routine Number
- Table Number
- Table Size
- Start Time (disabled for `ftgenonce` UI mode)
- Normalize table (positive GEN number vs negative GEN number behavior)

The backend validates GEN table settings before storage and again before Csound orchestra emission. Table Size can be



0 only for GEN01 ; other routines require a positive size. The maximum table size is 4194304 samples, and raw or structured GEN argument lists are capped at 512 entries. Raw argument text is limited to 8192 characters, with each token limited to 512 characters.

## Supported Routine Editors (Structured UI)

| Routine | Purpose                          | Main Parameters In Editor                                   |
|---------|----------------------------------|-------------------------------------------------------------|
| GEN10   | Harmonic sine partial amplitudes | Harmonic amplitude list                                     |
| GEN11   | Harmonic cosine partials         | harmonic count, lowest harmonic, multiplier                 |
| GEN02   | Literal value list               | Numeric value list                                          |
| GEN07   | Segments                         | Start value + repeated length/value rows                    |
| GEN08   | Cubic spline segments            | Start value + repeated length/value rows                    |
| GEN17   | Step table from x/y pairs        | x/y pairs (editable list)                                   |
| GEN20   | Window / distribution            | window type, max, optional parameter (for selected windows) |
| GEN01   | Load audio file into table       | uploaded asset, skip time, format, channel                  |

## GENpadsynth (Named Routine)

Orchestron supports a named GEN routine for padsynth (GENpadsynth).

- Select the GENpadsynth routine option.
- Enter padsynth parameters and partial amplitude/frequency pairs in the raw arguments editor.
- Argument order follows the Csound GENpadsynth documentation.

## Custom / Raw Routine Arguments

For routines without a dedicated structured editor:

- Use the raw arguments editor.
- Separate tokens by commas or new lines.
- Quote string literals manually (for example "file.wav").
- Prefix an argument with `expr:` to force raw Csound expression rendering without quotes.
- Keep raw argument lists within the backend validation limits: 512 tokens total, 512 characters per token, and 8192 characters for the raw text field.

Examples:

```
1, 0.5, expr:1024*2, "file.wav"
```

## GEN01 Audio File Workflow (Important)

The GEN01 editor loads sound files through uploaded backend assets:

### Uploaded Asset

- Click Upload Audio File
- Select an audio file (UI accepts common formats such as WAV/AIFF/FLAC/MP3/OGG and audio/\*)
- Orchestron stores the uploaded asset on the backend and references it in the GEN node config

Benefits:



- Instrument/performance exports can include the audio file automatically
- Imports can restore the asset automatically when using ZIP bundles
- Runtime compilation resolves the asset inside the configured backend asset directory
- Raw absolute paths and parent-traversal paths are rejected before Csound starts

## Additional GEN01 Parameters

- Skip Time
- Format
- Channel

## Upload Limits and Notes

- Default backend upload limit for GEN audio assets is 64 MiB per file (`VISUALCSOUND_GEN_AUDIO_ASSET_MAX_BYTES`)
- Persistent generated-asset storage is also capped by total bytes (`VISUALCSOUND_GEN_AUDIO_ASSETS_MAX_TOTAL_BYTES`, 1 GiB by default) and asset count (`VISUALCSOUND_GEN_AUDIO_ASSETS_MAX_COUNT`, 1024 by default)
- Garbage collection removes unreferenced generated assets older than `VISUALCSOUND_GEN_AUDIO_ASSET_GC_MIN_AGE_SECONDS` (24 hours by default)
- The backend streams GEN audio uploads and rejects requests once they exceed the configured cap
- ZIP bundle imports also enforce request, member-count, JSON, and total-uncompressed-size caps before importing bundled assets
- Deployments should set a matching maximum request size at the ASGI server, reverse proxy, or load balancer
- The preview/note panel explains that uploaded assets are required for GEN01 sample loading

## Preview Panel

The preview section shows:

- Effective GEN (including normalization sign / named routine behavior)
- Flattened argument list
- Rendered `ftgen` / `ftgenonce` line preview

Use this to verify the exact Csound line before compile.

## Editor Notes Worth Knowing

- `ftgenonce` behavior differs from `ftgen` and the editor shows notes about the differences.
- GEN01 asset-backed generation is resolved by the backend compiler to contained stored asset paths.
- GEN01 asset dependencies are automatically bundled in ZIP exports when referenced.

## Save Behavior

- Save GEN writes the GEN node configuration into patch UI metadata (`graph.ui_layout`).
- Closing without saving leaves the previous GEN node settings unchanged.

## Screenshots



### GEN Editor

NODE 169771C8-D3FE-4AC6-8682-17FD8BE4C78F

#### TABLE GENERATION MODE

OPCODE

ftgen

GEN ROUTINE

GEN10 - Harmonic Sine Partialsv

ROUTINE NUMBER

10

TABLE NUMBER

3

TABLE SIZE

16384

START TIME

0

☒ Normalize table (use positive GEN number)

GEN10 - Harmonic Sine Partialsv: Enter harmonic amplitudes (1st, 2nd, 3rd partial, ...).

#### ROUTINE PARAMETERS

HARMONIC AMPLITUDES

11

ADD

DEL

#### PREVIEW

EFFECTIVE GEN  
10

FLATTENED ARGS  
1

RENDERED LINE  
{ift} ftgen 3, 0, 16384, 10, 1

#### NOTES

GEN01 uses the uploaded asset if present. The backend compiler resolves it to the stored file path.

'ftgenonce' ignores Start Time and uses the same table parameter as the 'ftgenonce' first argument.

CANCEL

SAVE GEN

GEN editor in structured routine mode with live preview panel.

### GEN Editor

NODE 7D43E03D-BF8A-41BF-BF4A-3D6EB5148BA3

#### TABLE GENERATION MODE

OPCODE

ftgen

GEN ROUTINE

GEN01 - Audio Filev

ROUTINE NUMBER

1

TABLE NUMBER

5

TABLE SIZE

32768

START TIME

0

☒ Normalize table (use positive GEN number)

GEN01 - Audio File: Load a sound file into a function table (uploaded asset or custom path).

#### ROUTINE PARAMETERS

UPLOAD AUDIO FILE

CLEAR ASSET

Persisted asset: twopeaks.aiff  
cc978fdd-af9b-4805-b92c-d338f8a63bfd.aiff

FALLBACK SAMPLE PATH (OPTIONAL)  
/absolute/path/file.wav

SKIP TIME

0

FORMAT

0

CHANNEL

0

#### PREVIEW

EFFECTIVE GEN  
1

FLATTENED ARGS  
"twopeaks.aiff", 0, 0, 0

RENDERED LINE  
{ift} ftgen 5, 0, 32768, 1, "twopeaks.aiff", 0, 0, 0

#### NOTES

GEN01 uses the uploaded asset if present. The backend compiler resolves it to the stored file path.

'ftgenonce' ignores Start Time and uses the same table parameter as the 'ftgenonce' first argument.

CANCEL

SAVE GEN

GEN01 table setup using an uploaded backend audio asset.

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### GEN Editor

NODE 79AF0909-152A-4589-B9FC-6F9C704DC17C

CLOSE

#### TABLE GENERATION MODE

OPCODE

ftgen

GEN ROUTINE

GENpadsynth - padsynth algorithm

ROUTINE NUMBER

10

TABLE NUMBER

0

TABLE SIZE

16384

START TIME

0

☒ Normalize table (use positive GEN number)

**GENpadsynth - padsynth algorithm:** Named GEN routine. Enter padsynth parameters and partial amplitude/frequency pairs in Raw Arguments (see Csound GENpadsynth docs for argument order).

#### ROUTINE PARAMETERS

RAW ARGUMENTS

440, 50, 1, 2, 3, 0.6533, 0.64322, 0.23435331, 0.02323, 0.34343, 0.332

Use commas or new lines. Strings should be quoted. Prefix with **expr:** to emit an unquoted expression.

#### PREVIEW

EFFECTIVE GEN

"padsynth"

FLATTENED ARGS

440, 50, 1, 2, 3, 0.6533, 0.64322, 0.23435331, 0.02323, 0.34343, 0.332

RENDERED LINE

{ift} ftgen 0, 0, 16384, "padsynth", 440, 50, 1, 2, 3, 0.6533, 0.64322, 0.23435331, 0.02323, 0.34343, 0.332

#### NOTES

GEN01 uses the uploaded asset if present. The backend compiler resolves it to the stored file path.

`ftgenonce` ignores Start Time and uses the same table parameter as the `ftgenonce` first argument.

CANCEL

SAVE GEN

*GENpadsynth named routine using raw arguments input.*



# Configuration

The Configuration chapter covers both UI-visible settings and practical runtime/deployment behaviors that affect how Orchestron is used.

## What Is Covered

- GUI language switching and integrated help/docs behavior
- Config page engine settings ( `sr` , control rate, buffers, derived `ksmps` ) plus browser-clock latency controls when relevant
- MIDI setup and runtime input binding workflow
- Unified browser-clock playback and latency tuning from the Config page
- Internal loopback versus external host-bridge MIDI binding
- App-state persistence and default values

## Chapter Contents

- [GUI Language and Integrated Help](#)
- [Audio Engine Settings \(Config Page\)](#)
- [MIDI Setup and Inputs](#)
- [Browser-Clock Latency](#)
- [Persistence and Defaults](#)

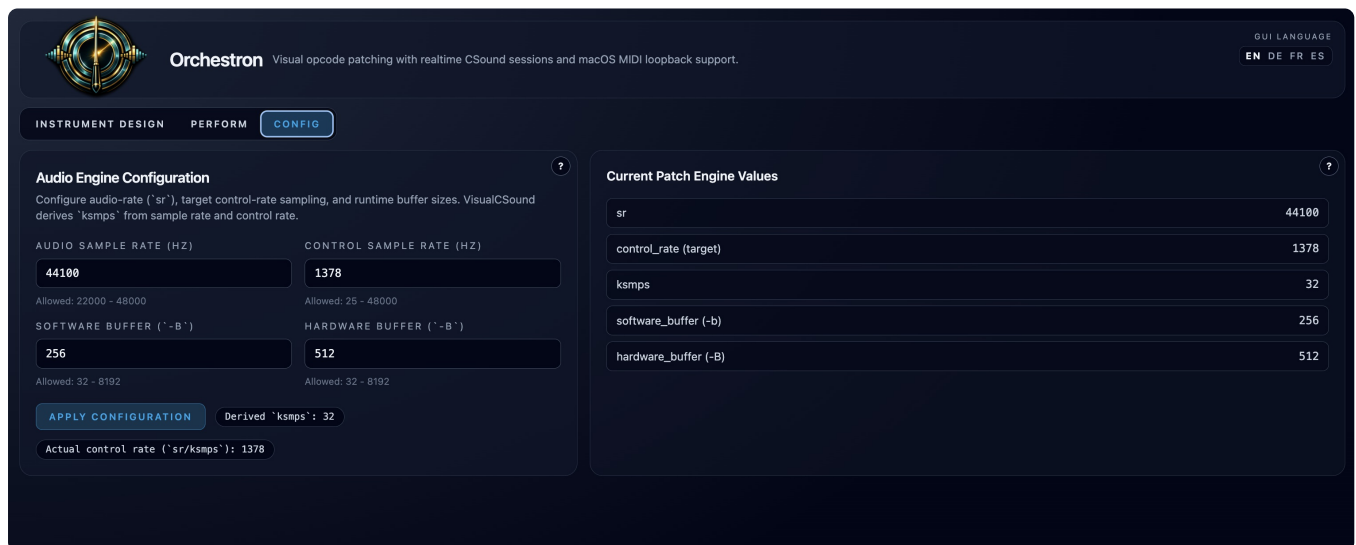
## Configuration Scope Clarification

Orchestron has three kinds of configuration:

- **Patch-level configuration** (stored with a patch, editable in the Config page)
- **Runtime/startup configuration** (browser-clock startup options and optional host MIDI bridge token)
- **Workspace runtime tuning** (browser-clock latency settings stored in app state)

Both are important for successful use.

## Screenshots



Configuration page overview of engine settings. The dedicated Browser-Clock Latency chapter shows the browser-clock-specific tuning section in detail.



## Audio Engine Settings (Config Page)

The Config page controls both patch-level Csound engine settings and browser-clock latency settings for the current workspace.

### Where These Settings Are Stored

Config page changes are stored in the current patch's `graph.engine_config`.

That means:

- different patches can use different engine settings
- switching patches can change the visible Config page values

Browser-clock latency settings are different:

- they apply to the active browser-clock session runtime
- they are stored in app state, not in the patch
- applying them updates the active browser-clock controller without changing patch data

### Editable Settings (Config Page)

#### Patch Engine Section

##### Audio Sample Rate ( `sr` )

- Field label: Audio Sample Rate (Hz)
- Allowed range: **22000 .. 48000**
- Integer only

##### Control Sample Rate ( `control_rate` target)

- Field label: Control Sample Rate (Hz)
- Allowed range: **25 .. 48000**
- Integer only

This is the target control-rate value used to derive `ksmps`.

##### Software Buffer ( `-b` )

- Field label: Software Buffer (`-b`)
- Allowed range: **32 .. 8192**
- Integer only

##### Hardware Buffer ( `-B` )

- Field label: Hardware Buffer (`-B`)
- Allowed range: **32 .. 8192**
- Integer only

### Derived Values Shown In The UI

The Config page previews derived values before applying:

- Derived `ksmps`
- Actual control rate ( `sr / ksmps` )



This helps you see the real runtime effect of your target control-rate choice.

## Current Patch Engine Values (Read-Only Panel)

The right-side panel shows the normalized engine values stored in the patch:

- `sr`
- `control_rate`
- `ksmps`
- `software_buffer`
- `hardware_buffer`

Use this panel to verify what will be used at compile/start time.

## Browser-Clock Latency Section

The Config page shows an additional section for browser-owned audio latency tuning whenever a browser-clock session is active.

See [Browser-Clock Latency](#) for the unified browser-clock workflow explanation and the full UI screenshot of this section.

Editable fields:

- steady queue low water / high water ( ms )
- startup queue low water / high water ( ms )
- underrun recovery boost / maximum underrun boost ( ms )
- maximum blocks per render request
- steady / startup / recovery parallel request limits
- immediate note render blocks
- immediate note render cooldown ( ms )

These values control the browser PCM queue depth and render request behavior, which are the dominant latency/stability knobs on the unified browser-clock path.

The patch engine section still matters because `sr`, `ksmps`, and the configured buffers are used by the backend render worker when it compiles and starts the headless Csound session.

### Field-By-Field Meaning

#### Steady Low Water / Steady High Water

- These define the normal browser PCM queue target after startup is complete.
- Lower values reduce live-play latency.
- Higher values make clicks and underruns less likely.

#### Startup Low Water / Startup High Water

- These define the larger queue target used while the stream is still priming after connect or after an underrun recovery phase.
- They are usually kept above the steady values so the stream can stabilize before dropping into lower-latency playback.

#### Underrun Recovery Boost / Max Underrun Boost

- When the browser detects underruns, VisualCsound temporarily adds extra queue headroom.
- Underrun Recovery Boost controls how much extra buffer is added per underrun event.
- Max Underrun Boost limits how far that temporary recovery buffer can grow.

#### Max Blocks Per Request



- Caps the maximum backend render chunk size requested in one browser-clock render request.
- Smaller values reduce head-of-line delay for live notes.
- Larger values reduce request overhead but can make live interaction feel less immediate.

### Steady / Startup / Recovery Parallel Requests

- These limits control how many render requests may remain in flight at once.
- More parallelism can keep the queue full more aggressively.
- Too much parallelism can also create larger bursts and less predictable latency.

### Immediate Note Render Blocks / Immediate Note Render Cooldown

- A live `note_on` sends a small urgent render request in addition to the normal refill logic.
- `Immediate Note Render Blocks` sets the size of that urgent render burst.
- `Immediate Note Render Cooldown` throttles how often these urgent requests can be sent.
- These controls help manual piano playing feel more responsive, but values that are too aggressive can reintroduce clicks.

### Practical Adjustment Order

1. Start with `steady low/high water`.
2. If clicks remain, raise the steady or startup watermarks slightly.
3. If latency still feels too high, reduce `max blocks per request` carefully.
4. Only then tune immediate note render and parallel request limits.

## Validation Behavior

The page validates:

- integer format
- allowed range
- high-water fields must be greater than their matching low-water fields

`Apply Configuration` is disabled until all visible values are valid.

## Apply Configuration

`Apply Configuration` writes the new engine settings into the current patch state.

Practical effect:

- future compile/start operations for the patch use the updated engine config
- the patch must still be saved if you want the change persisted in the patch library

`Apply Browser-Clock Settings` writes the browser-clock latency settings into app state and refreshes the active browser-clock controller if one is connected.

## Other Engine Fields Present But Not Edited In UI

The patch engine config also contains values such as:

- `nchnls` (default 2)
- `0dbfs` (default 1)

These are part of the patch model, but the current Config page focuses on the user-facing timing and buffer controls listed above.

## Tuning Tips



- Lower buffers reduce latency but increase glitch risk.
- Higher buffers improve stability but increase latency.
- Use the Runtime panel and live testing to find a stable setting for your machine and patch complexity.
- On the browser-clock path, start by adjusting browser-clock low/high water, render blocks, and parallel request limits before changing patch `-b/-B`.
- For live MIDI sessions, prefer `ksmps <= 32`. The runtime warns when a live session starts with a larger `ksmps` because MIDI execution is still quantized to the `k-period`.

## Screenshots

The screenshot shows the Orchestron Config page with the following settings:

| Parameter                | Value |
|--------------------------|-------|
| AUDIO SAMPLE RATE (HZ)   | 44100 |
| CONTROL SAMPLE RATE (HZ) | 1378  |
| SOFTWARE BUFFER (-B)     | 256   |
| HARDWARE BUFFER (-B)     | 512   |

Buttons: **APPLY CONFIGURATION**, **Derived `ksmps`: 32**, **Actual control rate (`sr/ksmps`): 1378**

Current Patch Engine Values:

| Parameter             | Value |
|-----------------------|-------|
| sr                    | 44100 |
| control_rate (target) | 1378  |
| ksmps                 | 32    |
| software_buffer (-b)  | 256   |
| hardware_buffer (-B)  | 512   |

Config page audio engine settings in a valid state.

The screenshot shows the Orchestron Config page with the following settings:

| Parameter                | Value |
|--------------------------|-------|
| AUDIO SAMPLE RATE (HZ)   | 44100 |
| CONTROL SAMPLE RATE (HZ) | 8800  |
| SOFTWARE BUFFER (-B)     | 256   |
| HARDWARE BUFFER (-B)     | 10240 |

Buttons: **APPLY CONFIGURATION**, **Derived `ksmps`: 5**, **Actual control rate (`sr/ksmps`): 8820**

Validation messages:

- Allowed: 22000 - 48000
- Allowed: 25 - 48000
- Allowed: 32 - 8192
- Hardware Buffer (-B) must be between 32 and 8192.

Config page validation messages with invalid values entered.



The dedicated [Browser-Clock Latency](#) chapter shows the browser-clock tuning section in detail.



# Persistence and Defaults

Orchestron persists both library data (patches/performances) and a separate app-state snapshot (workspace state).

Persistent JSON documents are bounded before storage so direct API clients cannot grow request memory or the SQLite database without limit. Defaults are 8 MiB for app state ( `VISUALCSOUND_APP_STATE_MAX_BYTES` ), 4 MiB for saved patch graphs ( `VISUALCSOUND_PATCH_GRAPH_MAX_BYTES` ), 1 MiB for patch UI layout metadata ( `VISUALCSOUND_PATCH_UI_LAYOUT_MAX_BYTES` ), 8 MiB for performance configs ( `VISUALCSOUND_PERFORMANCE_CONFIG_MAX_BYTES` ), and 64 KiB for any single nested JSON string ( `VISUALCSOUND_PERSISTED_JSON_STRING_MAX_BYTES` ).

## Two Persistence Layers (Important)

### 1. Library Data (Explicit Save)

These are stored when you click save actions:

- Patches (Instrument Design Save )
- Performances (Perform page Save Performance )

### 2. App-State Snapshot (Automatic)

Orchestron also persists a workspace/app-state snapshot in the backend to restore your working context after reload.

Persisted app state includes (current implementation):

- active page ( Instrument Design / Perform / Config )
- GUI language
- instrument tabs (including editable patch snapshots)
- active instrument tab
- sequencer state (tracks, controller sequencers, arpeggiators, piano rolls, MIDI controllers)
- performance workspace metadata ( `currentPerformanceId` , name, description)
- sequencer instrument bindings (stable rack instance IDs)
- explicit audio routes, Master selection, insert ownership and mixer strip/send values (app-state version 2)
- active MIDI input selection reference
- browser-clock latency settings for the current workspace/runtime path

New sessions default that MIDI input reference to `internal:loopback` unless a different external input is explicitly bound.

## What This Means In Practice

- You can reload the app and continue from a similar workspace state.
- Unsaved edits may reappear because they exist in the app-state snapshot.
- Unsaved edits are **not** the same as saved library data.

Always use explicit save actions when you want stable, shareable library entries.

## Default New Patch Values

New opens the [built-in/template chooser](#). The selected starter supplies its name, graph, Activation and audio interface.

Choose Empty patch for an empty graph ( `nodes = []` , `connections = []` ). All starters use these default engine settings:

- `sr = 48000`
- `control_rate = 1500`
- `ksmps = 32`
- `nchnls = 2`
- `software_buffer = 128`



- `hardware_buffer = 512`
- `0dbfs = 1`

## Saved mixes and older performances

Save Performance, Load Performance, Clone and native performance bundles retain version 11 audio routing and mixer settings, including Master, dedicated inserts, sends and pre/post selection. App-state restore also retains the current mix. These are snapshots; recording mixer automation is not included.

Opening versions 1–10 converts the old Level values to audio gain in dB and resolves legacy connections into explicit routes. Missing Level means 0 dB. The migration notice explains that Level no longer scales MIDI velocity, so velocity-sensitive instruments may change timbre. Check the mix and save the migrated performance. See [Mixer persistence and compatibility](#).

## Compile Status vs Persistence

The compile status badge (`compiled`, `pending changes`, `errors`) indicates compile state for the current patch snapshot. It does **not** indicate whether the patch has been saved to the patch library.

## Runtime Defaults

- runtime audio mode is fixed to `browser_clock`
- the default session MIDI binding is `internal:loopback`
- external host MIDI is disabled unless `VISUALCSOUND_HOST_MIDI_TOKEN` is configured and a helper connects
- runtime session creation is bounded by `VISUALCSOUND_SESSION_MAX_ACTIVE`, `VISUALCSOUND_SESSION_MAX_ACTIVE_PER_CLIENT`, `VISUALCSOUND_SESSION_CREATE_RATE_PER_MINUTE`, and `VISUALCSOUND_SESSION_CREATE_RATE_BURST`; stopped idle sessions are deleted after `VISUALCSOUND_SESSION_IDLE_TIMEOUT_SECONDS`
- session event WebSocket observers are bounded by `VISUALCSOUND_SESSION_EVENT_WS_MAX_SUBSCRIPTIONS_TOTAL`, `VISUALCSOUND_SESSION_EVENT_WS_MAX_SUBSCRIPTIONS_PER_SESSION`, `VISUALCSOUND_SESSION_EVENT_WS_CONNECT_RATE_PER_MINUTE`, and `VISUALCSOUND_SESSION_EVENT_WS_CONNECT_RATE_BURST`

## Recommended Habits

1. Use app-state restore for convenience.
2. Use `Save / Save Performance` for intentional versions.
3. Use `Export` bundles for backups or sharing across machines.